Indira Gandhi
National Open University
School of Computer and
Information Sciences

MCS-202
Computer Organisation

Block

2

Memory and Input/Output Organisation

UNIT 5
The Memory System

UNIT 6
Advance Memory Organisation

UNIT 7
Input/Output Organisation

UNIT 8
Device Technology

FACULTY OF THE SCHOOL

Prof P. V. Suresh, Director
Dr Shashi Bhushan
Mr M. P. Mishra
Prof. V. V. Subrahmanyam
Mr Akshay Kumar
Dr Sudhansh Sharma

PROGRAMME/COURSE DESIGN COMMITTEE

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Amity School of Computer Sciences,
Noida Shri Amrit Nath Thulal
Amity School of Engineering and Technology
New Delhi
Dr. Om Vikas(Retd),
Ministry of ICT, Delhi
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SOCIS, IGNOU, New Delhi
Shri M. P. Mishra,
SOCIS, IGNOU, New Delhi
Dr. Sudhansh Sharma,
SOCIS, IGNOU, New Delhi

BLOCK PREPARATION TEAM

Dr Zahid Raja *(Content Editor)*
Jawaharlal Nehru University
New Delhi

Mr Vikas Mittal *(Course Writer – Unit 5 & 6)*
Maharaja Agrasen College,
University of Delhi
Delhi

Dr Mohammad Sajid (*Course Writer – Units 7 & 8)*
Department of Computer Science
Aligarh Muslim University
Aligarh

(Language Editor) School of Humanities
IGNOU

Course Coordinator: Mr Akshay Kumar

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BLOCK 2 INTRODUCTION

In the first block this course, you are introduced to basic computer architectures, instruction execution, data representation and logic circuits of a computer system. This Block covers two of the most important units that are part of computer system architecture, viz. the memory system and Input/output system. This block consists of 4 units (unit 5 to unit 8).

Unit 5 explains the basic memory hierarchy of a computer system. It also explains different kinds of memories used in a computer system. The logic of RAM and ROM has also been explained in details. The unit also discusses secondary memories like magnetic and optical disks in details.

Unit 6 provides details on advance memory organisation. It provides details on different aspects of cache memory and main memory to cache mapping schemes. This unit also discusses the concept of interleaved memory, associative memories and virtual memories used in a computer system.

Unit 7 explains the basic interfaces and mechanisms that are used to perform input and output. This unit also presents the concept of DMA and input/output processor.

Unit 8 introduces you to basic technology of some of the popular input/output devices including keyboard, mouse, monitor, printer etc.

A course on computers can never be complete because of the existing diversities of the computer systems. Therefore, you are advised to read through further readings to enhance the basic understanding that you will acquire from the block.

Further Readings For The Block

1) Mano M Morris, *Computer System Architecture*, 3rd Edition/Latest Edition, Prentice Hall of India Publication, Pearson Education Asia

2) Stallings W., *Computer Organization & Architecture: Designing For Performance*, 10th/11th Edition, Pearson Education Asia

3) Hennessy/Patterson, *Computer Organization and Design : The Hardware/ Software Interface*; 5th/6th Edition, Morgan Kaufmann.

UNIT 5 THE MEMORY SYSTEM

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5.0 INTRODUCTION

In the previous block, fundamentals of a computer system were discussed. These fundamentals included discussion on von-Neumann architecture based machines, instruction execution, representation of digital data and logic circuits etc. This Block explains the most important component of memory and Input/output systems of a computer. This unit covers the details of the Memory. This unit discusses issues associated with various components of the memory system, the design issues of main memory and the secondary memory. Various characteristics of secondary memory and its types that are used in a computer system, would also be discussed. The unit also defines how multiple disks can be used to create a redundant array of disks that can be used to provide a faster and reliable storage.

5.1 OBJECTIVES

After going through this Unit, you will be able to:

- explain the key characteristics of various types of memories and memory hierarchy;
- explain and differentiate among various types of random access memories;
- explain the characteristics of secondary storage devices and technologies;
- explain the latest secondary storage technologies;
- identify the various levels of RAID technologies

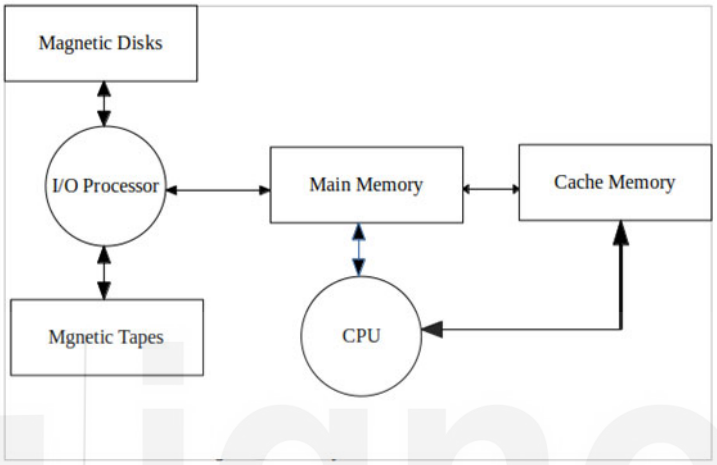
5.2 THE MEMORY HIERARCHY

In computers, memory is a device used to store data in binary form. Smallest unit of binary data is called ‘bit’. Each bit of binary data is stored in a different cell or storage unit and collection of these cells is defined as the memory. A memory system is composed of a memory of fixed size and procedures which tells how to access the data stored in the memory. Based on the persistence of the stored data, memory is classified into two categories:

- Volatile memory: which loses its data in the absence of power.
- Non-volatile memory: Do not lose data when power is switched off.

Another classification of memory devices, which is also the objective of this unit is based on the way they interact with the CPU which can be determined from figure 5.1 Main/ Primary memory interact directly with the CPU e.g. RAM and ROM.

Auxiliary/ secondary memory need I/O interface to interact with the CPU e.g. magnetic disks and magnetic tapes. There are other memories like cache and registers, which directly interacts with the CPU. Such memories are used to speed up the program execution. For execution, a program must be loaded into the main memory and should be stored on the secondary storage when it completes its execution. Auxiliary memory is used as a backup storage, whereas main memory contains data and program only when it is required by the CPU.



Various memory devices in a computer system forms a hierarchy of components which can be visualised in a pyramidal structure as shown in Figure 5.2. As you can observe in the Figure 5.2 that at the bottom of the pyramid, you have magnetic tapes and magnetic disks; and registers are at the top of the pyramid. Main memory lies at the middle as it can interact directly with the CPU, cache memory and the secondary memory. As you go up in the pyramid, the size of the memory device decreases, the access speed, however, increases and cost per bit also increases. Different memories have different access speeds. CPU registers or simply registers are fastest among all and are used for holding the data being processed by the CPU temporarily but because of very high cost per bit they are limited in size. Instruction execution speed of the CPU is very high as compared to the data access speed of main memory. So, to compensate the speed difference between main memory and the CPU, a very high speed special memory known as cache is used. The cache memory stores current data and program plus frequently accessed data which is required in ongoing instruction execution.

You may note the following points about memory hierarchy:

- ✓ The size of the memory increases as you go down the memory hierarchy.
- ✓ The cost of per unit of memory increases as you go up in the memory hierarchy i.e. Memory tapes and auxiliary memory are the cheapest and CPU Registers are the costliest amongst the memory types.
- ✓ The amount of data that can be transferred between two consecutive memory layers at a time decreases as you move up in the pyramid. For example, from main memory to Cache transfer one or few memory words of size in Kilobytes are accessed at a time, whereas in a hard disk to main memory transfer, a block data of size of 1 Megabyte is transferred in a single access.
- ✓ One interesting question about the memory hierarchy is why having faster smaller memories does not slow down the computer? This is primarily due to the

fact that there is very high probability that a program may access the instructions and data in the closed vicinity of presently executing instruction and data. This concept is further explained in next unit.

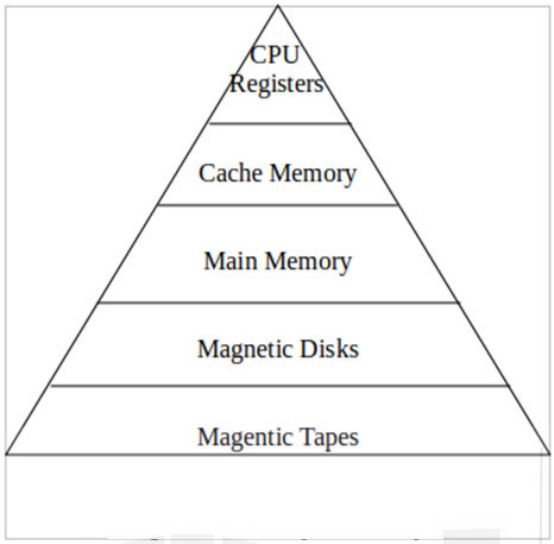


Figure 5.2: Memory Hierarchy

In subsequent sections and next unit, we will discuss various types of memories in more detail.

5.3 SRAM, DRAM, ROM, FLASH MEMORY

The main memory is divided into fixed size memory blocks called *words*. Size of the memory word may be limited by the communication path and the processing unit size. As word size/ length denotes the amount of bits that can be processed by the processor at one time. Each memory word is addressed uniquely in the memory. A 32-bit processor uses a word size of 32 bits whereas 64-bit processor uses a word of 64 bits. RAM (random access memory) is a volatile memory i.e. content of the RAM vanishes when power is switched off. RAM is a major constituent of the main memory. Both read and write operations can be performed on RAM, therefore, it is also known as read-write memory. Access time of each memory word is constant in random access memory. RAM can be constructed from two types of technologies - Static Random Access Memory (SRAM) and Dynamic Random Access Memory (DRAM). The main difference being that DRAM loses its content even if power is on, therefore requires refreshing of stored bits in DRAM. Thus, DRAM is slower than SRAM, however, the DRAM chips are cheaper. In general, DRAM is used as the main memory of the computer, while SRAM is used as the Cache memory, which is discussed in details in the next unit.

SRAM

SRAM can be constructed using flip-flops. It is a sequential circuit. A SRAM cell using SR flip flop is shown in figure 5.3. As you can observe, this sequential circuit has three inputs: *select, read/write, and input* and single output: *output*. When s*elect* input is high “1” circuit is selected for read/write operation and when *select* input is low “0” neither read nor write operation can be performed by the binary cell. Thus, *select* input must be high in order to perform read/write operation by the binary cell. Binary cell reads a bit when *read/write* input is low “0” and writes when read/write input is high “1”. Third input *input* is used to write into the cell. The only caution over here is that when *read/write* input is low “0” i.e. we want to perform a read operation, then read operation must not be affected by the input *input.* This is ensured by

inverted *input* to the first AND gate which guarantees the input to both R and S to be low and thus prevents any modification to the flip flop value. The characteristic table of SR flip flop is given in Unit 4 Block 1 for better understanding of the functioning of the binary cell.

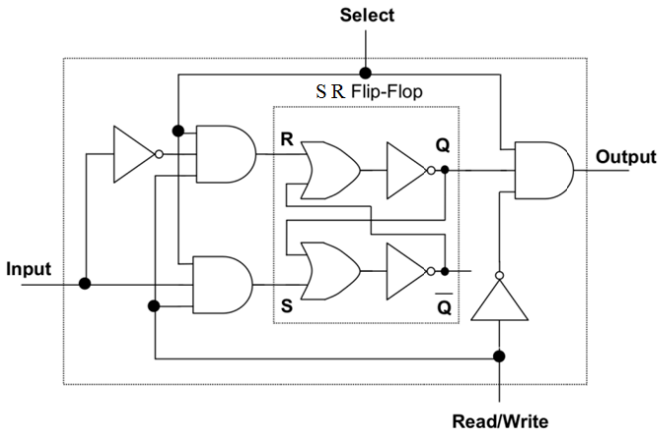


Figure 5.3: Logic Diagram of RAM cell

Read operation: *select* is high "1", *read/write* is low "0" and *input* is either low "0" or high "1" then input to R and S will be 0 and flip flop will keep its previous state and that will be the output.
Write operation: select is high "1", *read/write* is high "1" and if *input* is low "0" then R will be high "1" and S will be low "0" and flip flop will store "0" and if *input* is high "1" then R will go low"0" and S will go high "1" and flip flop will store "1".

A RAM chip is composed of several read/write binary cells. A block diagram of 2^mx n RAM is shown in Figure 5.4. The RAM shown has a total capacity of 2^m words and each word is n bits long e.g. in 64 x 4 RAM, the RAM has 64 words and each word is 4 bits long. To address 64 i.e. 2^6words, we need 6 address lines. So in a $2^m \times n$ RAM, we have 2^m words where each word has n bits and RAM has m-bit address which requires m address lines. The RAM is functional only when chip select (CS1) signal =1 and $\overline{CS2}$ = 0. If chip select signal is not enabled or chip select signal is enabled and neither read nor write input is enabled then data bus will in high impedance state and no operation can be performed. During high impedance state, other input signals will be ignored which means output has no logical significance and does not carry a signal

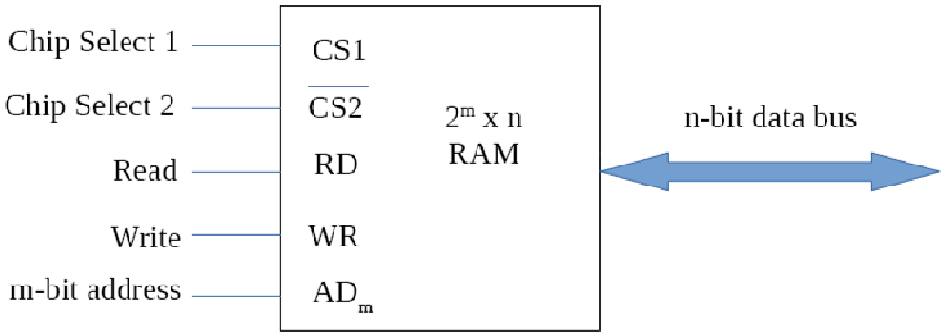


Figure 5.4: Block Diagram of 2^mx n RAM

DRAM

Dynamic Random Access Memory (DRAM) is a type of RAM which uses 1 transistor and 1 capacitor (1T1C cell) for storing one bit. A block diagram of a single DRAM cell is shown in Figure 5.5. In DRAM, transistor is used as a gate which opens and closes the circuit and thus stops and allows the current to flow. Charging level of the capacitor is used to represent the bit "1" and bit "0". As capacitors tends to discharge in a very short time period DRAM cells need to be refreshed periodically to store the binary information despite continuous power supply. Hence they are called dynamic random access memory. With low power consumption and very compact in size because of 1T1C architecture DRAM offers larger storage capacity in a single chip. Each DRAM cell in the memory is connected with Word Line (Rows) and Bit Line (Columns) as shown in Figure 5.5. Word line (rows) controls the gates of the transfer lines while Bit lines (columns) are connected to sense amplifiers i.e. to determine "0" or "1".

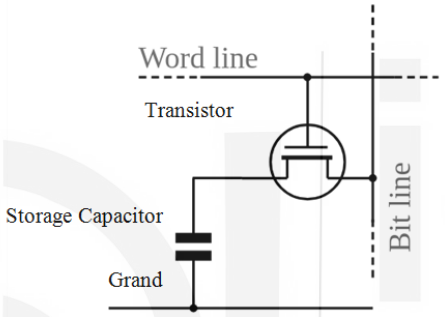


Figure 5.5: A DRAM cell

Figure 5.6 presents the general block diagram of $2^M \times 2^M \times N$ DRAM, where binary cells are arranged in a square of $2^M \times 2^M$ words of N bit each. For example, 4 megabit DRAM is represented in a square arrangement of (1024×1024) or $(2^{10} \times 2^{10})$ words of 4 bit each. Thus, in the given example we have 1024 horizontal/ word lines and 1024×4 column/ bit lines. In other words, each element, which consists of 4 bits of array, is connected by horizontal row lines and vertical column lines.

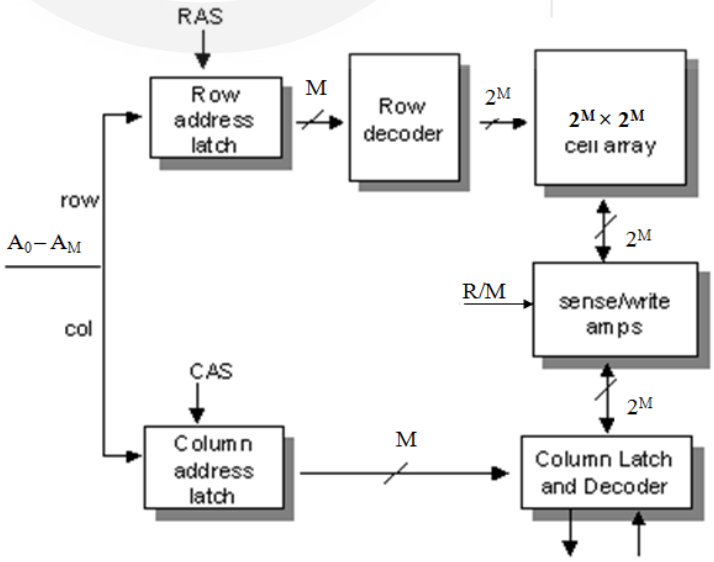


Figure 5.6: Block Diagram of DRAM

Selection and role of various signals for read and write operation is as follows:

1. RAS (Row Address Strobe): On the falling edge of RAS signal, it opens or strobe the address lines (rows) to be addressed.

2. /CAS (Column Address Strobe): Similar to /RAS, on the falling edge, this enables a column to be selected as mentioned in the column address from the rows opened by the /RAS to complete the read-write operation.

3. R/(/W), Write enable: This signal determines whether to perform a read operation or a write operation. While the signal is low, write operation is enabled and data input is also captured on falling edge of /CAS whereas high enables the read operation.

4. Sense amplifier compares the charge of the capacitor to a threshold value and returns either logic "0" or logic "1".

For a read operation once the address line is selected, transistor turns ON and opens the gate for the charge of the capacitor to move to the bit line where it is sensed by the sense amplifier. Write operation is performed by applying a voltage signal to the bit line followed by the address line allowing a capacitor to be charged by the voltage signal.

ROM (Read-Only Memory)

Another constituent of the main memory is ROM (read only memory). Unlike RAM, which is read-write memory and volatile, ROM's are read only and non-volatile memory i.e. content of the ROM persist even if power is switched-off. Once data is stored at the time of fabrication, it cannot be modified. This is why, ROM is used to store the constants and the programs that are not going to change or get modified during their lifetime and will reside permanently in the computer. For example, bootstrap loader, which loads the part of the operating system from secondary storage to the main memory and starts the computer system when power is switched on, is stored in ROM.

A block diagram of $2^m \times$ n ROM looks similar to that of RAM. As ROM is a read-only memory there is no need of explicit read and write signals. Once the chip is selected using chip select signals a data word is read and placed on to the data bus. Hence, in the case of ROM, you need an unidirectional data bus i.e. only in output mode as shown in figure 5.7. Another interesting fact about ROM is that, ROM offers more memory cells and thus, memory as compared to the RAM for same size chip.

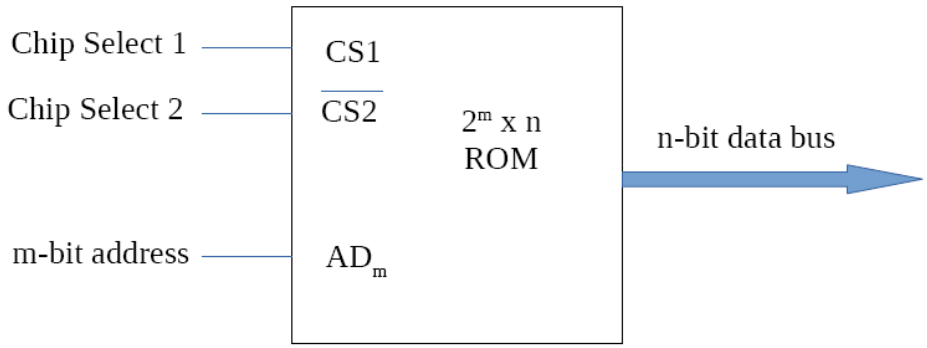


Figure 5.7: Block Diagram of 2^mx n ROM

As shown in Figure 5.7, $2^m \times$ n ROM has 2^m words of n bits each for which it has *m* address lines and *n* output data lines. For example, in 128 × 8 ROM, you have 128 memory words of 8-bit each. For 128 × 8 ROM i.e. $2^m = 2^7$, m = 7, you need 7 address lines (minimum number of bits required to represent 128) and 8-bit output data bus. Figure 5.8 shows a 32×8 ROM.

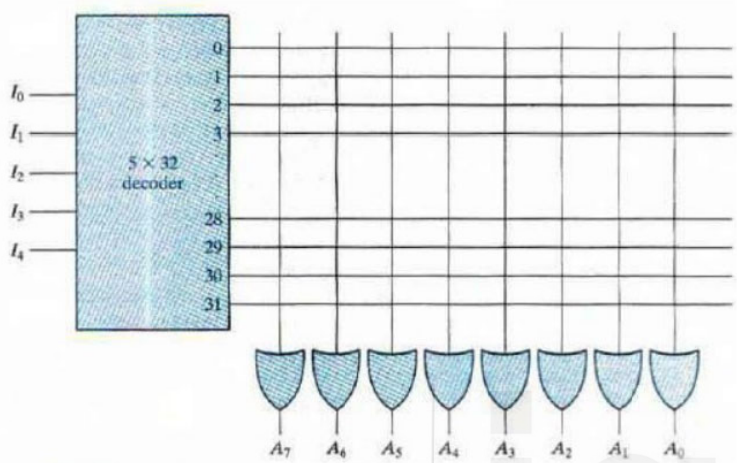


Figure 5.8: Internal diagram of 32x8 ROM

Unlike RAMs, which are sequential circuits, ROMs are combinational circuits. Typically, to design a RAM of specific size you need a decoder and OR gates. For example, to design a ROM of size 32 x 8 bits you need a decoder of size 5×32 and 8 OR gates. 5×32 decoder will have 5 input lines, which will act as 5 address lines of the ROM, the decoder will convert 5-bit input address to 32 different outputs. Figure 5.8 shows the construction of 32 × 8 ROM using 5×32 decoder and eight OR gates for data output. ROMs of other sizes can be constructed similarly. For example, to construct a ROM of 64 × 4 ROM, you need a 6×64 decoder and four OR gates and to construct a ROM of size 256×8, you need 8×256 decoder and 8 OR gates.

As discussed, ROMs are non-volatile memory and content of the ROM once written, cannot be changed. Therefore, ROMs are used to store the look-up tables for constants to speed up the computation. In addition, ROM can store the boot loader programs and gaming programs. All this requires, zero error in writing of such programs and therefore, ROM device fabrication requires very high precision. Constructing a ROM, as shown in figure 5.8, requires decision about which interconnections in the circuit should be open and which interconnections should be closed. There are four ways you can program a ROM which are as follows:

1. Mask ROM (MROM): Masking of ROM is done by the device manufacturer in the very last phase of the fabrication process on customers special request. Mask ROMs are customised as per the user requirements, thus, are very costly as different masks are required for different specifications. Because of very high cost of masking, this customization is generally used in manufacturing of ROM at very large scale.
2. Programmable ROM (PROM): MROMs are not cost effective for small productions, PROMs are preferred for small quantities. PROMs are programmed using a special hardware which blow fuses with a very high voltage to produce logic “0” and intact fuse defines logic “1”. The content of PROM is irreversible once programmed.
3. Erasable PROM (EPROM): EPROMs are third type of ROMs which are restructured or reprogrammed using shortwave radiations. An ultraviolet light for

a specific duration is applied to the EPROM, which destroys/ erases the internal information and after which EPROMs can be programmed again by the user.

4. Electrically EPROM (EEPROM) : EEPROMs are similar to EPROMs except of using ultraviolet radiations for erasing PROM, EEPROM uses electrical signals to erase the content. EEPROM can be erased or reprogrammed by the user without removing them from the socket.

Flash Memory

Flash memory is a non-volatile semiconductor memory which uses the programming method of EPROM and erases electrically like EEPROM. Flash memory was designed in 1980s. Unlike, EEPROM where user can erase a byte using electrical signals, a section of the memory or a set of memory words can be erasable in flash memory and hence the name flash memory i.e. which erases a large block of memory at once. Flash memory is easily portable and mechanically robust as there is no mechanical movement in the memory to read-write data. Flash memory is widely used in USB memory, SD and micro SD memory cards used in cameras and mobile phones respectively.

There are two types of flash memory, viz. NAND flash memory, where read operation is performed by paging the contents to the RAM i.e. only a block of data is accessed not an individual byte or word; and NOR flash memory, which are able to read an individual memory byte/word or cell.

The features of various semiconductor memories are summarised in the Table 1.

Memory	Type	Erase Mechanism/ Level	Write Mechanism	Volatile/ Non- Volatile
Random-access Memory (RAM)	Read–Write	Electrical/ Byte	Electrical	Volatile
Read –only Memory (ROM)	Read–Only	Not Applicable	Masks	Non-volatile
Programmable ROM (PROM)	Read–Only	Not Applicable	Electrical	Non-volatile
Erasable PROM (EPROM)	Read-mostly	UV light/ Chip	Electrical	Non-volatile
Electrically Erasable (EEPROM)	Read-mostly	Electrical/ Byte	Electrical	Non-volatile
Flash memory	Read-mostly	Electrical/ Block	Electrical	Non-volatile

Table 1: Features of Semiconductor Memories

Check Your Progress 1

1. Differentiate among RAM, ROM, PROM and EPROM.

   ……

   ……

2. What is a flash memory? Give a few of its typical uses.

   ……

   ……

3. A memory has a capacity of 16K × 16
   (a) How many data input and data output lines does it have?
   (b) How many address lines does it have?

   ……

   ……

4. A DRAM that stores 4K bytes on a chip and uses a square register array. Each array is of size 4 bits. How many address lines will be needed? If the same configuration is used for a chip which does not use square array, then how many address lines would be needed?

   ……

   ……

5. How many RAM chips of size 256K × 4 bit are required to build 1M Byte memory?

   ……...

   ……...

5.4 SECONDARY MEMORY AND CHARACTERISTICS

In previous section, we have discussed various types of random access and read only memories in detail. RAM and ROM together make the main memory of the computer system. You know that a program is loaded into main memory to complete its execution. Computational units or CPU can directly interact with the main memory. Hence, faster main memory, which can match with the speed of CPU, is always desirable. In the previous section configuration of two types of RAMs, viz SRAM and DRAM were discussed. As you may observe the SRAM consists of flip-flop based circuits, therefore, is quite fast in comparison to DRAM However, the cost per bit of DRAM is much less than the SRAM. Thus, you may observe the size of main memory is much more than cache. It is discussed in more details in the next unit. To achieve high speed, cost per bit of main memory is generally high which also limits its size.

On the other hand, as we have discussed, RAM, which is a major constituent of the main memory, is volatile i.e. content of the main memory is lost when power is switched off. Because of above mentioned issues, you need a low cost and high capacity, non-volatile memory to store program files and the data for later use. Secondary memory devices which are at the bottom of the memory hierarchy pyramid are ideal for the said purpose. We will discuss various secondary storage devices in this section.

5.4.1 Hard Disk Drive

In the era of Big Data, in which variety of data is generated rapidly, large secondary storage has become an important component of every computer system. Today, hard disk drives (HDD) is the primary type of secondary storage. The size of hard disk drives in modern computer system ranges from Gigabytes (GB) to Terabytes (TB). Internal hard drives extends the internal storage of a computer system whereas external hard drives are used for back up storage.

HDD are electro-mechanical storage devices, which store digital data in the form of small magnetic fields induced on the surface of the magnetic disks. Data recorded on the surface of magnetic disks is read by disks read/write head, which transforms magnetic signal to electrical signal for reading and electrical signal to magnetic field

for writing. HDD is composed of many concentric magnetic disks mounted on a central shaft as shown in Figure 5.8.

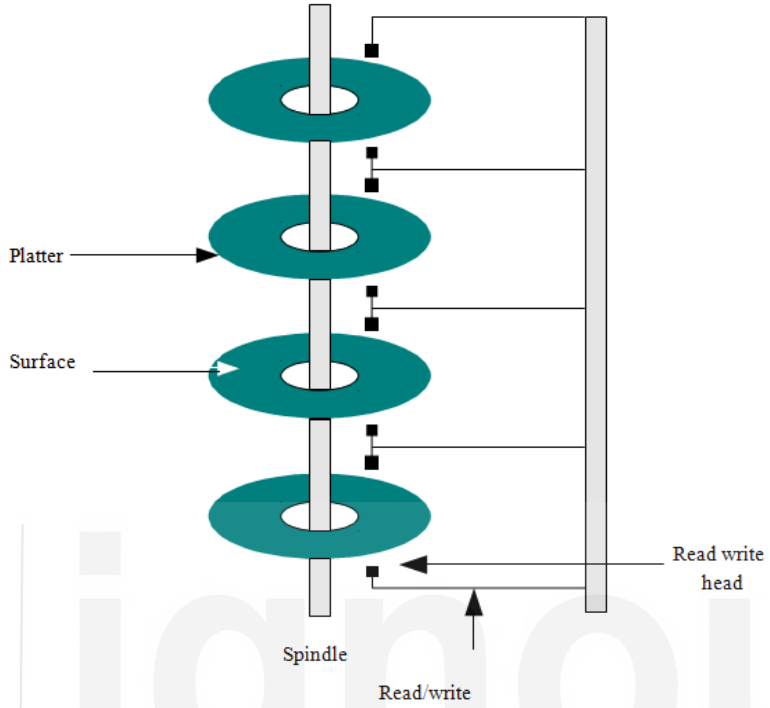


Figure 5.8: Internal structure of Hard disk drives (HDD)

Figure 5.8 shows the internal structure of an HDD. An HDD is made of several concentric magnetic disks mounted on a central shaft called spindle. Each magnetic disk is made of either glass or an aluminium disk called platter. Each platter is coated with ferromagnetic material for storing data. Platter itself is made of non-ferromagnetic material so that its own magnetic field should not interfere the magnetic field of the data. Generally, both sides of the platter is coated with magnetic material for good storage capacity at low cost.

Data recorded on the disk is accessed through a read/write head. Each side of the disk has its own read write head. Each read/write head is positioned at a distance of tens of nanometer called flying height to the platter so that it can easily sense or detect the polarization of the magnetic field.

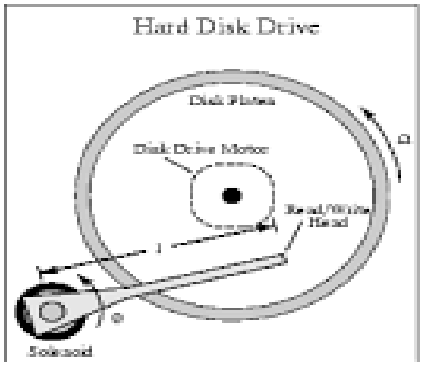


Figure 5.9: Read/ Write Head

Two motors are used in HDD. First one is called the spindle motor, which is used to rotate the spindle on which all the platters are mounted. Second motor is used to move

the read/write heads across the entire surface of the platter radially and is called actuator or access arm.

Magnetic Read and Write Mechanisms

During a read/ write operation, read/write head is kept stationary while platter is rotated by the spindle motor. As you know, data on the disk is recorded in the form of magnetic field. The current is passed through the read/write head which induces a magnetic field on the surface of platter and thus, records a bit on the surface. Different directions of current generates magnetic fields with different polarities and hence are used for storing "1" and "0". Similarly, to read a bit from the surface, the magnetic field is sensed by the read/write head which produces an electric current of the same polarity and hence the bit value is read.

Data Organization and Formatting

As discussed and shown in figure 5.8, hard disk drives consists of number of concentric platters which are mounted on a spindle forming a cylindrical structure. Data is written in the form of magnetic fields on both surfaces of these platters and is read by read/write head which is connected to an actuator. In this section, we will discuss structure of magnetic disk in detail.

Structure of the disk is shown in figure 5.10. As you know, each magnetic disk is a circular disk mounted on a common spindle but entire disk space is not used for data. Disk surface is divided in to thousands of concentric circular regions called *tracks.* The width of every track is kept the same. Data is stored in these tracks. Magnetic field of one track should not affect the magnetic region in the other track thus two tracks are kept apart with each other by a constant distance. Further, each track is divided into number of sectors and two sectors are kept apart using inter-sector gap. Data is stored in these sectors. Each track forms a cylindrical structure with other tracks on other platters below or above it. For example, an outer most cylinder will have outer most track of all the platters. So, if we have n tracks in a platter then there will be n concentric cylinders too.

Components of the drive are controlled by a disk controller. Now a days, disk controllers are built in to the disk drive. A new or blank magnetic disk is divided into sectors. Each sector has three components: header, 512 byte (or more) data area and a trailer. This process of is called physical / low level formatting. Header and trailer contains metadata about the sectors e.g. sector number, error correcting code etc. Disk controller uses this information whenever it writes or reads a data item on to a sector.

Data is stored in series of logical blocks. The disk controller maps the logical blocks on to the physical disk space and also manages sectors which have been used for storing data and which are still free. This is done by the operating system after partitioning the disk in to one or more groups of cylinders. Disk controller stores the initial data structure file of every sector on to the disk. This data structure file contains a list of used and free sectors, list of bad sectors etc. Windows uses File Allocation Table (FAT) for the said purpose.

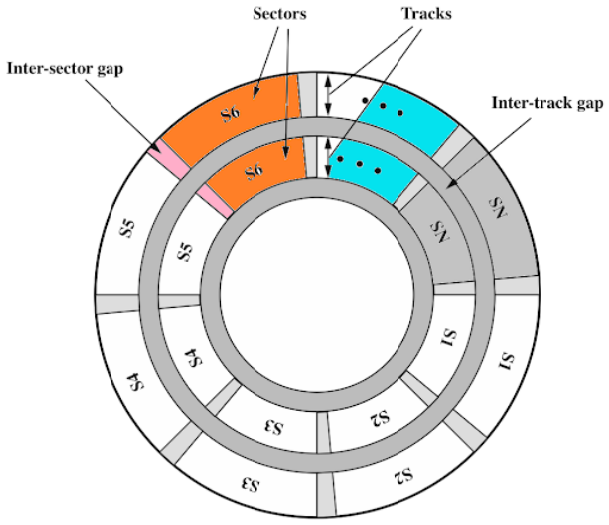


Figure 5.10: Magnetic Disk Structure of CAV

There are two arrangements with which platters are divided into tracks and sectors. The first arrangement is called as *constant linear velocity (CLV)*, in which the density of bits per track is kept uniform, i.e. outer tracks are longer than the inner tracks and hence contains more number of sectors and data. Outermost tracks are generally 40% longer than the innermost track. In this arrangement, in order to maintain uniform bit/ data rate among tracks, the rotation speed is increased from outermost to inner most track. This approach is used by CD-ROM and DVD-ROM drives.

In another approach called as *constant angular velocity (CAV)*, the density of bits / data per track is decreasing as we move from innermost track to outermost track by keeping the disk rotation speed constant. As disk is moving at a constant speed, the width of the data bits increases in the outer tracks, which results in the constant data rate. Figure 5.10 shows that the width of sectors in outer tracks is increasing and density of bits is decreasing.

Disk Performance

Data is read and written on the disks by the operating system for usage at later stage. A disk stores the programs and related data. However, disk is a much slower device and the programs stored on it cannot be executed by the processing unit directly. Therefore, the programs and its related data, which are not in the main memory, are loaded in the main memory from the secondary storage. Since, the speed of disk read/write is very slow in compared to RAM, time to read or write a byte from or on to the disk affects the computer overall efficiency. Therefore, *in a single read/write operation on disk data of one or more sectors is transferred to/from the memory*. An operating system, in general, request for read/write to one or more sectors on the disk. The time taken by the disk to complete a read/ write request of the operating system is known as disk access time. There are number of factors which affect the performance of the disk. These factors are:

1. Seek Time: It is defined as a time taken by the read/write head, or simply as a head, to reach the desired track on which the requested sector is located. Head should reach the desired track in minimum time. Shorter seek time leads to faster I/O operation.
2. Rotational Latency: Since, every track consists of a number of sectors, therefore, the read/write operation can be completed only when the desired sector is available under the read/write head for the I/O operation. It depends on the

rotational speed of the spindle and is defined as a time taken by a particular sector to get underneath the read/write head.

3. Data Transfer Rate: Since, large amount of data is transferred in one read/write operation, therefore, the data transfer rate is also a factor for I/O operation. It is defined as the amount of data read or written by the read/write head per unit time.
4. Controller overhead: It is the time taken by the disk controller for mapping logical blocks to physical storage and keep track of which sectors are free and which are used.
5. Queuing Delay: time spent waiting for the disk to be free.

The disk access time is defined as the summation of seek time, rotational latency, data transfer rate, controller overhead and queuing delay and is given by the equation.

$$access_{time} = seek_{time} + rotational_{latency} + data_{tranferrate} + controller_{overhead} + queuing_{delay}$$

Out of the five parameters mentioned in the above equation, most of the time of the disk controller goes in moving the read/write to the desired location and thus seeking the information. If the disk access requests are processed efficiently then performance of the system can be improved. The aim of disk scheduling algorithm is to serve all the disk access requests with least possible head movement. There are number of disk scheduling algorithms which are presented here in brief.

First Come First Serve (FCFS) scheduling: This approach serves the disk access request in the order they arrived in the queue.

Shortest Seek Time First (SSTF) scheduling: Shortest Seek Time First disk scheduling algorithm selects the request from the queue which requires least movement of the head.

SCAN scheduling: The current head position and the head direction is the necessary input to this algorithm. Disk access requests are serviced by the disk arm as disk arm starts from one end of the disk and moves towards the other end. On reaching the other end the direction of the head is reversed and requests are continued to be serviced.

C-SCAN scheduling: Unlike SCAN algorithm, C-SCAN does not serve any request in the return trip. Instead, on reaching to the end, it reverses back to the beginning of the disk and then serves the requests.

LOOK scheduling: LOOK is similar to SCAN algorithm with only a single difference, after serving the last request, LOOK algorithm does not go till the end instead it immediately reverses its direction and moves to the beginning of the other end.

5.4.2 Optical Memories

So far, the storage devices you have studied are based on either electric charge or magnetic field. Magnetic memories are primarily used as a secondary storage device, but they can easily be damaged. However they have lower cost per bit than solid state devices.

First laser based memory was developed in 1982 by Phillips and Sony. Laser based storage devices uses a laser beam to read or write data and are called as optical memories or optical storage devices. As laser beams can be controlled more precisely and accurately than magnetic read/write heads. Data stored on optical drives remains unaffected by the magnetic disturbances in its surrounding.

Initially, these optical storage devices commonly known as compact disk (CD) or CD-DA (Digital Audio) were used to store only audio data of 60 minute duration. Later, huge commercial success of CD lead to development of low cost optical disk technology. These CDs can be used as auxiliary storage and can store any type of digital data. A variety of optical-disk devices have been introduced. We briefly review some of these types.

Compact Disk ROM (CD-ROM)

Compact Disk or CD-ROM are made of a 1.2 mm thick sheet of a polycarbonate material. Each disk surface is coated with a reflective material generally aluminium. The standard size of a compact disk is 120 mm in diameter. An acrylic coat is applied on top of the reflective surface to protect the disk from scratches and dust.

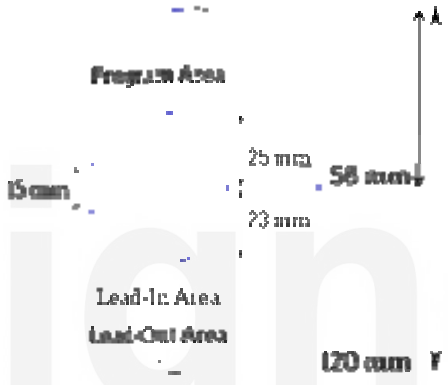


Figure 5.11: Outer Layout of a CD

Unlike magnetic disks, data on an optical disk is recorded in a spiral shape tracks. Each track is separated by a distance of 1.6 mm. Data in a track is recorded in the form of land and pit as shown in Figure 5.13. When a focused laser beam in incident on to the optical disk, the disk is burned as per the digitally recorded data forming a pit and land structure. The data is read from the surface by measuring the intensity of the reflected beam. The pit area scatters the incident beam, whereas the land reflects the incident beam, which are read as "0" and "1" respectively.

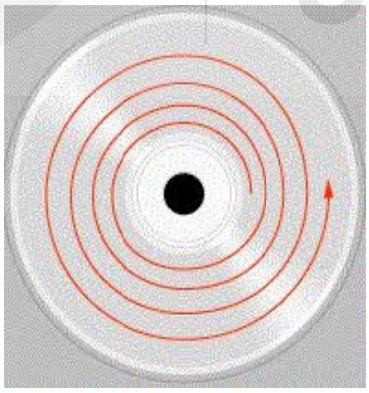


Figure 5.12: Spiral track of CD

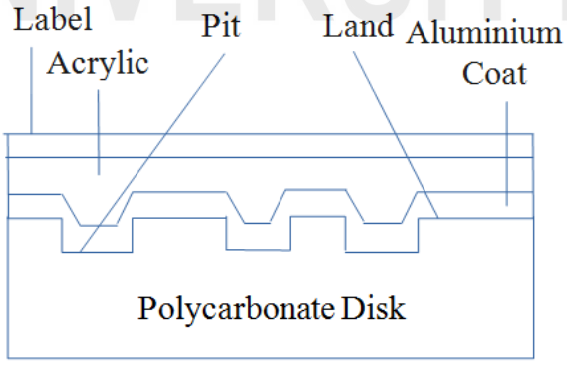


Figure 5.13: Land & Pit formation in CD track

As shown in Figure 5.12, the tracks in CD are in spiral shape. The tracks in CDs are further divided into sectors. All sectors in CDs are equal in length. This means that density of data recorded on the disk is uniform across all the tracks. Inner tracks have less number of sectors whereas outer tracks have more sectors. CD-ROM devices uses constant linear velocity (CLV) method for reading the disk content. In this method, the

disk is rotated at lower velocity as we move away from the center of the disk. This ensures a constant linear velocity at each track of the CD. The format of a sector of CD is show in Figure 5.14.

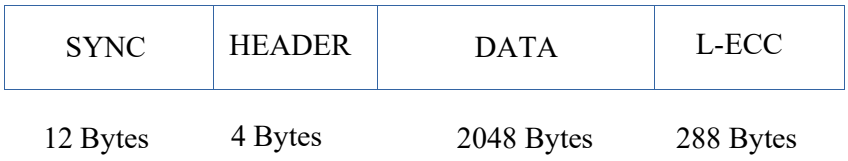


Figure 5.14: Sector format of CD

Data on the CD-ROM are stored in a track as a sequence of sectors. As shown in the Figure 5.14 each sector has four fields *viz.* sync, header, user data followed by error correcting codes. Each part of the sector is described below:

- Sync: It is the first field in every sector. The sync field is 12 byte long. The first byte of sync field contains a sequence of 0s followed by 10 bytes of all 1s and 1 byte of all 0s.

- Header: Header is four byte field in the sector. Three bytes are used to represent the sector address and one byte is used to represent the mode i.e. how subsequent fields in the sector are going to use. There are 3 modes:

    - Mode Zero: Specifies a no user data i.e. blank data field.
    - Mode One: Specifies an user data of 2048 bytes followed by 288 bytes of error correcting code.
    - Mode Two: No error correcting code will be used thus subsequent field will contain 2336 bytes of user data.

- Data: Data field contains the user 2048 byte of user data when mode is 1 or mode 2.

- L-ECC: Layered error correcting code field is 288 byte long field which is used for error detection and correction in mode 1. In mode 2, this field is used to carry an additional 288 bytes of user data.

Compact Disk Recordable (CD-R)

CD-Recordable are the compact disks which are capable of storing any type of digital data. The physical structure of CD-R is same as that of CD-ROM as discussed in previous section except that polycarbonate disk has a very thin layer of an organic dye before the Aluminum coating. CD- R can record user data only once but user can read the data many times thus these are also known as CD-WO (write once), or WORM (write once read many). Many CD writers allow the users to write CD-R in multiple session until CD is full. In each writing session, a partition is created in the CD- R. But once written, data on CD-R cannot be changed or deleted. There are three types of organic dyes used in CD-R.

Cyanine dyes are the most sensitive dye amongst the three types. CD-Rs have cyanine dyes are green in color. Very sensitive to UV rays and even can lose the data if exposed to direct sunlight for few days.

Phthalocyanine dye does not need a stabilizer as compared to cyanine dyes. They are silver, gold or light green in color. They are very less sensitive as compared to cyanine dyes but if exposed to direct sunlight for few weeks, it may lose the data.

Azo dye is the most stable among all types. It is most resistant to UV rays but if exposed to direct sunlight for 3-4 weeks, the CD -R may lose the data.

Compact Disk Rewritable (CD-RW)

The CD-RW are re-writable optical disks. The data on CD-RW can be read or written multiple times. But for writing again on the already written CD-RW, the disk data must be erased first. There are two approaches of erasing the data written on CD-RW. In the first approach, the entire disk data is erased completely i.e. all traces of any previous data is erased. This is called full blanking. Whereas in another approach called as fast blanking, only the meta data is erased. The later approach is faster and allows rewriting the disk. The first approach is used for confidentiality purposes.

The phase change technology is used in CD-RW. The phase change disk uses a material that has significantly different reflectivity in two different phase states. There is an amorphous state, in which the molecules exhibit a random orientation and which reflects light poorly; and a crystalline state, which has a smooth surface that reflects light well. A beam of laser light can change the material from one phase to the other. The phase change technology of CD-RW uses a 15-25 % degree of reflection whereas CD-R works on 40-70 % degree of reflection.

Digital Versatile Disk (DVD)

Digital versatile disk commonly known as DVD is also an optical storage device like CD, CD-R, CD-WR. Among the three DVDs have highest storage capacity ranges from 1.4 GB to 17 GB on a single side. The higher storage capacity is enabled by the use of laser beams of shorter wavelength as compared to compact disks. DVD uses a laser beam of 650 nm whereas compact disk uses a laser beam of 780 nm. Shorter wavelength laser beam creates shorter pits on the polycarbonate disk, thus offers higher storage capacity for similar dimensions. DVD-Audio and DVD-video are a standard format for recording audio and video data on DVDs. Like compact disks, DVD also comes in various variants like DVD-ROM, DVD-R, DVD-WR etc.

Blue Ray Disk

A blue ray disk is a digital disk that can store several hours high definition videos. Blue ray disks is of the same size of DVD, but can store 25 GB to 128 GB of data. A blue ray disk is designed to replace DVD technology. It has its applications in gaming applications, which uses very high quality animations.

5.4.3 Charge-coupled Devices, Bubble Memories and Solid State Devices

Charge-coupled Devices CCDs (CCDs)

Charge couple devices are photo sensitive devices which are used to store digital data. CCD is an integrated circuit of MOS-capacitors called cells, which are arranged in an array like structure in which each cell is connected with its neighbouring cell. Each capacitor can hold the charge which is used to represent the logic “1”. While reading the array of capacitors, the capacitor moves its charge to the neighbouring capacitor with next clock pulse. CCD arrays are mainly used in representing images and video data, where presence and absence of charge in the capacitor represents the corresponding pixel intensity.

As mentioned, CCD are highly photo-sensitive in nature and thus, produces a good quality picture even if light is dim or in low illumination intensity. Now a days, CCDs are widely used in digital cameras, satellite imagery, radar images and other high resolution imagery applications.

Magnetic Bubble Memories

Working principle of magnetic bubble memory is similar to that of charge coupled devices (CCD) discussed in the previous section. Magnetic bubble memory is an

arrangement of small magnetic area called bubble on a series of parallel track made of magnetic material. Each bubble represents a binary "1" and absence of a bubble on magnetic material is interpreted as "0". Binary data is read from the memory by moving these bubbles towards the edge of a track under the influence of external magnetic field. Magnetic field produces as bubbles remain persistent and do not demagnetise by its own. So, magnetic bubble memories are non-volatile type memories.

Solid State Devices (SSD)

Solid state drives also known as solid state storage devices are based on flash memory. As discussed, flash memory, a non-volatile type memory uses semiconductor devices to store the data. The major advantage of SSD is that it is purely an electronic device i.e. unlike HDD, SSD does not have mechanical read/ write head other mechanical components. Hence, reading and writing through SSD is faster than HDD. Now a days, SSD have replaced HDD in computer systems, however, SSD disks are more expensive than HDDs.

Check Your Progress 2

1. What will be the storage capacity of a disk, which has 8 recording surfaces, 32 tracks with each track having 64 sectors. Also, what would be the size of one cylinder of the disk? You may assume that each sector can store 1 MB data.

   ……

   ……

2. What would be the rotation latency time for the disk specified above, if it has a rotational speed of 6000 rpm?

   ……

   ……

3. What are the advantages and disadvantages of using SSD over HDD?

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   ……

4. What are the differences between CLV and CAV disks?

   ……...

   ……...

5.5 RAID AND ITS LEVELS

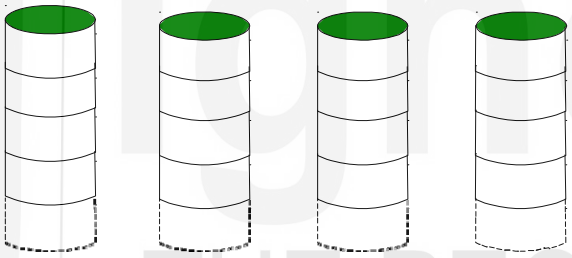
Continuous efforts have been made by researchers to enhance the performance of the secondary storage devices. As pointed out in previous sections performance of the secondary storage is inversely affected by disk access time. Lower the disk access time higher would be the performance. What about an idea of providing parallel access to a group of disks? With the use of parallel access the amount of data that can be accessed per unit time can be enhanced by a significant factor. A mechanism which splits the data on multiple disk is known as data striping. Data access through parallel access allows users to access data stored at multiple disks simultaneously, thus reduces effective reading time. Does data striping ensure protection of data against disk failure?

Another important factor for secondary storage is the reliability of data storage system. Storing same data on more than one disks enhances reliability. If one disk fails, then data can be accessed through another disk. Replicating data on multiple disks is called mirroring. Mirroring brings redundancy in data. So many schemes have been employed to enhance the performance and reliability of data and collectively they are called as redundant arrays of inexpensive disks (RAID). Based on the trade-off between reliability and performance RAID schemes have been categorises into various RAID levels.

Data striping increases the data transfer speed as different data bytes are accessed in parallel from different disks in a single disk access time. Whereas mirroring protects data from disk failures. If one disk fails then same data is accessed from the copy of the data stored in other disk.

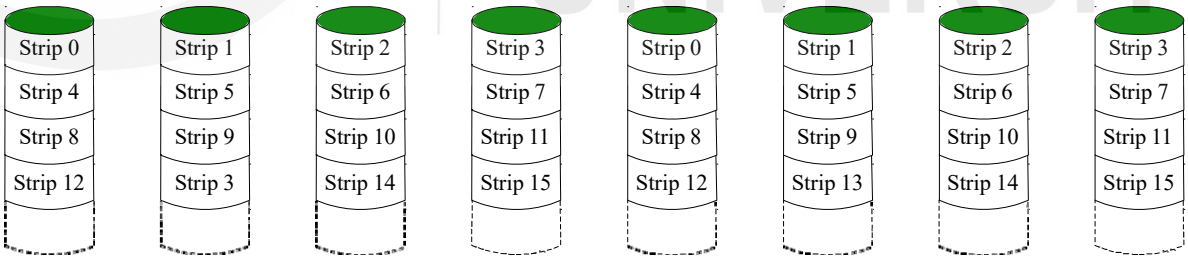
RAID Levels

RAID Level-0: RAID level-0 implements block splitting of data with no protection against disk failures. In block splitting, each block is stored in a different disk in the array. For example, i^{th} block of a file will be store in (*i mod n)* + *1* disk, where n is the total number of disks in the array. In this case, a significant enhancement on the performance can be observed as *n blocks* can be accessed (one each from each disk) in a single disk access time.



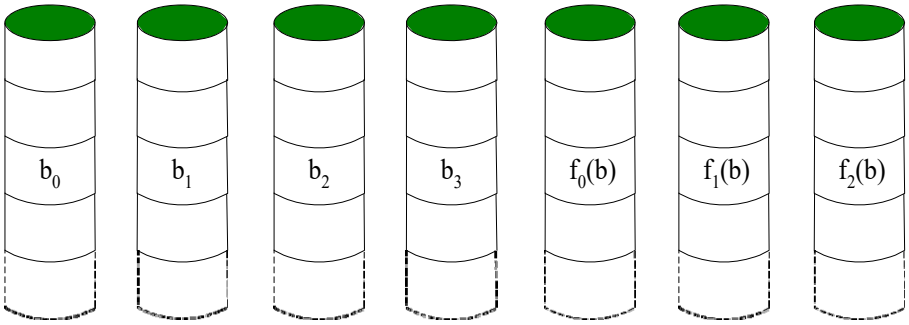
(a) RAID Level 0

RAID Level-1: This level protects data by implementing mirroring. If a system has 2 disks then each block of information will be stored in both of the disks. This ensures, if one disk fails then same copy of the block is accessed from the second disk. Mirroring introduces redundancy unlike level-0 which increases the data transfer rate.



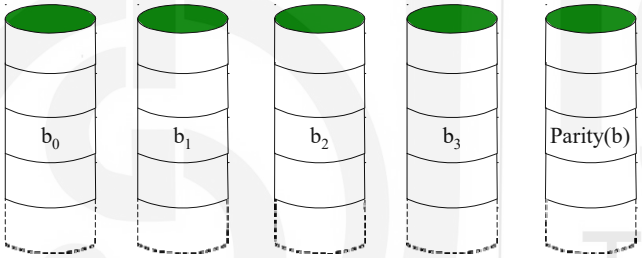
(b) RAID Level 1

RAID Level-2: This level uses error detection and correction bits, which are extra bits used for detection and correction of a single bit error in a byte. This is why this level is also known as memory-style error correction code organization. If one of the disk fails then parity bits and remaining bits of the byte are used to recover the bit value.



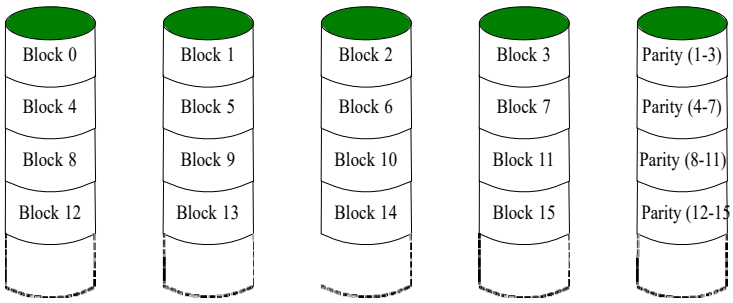
(c) RAID 2 (Redundancy through Hamming Code)

RAID Level-3: Single parity disk is used in this scheme. Parity bit for a sector is computed and stored in a parity disk. During the access, parity bit of the sector is computed and if computed parity bit is equal to the stored parity, the missing bit is 0 otherwise it is 1. This RAID level is also known as bit-interleaved parity organization. Thus has an advantage over level-2 that only single parity disk is used as compare to number of parity disks in level-2. The biggest drawback of this approach is that all the disks are used for single I/O operation in computation of the parity bit which slows down the disk access and also restricts parallel access.



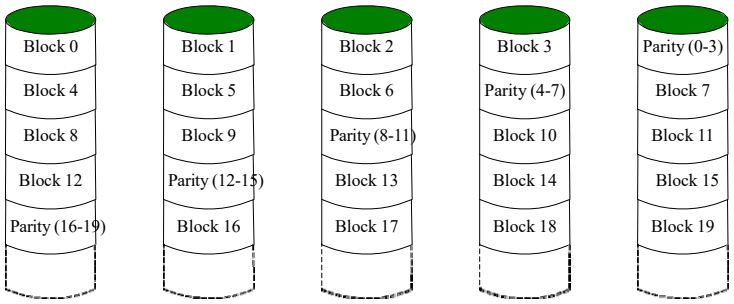
(d) RAID Level 3

RAID Level-4: This level uses block striping and one disk is used to keep parity block. This is also called block-interleaved parity organization. The advantage of block interleaving is that parity block along with corresponding blocks on other disks is used to retrieve the damaged block or the blocks of the failed disk. Unlike in level-3, block access reads one disk which allows parallel access to other blocks stored in other disks in the array.



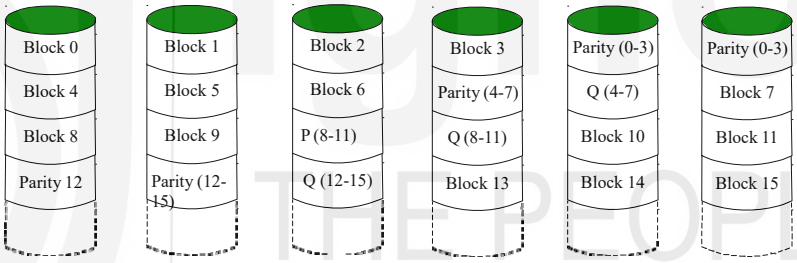
(e) RAID 4 (Block level Parity)

RAID Level-5: This level stores block of data and parity in all the disks in the array. One disk store the parity while data is spread out on different disks in the array. This structure is also known as block-interleaved distributed parity.



(f) RAID 5 (Block-level Distributed Parity)

RAID Level-6: Level-6 uses error correcting codes for recovery of damaged data while other levels uses parity. It also provides protection against multiple disks failures. For the recovery purposes, this arrangement is used to store redundant data on some of the disks, hence it is also called as p + q redundancy scheme. Here, p is the number of disks that store the error correcting codes while q is the number of disks that store redundant data.



(g) RAID Level 6

Table below summarises characteristics of various RAID levels.

RAID Level	Category	Features	I/O Request Rate (Read /write)	Data Transfer Rate (Read /write)	Typical Application
0	Striping	a) The disk is divided into blocks or sectors. b) Non-redundant.	Large Blocks: Excellent	Small Blocks: Excellent	Applications which requires high performance for non-critical data
1	Mirroring	a) Mirror disk which contains the same data is associated with every disk. b) Data Recovery is simple. On failure, data is recovered from the mirror disk.	Good / fair	Fair /fair	May be used for critical files

2	Parallel Access	a) All member disks participate in every I/O request. b) Synchronizes the spindles of all the disks to the same position. c) The blocks are very small in size (Byte or word). d) Hamming code is used to detect double-bit errors and correct single-bit error.	Poor	Excellent	Not useful for commercial purposes.
3	Parallel Access	a) Parallel access as in level 2, with small data blocks. b) A simple parity bit is computed for the set of data for error correction.	Poor	Excellent	Large I/O request size application, such as imaging CAD
4	Independent access	a) Each member disk operates independently, which enables multiple input/output requests in parallel. b) Block is large and parity strip is created for bits of blocks of each disk. c) Parity strip is stored on a separate disk.	Excellent/ fair	Fair / poor	Not useful for commercial purposes.
5	Independent access	a) Allows independent access as in level 4. b) Parity strips are distributed across all disks. b) Distribution avoids potential input/output bottleneck found in level 4.	Excellent / fair	Fair / poor	High request rate read intensive, data lookup
6	Independent access	Also called the p+q redundancy scheme, is much like level 5, but stores extra redundant information to guard against multiple disk failures.	Excellent/ poor	Fair / poor	Application requiring extremely high availability

Check Your Progress 3

1. What is the need of RAID?

   ……

   ……

2. Which RAID levels provide good data transfer rate?

   ……

……

3. Which RAID level is able to fulfil large number of I/O requests?

……

……

5.6 SUMMARY

This unit introduces the concept of memory hierarchy, which is primarily required due to the high cost per bit of high speed memory. The processing unit have register, cache, main memory and secondary or auxiliary memory. The main memory consists of RAM or ROM. This unit explains the logic circuit and organisation of RAM and ROM. The unit also explains several different types of secondary storage memories. The unit provide details on hard disk and its characteristics. It also gives details of different kind of optical disk. The concept of access time and constant linear and angular velocity has also been explained in details. For larger computer systems simple hard disk is not sufficient, rather an array of disks called RAID are used for such systems to provide good performance and reliability. The concept of RAID and various levels of RAID has been defined in this unit. The next unit will introduce you to the concept of high speed memories.

5.7 ANSWERS

Check Your Progress 1

1. RAM is a sequential circuit, volatile, requires refreshing (DRAM) and is a read/ write memory; ROM, PROM and EPROM are mostly non-volatile memories. ROM is a combinational circuits. All these ROMs are written mostly once and read many times.
2. Flash memory is a non-volatile semiconductor memory, where a section of the memory or a set of memory words can be erased. They are portable and mechanically robust as there is no mechanical movement in the memory to read-write data. Flash memory is used in USB memory, SD and micro SD memory cards used in cameras and mobile phones respectively.
3. (a) Since a word of data is 16 bits, it will have 16 data input and 16 data output lines, if not multiplexed.
   (b) The number of words are 16K, which is 2^{14}. Thus, 14 address lines would be required.
4. The memory must select one of the 4K bytes, which is 2^{12}. In case a square array is used (as shown in Figure 5.6), then 6 row address and 6 column address lines would be needed, which can be multiplexed. So just 6 address lines be sufficient. However, for a non square memory you may require all 12 address lines.
5. Two chips will be required to make 256 × 8 memory. 4 such combinations would be required to make 1 MB memory. Thus, you will require 8 such chips.

Check Your Progress 2

1. Storage capacity of a disk = recording surfaces × tracks per surface × sectors per track × size of each sector

   Storage capacity of a disk = 8× 32 × 64 × 1 MB = $2^3 \times 2^5 \times 2^6 \times 2^{20} = 2^{34}$ = 16 GB

   One cylinder will have = = 8× 64 × 1 MB = $2^3 \times 2^6 \times 2^{20}$ = 512 MB

2. The time of one rotation = 1/6000 min = 60/6000 sec = 1/100 sec= 10 millisec
   Rotational latency = on an average time of half rotation = 5 ms

3. SSD drives does not require any mechanical rotation, therefore are less prone to failure. In addition, they are much faster than HDD. But they are more expensive than HDD

4. The size of sectors on CLV disks is same on the entire disk, therefore, these disks are rotated a different speed. Density of data is same in all the sectors. In CAV disks the rotation speed is same, thus, sector size is more in the outer tracks. However, reading/writing process, in general, is faster.

Check Your Progress 3

1. RAID are a set of storage devices put together for better performance and reliability. Different kind of RAID levels have different objectives.

2. Good data transfer rate are provided by RAID level 0, 2 and 3.

3. Large number of I/O requests are fulfilled by RAID level 0, 1, 4,5 ,6.

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UNIT 6 ADVANCE MEMORY ORGANISATION

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6.7 Virtual Memory
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6.9 Answers

6.0 INTRODUCTION

In the last unit, the concept of Memory hierarchy was discussed. The Unit also discussed different types of memories including RAM, ROM, flash memory, secondary storage technologies etc. The memory system of a computer uses variety of memories for program execution. These memories vary in size, access speed, cost and type, such as volatility (volatile/ non-volatile), read only or read-write memories etc. As you know, a program is loaded in to the main memory for execution. Thus, the size and speed of the main memory affects the performance of a computer system. This unit will introduce you to concepts of cache memory, which is small memory between the processing unit and main memory. Cache memory enhances the performance of a computer system. Interleaved memory and associative memories are also used as faster memories. Finally, the unit discusses the concept of virtual memory, which allows programs larger than the physical memory.

6.1 OBJECTIVES

After going through this Unit, you will be able to:

- explain the concept of locality of reference;
- explain the different cache organisation schemes;
- explain the characteristics of interleaved and associative memories;
- explain the concept of virtual memory.

6.2 LOCALITY OF REFERENCE

Memory system is one of the important component of a computer. A program is loaded in to the main memory for execution. Therefore, a computer should have a main memory, which should be as fast as its processor and should have large size. In general, the main memory is constructed using DRAM technology which is about 50 to 100 times slower than the processor speed. This may slow down the process of instruction execution of a computer. Using SRAM may change this situation as it is almost as fast as a processor, however, it is a costly memory. So, what can you do? Is it possible to use large main memory as DRAM, but use a faster small memory between processor and main memory? Will such a configuration enhance performance of a computer? This section will try to answer these questions.

The important task of a computer is to execute instructions. It has been observed that on an average 80-85 percent of the execution time is spent by the processor in accessing the instruction or data from the main memory. The situation becomes even worst when instruction to be executed or data to be processed is not present in the main memory.

Another factor which has been observed by analysing various programs is that during the program execution, the processor tends to access a section of the program instructions or data for a specific time period. For example, when a program enters in a loop structure, it continues to access and execute loop statements as long as the looping condition is satisfied. Similarly, whenever a program calls a subroutine, the subroutine statements are going to execute. In another case, when a data item stored in an array or array like structure is accessed then it is very likely that either next data item or previous data item will be accessed by the processor. All these phenomenons are known as *Locality of Reference* or *Principle of Locality*.

So, according to the principle of locality, for a specific time period, the processor tends to make memory references closed to each other or accesses the same memory addresses again and again. The earlier type is known as *spatial locality*. Spatial locality specifies if a data item is accessed then data item stored in a nearby location to the data item just accessed may be accessed in near future. There can be special case of spatial locality, which is termed as sequence locality. Consider a program accesses the elements of a single dimensional array, which is a linear data structure, in the sequence of its index. Such accesses will read/write on a sequence of memory locations one after the other. This type of locality, which is a case of spatial locality, is referred to as sequence locality.

Another type of locality is the *temporal locality*, if a data item is accessed or referenced at a particular time, then the same data item is expected to be accessed for some time in near future. Typically it is observed in loop structures and subroutine call.

As shown in Figure 6.1, when the program enters in the loop structure at line 7, it will execute the loop statements again and again multiple times till the loop terminates. In this case, processor needs to access instructions 9 and 10 frequently. On the other hand, when a program accesses a data item store in an array, then in the next iteration it accesses a data item stored in an adjacent memory location to the previous one.

```
1    int main()
2    {
3      int i, num;
4      int fact = 1, sum = 0;

5      cout<<"\n Please Enter any number to Find Factorial\n" ;
6      cin>>>num ;

7      for (i = 1; i <= num; i++)
8       {
9                fact = fact * i;
10               sum = sum + fact;
11      }

12     cout<<"\nFactorial of "<<num<<"is" <<fact;
13     cout<<"\nsum is" <<sum;
14     return 0;
15   }
```

Figure 6.1: Loop structure

The locality of reference, be it spatial or temporal, suggests that in most cases accesses to a program instruction and data confines to a locality, hence, a very fast memory that captures the instructions and data nearer to the current instructions and data accesses can potentially enhance the overall performance of a computer. Thus,

attempts are continuously made to utilize the precious time of the processor efficiently A high speed memory, called cache memory, was developed. Cache memory utilises the principle of locality to reduce the memory references to the main memory by keeping not only the currently referenced data item but also the nearby data items. The cache memory and its organisation is discussed in the next sections.

6.3 CACHE MEMORY

A processor makes many memory references to execute an instruction. A memory reference to the main memory is time consuming as main memory is slower compared to the processing speed of the processor. These memory references, in general, tends to form a cluster in the memory, whether it is a loop structure, execution of a subroutine or an access to a data item stores in an array.

If you keep the content of the cluster of expected memory references in a small, extremely fast memory then processing time of an instruction can be reduced by a significant amount. Cache memory is a very high speed and expensive memory as compared to the main memory and its access time is closer to the processing speed of the processor. Cache memory act as a buffer memory between the processor and the main memory.

Because cache is an expensive memory so its size in a computer system is also very small as compared to the main memory. Thus, cache stores only those memory clusters containing data/ instructions, which have been just accessed or going to be accessed in near future. Data in the cache is updated based on the principle of locality explained in the previous section.

How data access time is reduced significantly by using cache memory?

Data in main memory is stored in the form of fixed size blocks/pages. Cache memory contains some blocks of the main memory. When processor wants to read a data item from the main memory, a check is made in the cache whether data item to be accessed is present in the cache or not. If data item to be accessed is present in the cache then it is read by the processor from the cache. If data item is not found in the cache, a memory reference is made to read the data item from the main memory, and a copy of the block containing data item is also copied into the cache for near future references as explained by the principle of locality. So, whenever processor attempts to read the data item next time, it is likely that the data item is found in the cache and saves the time of memory reference to the main memory.

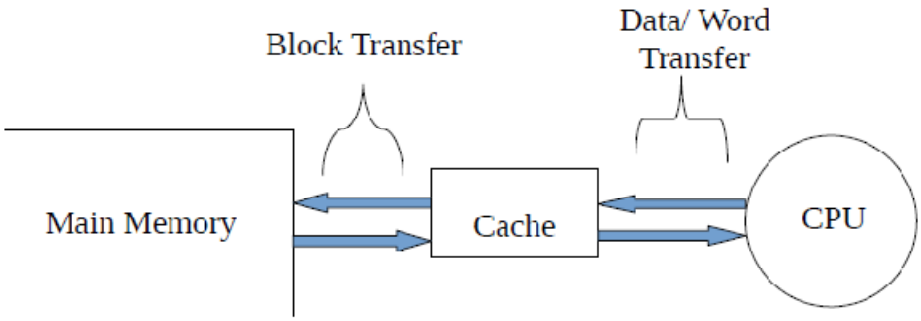


Figure 6.2: Cache Memory

As shown in the Figure 6.2, if requested data item is found in the cache it is called as ***cache hit*** and data item will be read by the processor from the cache. And if requested data item is not found in cache, called ***cache miss***, then a reference to the main memory is made and requested data item is read and block containing data item will also be copied into the cache.

Average access time for any data item is reduced significantly by using cache then without using cache. For example, if a memory reference takes 200 ns and cache takes 20 ns to read a data item. Then for five continuous references will take:

Time taken with cache : 20 (for cache miss) + 200 (memory reference)
+ 4 x 20 (cache hit for subsequent access)
= 300 ns

Time without cache : 5 x 200 = 1000 ns

In the given example, the system first looks into the cache for the requested data item. As it is the first reference to the data item it will not be present in the cache, called as cache miss, and thus, requested data item will be read from the main memory. For subsequent requests of the same data item, the data item will be read from the cache only and no references will be made to the main memory as long as the requested data remains in the cache.

Effective access time is defined as the average access time of memory access, when a cache is used. The access time of memory access is reduced in case of a cache hit, whereas it increases in case of cache miss. In the above mentioned example processor takes 20 + 200 ns for a cache miss, whereas it takes only 20 ns for each cache hit. Now suppose, we have a hit ratio of 80%, i.e. 80 percent of times a data item would be found in the cache and 20 % of the times it would be accessed from the main memory. So effective access time (EAT) will be computed as :

effective access time = (cache hit x data access time from cache only)
+(cache miss x data access time from cache and main memory)

effective access time = 0.8 (hit ratio) x 20 (cache hit time)
+ 0.2(miss ratio) x 220 (cache miss and memory reference)

effective access time = 0.8 x 20 + 0.2 x 220
= 16 + 44
= 60 ns

From the example it is clear that cache reduces the average access time and effective access time for a data item significantly and enhance the computer performance.

Check Your Progress 1

1. What is the importance of locality of reference?

   ……

   ……

   ……

2. What is block size of main memory for cache transfer?

   ……

   ……

   ……

3. Hit ration of computer system is 90%. The cache has an access time of 10ns, whereas the main memory has an access time of 50ns. Computer the effective access time for the system.

   ……

   ……

   ……

6.4 CACHE ORGANISATION

The main objective of using cache is to decrease the number of memory references to a significant level by keeping the frequently accessible data/ instruction in the cache. Higher the hit ratio (number of times requested data item found in cache / total number of times data item is requested), lower would be the references to the main memory. So there are number of questions that need to be answered while designing the cache memory. These Cache design issues are discussed in the next subsection.

6.4.1 Issues of Cache Design

In this section, we present some of the basic questions that should be asked for designing cache memory.

What should be the size of cache memory?

Cache is an extremely fast but very expensive memory as compared to the main memory. So large cache memory may shoot up the cost of the computer system and too small cache might not be very useful in real time. So, based on various statistical analyses, if a computer system has 4 GB of main memory then the size of the cache may go up to 1MB.

What would be the block size for data transfer between cache and main memory?

Block size directly affects the cache performance. Higher block size would ensure only fewer number of blocks in cache, whereas small block size contains fewer data items. As you increase the block size, the hit ratio first increases but it decreases as you further increase the block size. Further increase in block size will not necessarily result in access of newer data items, as probability of accessing data items in the block with larger number of data items tends to decrease. So, optimal size of the block should be chosen to maximise the hit ratio.

How blocks are going to be replaced in cache?

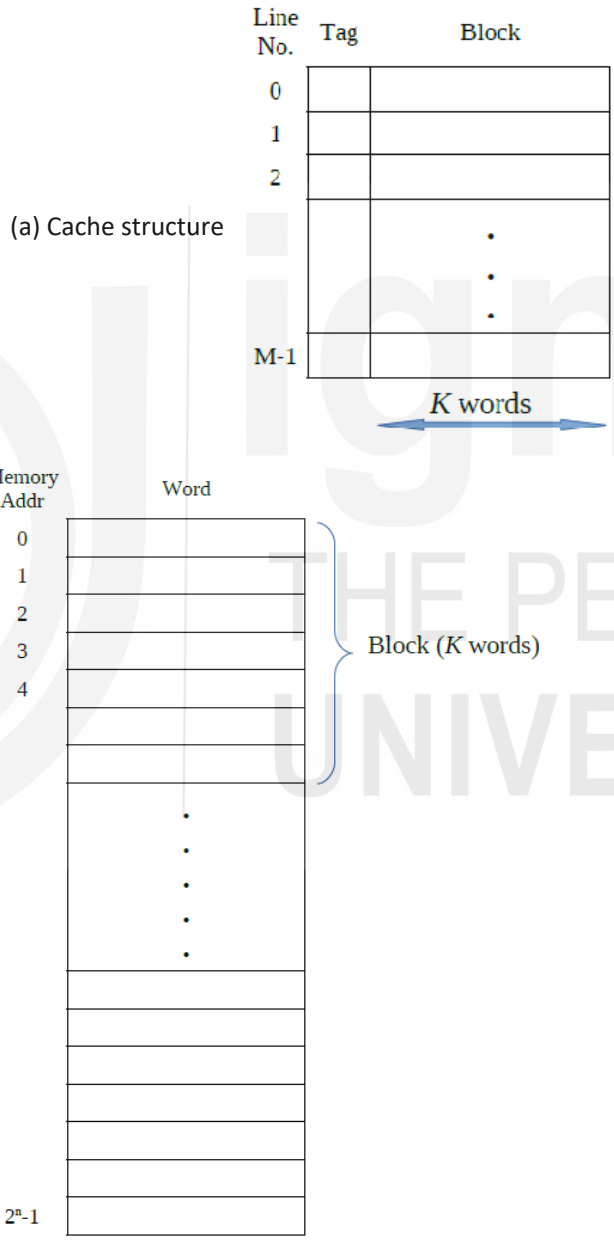
As execution of the process continues, the processor requests for new data items. For new data items and thus, new blocks to be present in the cache, the blocks containing old data items must be replaced. So there must be a mechanism which may select the block to be replaced which is least likely to be needed in near future.

When changes in the blocks will be written back on to the main memory?

During the program execution, the value of a data item in a cache block may get changed. So the changed block must be written back to the main memory in order to reflect those changes to ensure data consistency. So there must be a policy, which may decide when the changed cache block is written back to the main memory.

In certain computer organisations, the cache memory for data and instruction are placed separately. This results in separate address spaces for the instructions and data. These separate caches for instructions and data are known as **instruction cache** and **data cache** respectively. If processor requests an instruction, then it is provided by the instruction cache, whereas requested data item is provided by the data cache. Using separate cache memories for instruction and data enhances computer performance. While some computer systems implements different cache memories for data and instructions other implements multiple level of cache memories. Two level cache popularly known as **L1 cache** and **L2 cache** is most commonly used. Size of level 1 cache or L1 cache is smaller than the level 2 or L2 cache. Comparatively more frequently used data/ instructions are stored in L1 cache.

As discussed earlier, the main memory is divided into blocks/ frames/ pages of *k words* each. Each word of the memory unit has a unique address. A processor requests for read/write of a memory word. When a processor's request of a data item cannot be serviced by cache memory, i.e. a *cache miss* occurs, the block containing requested data item is read from the main memory and a copy of the same is stored in cache memory. A cache memory is organised as a sequence of line. Each cache line is identified by a cache line number. A cache line stores a tag and a block of data. Cache and main memory structure is shown in Figure 6.3. General structure of cache memory having M lines and $N=2^n$ main memory size is shown in figure 6.3(a) and figure 6.3(b) respectively.



(a) Cache structure

(b) Main Memory structure

Figure 6.3: Structure of Cache and Main Memory

An example of cache memory of size 512 words is shown in Figure 6.4. The example shown in Figure 6.4 has a main memory of 64 K words of 16 bits each and cache

memory can have 512 words of 16 bits each. To read a data item processor sends a 16 bit address to the cache and if cache misses then the data item/ word is fetched from the main memory and accessed data item/ word is also copied into the cache. Please note that the size of block is just 1 memory word in this case.

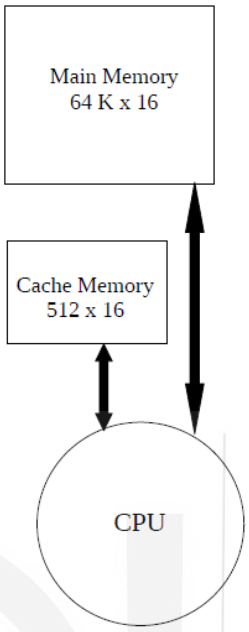


Figure 6.4: Example of Cache and Main Memory

6.4.2 Cache Mapping

As discussed earlier, a request of processor to access a main memory is to checked in cache memory. Where can a block of main memory be places in cache and how processor can determine, if the data requested is present in cache? Answer to these questions are provided by cache mapping scheme. A mapping mechanism maps the block from the main memory to a cache line. The mapping is required as cache is much smaller in size than the size of the main memory. So only few blocks from the main memory can be stored in cache. There are three types of mapping in cache:

Direct Mapping:

Direct mapping is the simplest amongst all three mapping schemes. It maps each block of main memory into only one possible cache line. Direct mapping can be expresses as a *modulo M* function, where *M* is the total number of cache lines as shown:

$$i = j \ modulo \ M$$

where, i = number of cache line to which main memory block would be mapped.

j = the block address of main memory, which is being requested

M = total number of cache lines

So, line 0 of the cache will store block 0, M, 2M….. and so on. Similarly, line 1 of cache will store block 1, M + 1, 2M + 1, and so on.

An address of main memory word, as shown in Figure 6.3(b), consists of *n bits*. This address of each word of main memory has two parts: block number (*n-k bits*) and word number within the block (*k bits*). Here, each block of the main memory contains 2^k number of words. The cache memory interprets *n-k bit* block number in to two parts

as: tag and line number. As indicated in Figure 6.3(a), the cache memory contains M lines. Assuming *m* address bits (2^m = M) are used to identify each line, then most significant *(n-k) - m* bits of *(n-k) bit* block number are interpreted as tag and *m* bits are used as line number of the cache. Please note the tag bits are used to identify, which of the main memory block is presently in that cache line. The following diagram summarizes the bits of main memory address and related cache address:

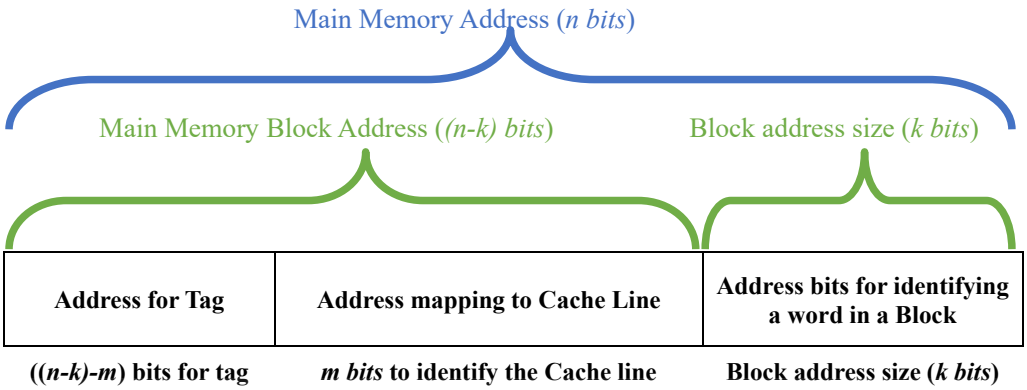


Figure 6.5: Main Memory address to Cache address mapping in Direct mapping Scheme

Please note the following points in the diagram given above.

The size of Main Memory address: n bits

Total number of words in main memory: 2^n

The size of a Main memory block address: most significant (n-k) bits

Number of words in each block: 2^k

In case, a referenced memory address is in cache, bits in the tag field of the main memory address should match the tag filed of the cache.

***Example*:** Let us consider a main memory of 16 MB having a word size of a byte and a block size is of 128 bits or 16 words (one word is one byte in this case). The cache memory can store 64 KB of data. Determine the following:

a) Size of various addresses and fields

c) how will you determine that a hexadecimal address is in cache memory, assuming a direct mapping cache is used?

Solution:

a) Size of main memory= 16MB = 2^{24} Bytes

Each block consists of 16 words, thus total number of blocks in main memory would be = $2^{24}/ 16 = 2^{20}$ blocks, thus $n = 24$ and $k = 4$ (as 2^4=16). Therefore, main memory block address $(n\text{-}r) = 20$

Data size of cache is 64 KB = 2^{16} Bytes

Total number of cache lines (M) = cache size/ block size = $2^{16} / 2^4 = 2^{12}$, therefore, number of cache lines = 2^{12} and m = 12

Length of address field to identify a word in a Block (k) = 4 bits

Length of address to identify a Cache line (m) = 12 bits

Length of Tag field = (24- 4) - 12 = 8 bits.

Thus, a main memory address to cache address mapping for the given example would look like this:

Main Memory Address n = 24 bits		
Address of a Block of data = 20 bits		***k=4 bits***
Address mapping for direct cache mapping scheme		
Tag = 8 bits	***Cache line number address m = 12 bits***	***k=4 bits***

Figure 6.6: Direct Cache mapping

b) Consider a 24 bit main memory address in hexadecimal as FEDCBA. The following diagram will help in identifying, if this address is in the cache memory or not in case direct mapping scheme is used.

Main Memory Address n = 24 bits = 6 hex digits					
F	E	D	C	B	A
Address of a Block of data = 20 bits					***k=4 bits***
F E D C B					A
Address mapping for direct cache mapping scheme					
Tag = 8 bits		***Cache line number address m = 12 bits***			***k=4 bits***
F E		D C B			A
1111 1110		1101 1100 1011			1010

Figure 6.7: Direct Cache mapping example

Now, the following steps will be taken by the processing logic of processing unit and hardware of Cache memory:

1. The tag number (FE in this case) is compared against the Tag number of data stored in the cache line (DCB in this case).
2. In case both are identical
   then (this is the case of cache hit): A^{th} word from the cache line DCB is accessed by the processing logic.
   else (this is a case of cache miss): The cache line 16 words data is read to cache memory line (DCB) and its tag number is now FE. The required A^{th} word is now accessed by the processing logic

Direct mapping is very easy to implement but has a disadvantage as location in which a specific block is to be stored in cache is fixed. This arrangement leads to low hit ratio as when processor wants to read two data items belongs to two different blocks, which map to single cache location, then each time other data item is requested, the block in the cache must be replaced by the requested one. This phenomenon is also known as *thrashing.*

Associative Mapping:

Associative mapping is the most flexible mapping in cache organisation as it allows to store any block of the main memory in any of the cache line/or location. It uses complete (*n-k*) *bits* of block address field as a tag field. Cache memory stores (*n-k*) *bits* of Tag and ($2^k \times$ Word Size in bit) data. When a data item/ word is requested, (*n-k*) *bit* tag field is used by the cache control logic to search the all the tag fields stored in the cache simultaneously. If there is a match (cache hit) then corresponding data item is read from the cache, otherwise (cache miss) the block of data that contains the word to be accessed is read from the main memory. It replaces any of the cache line. In addition, the block address of the accessed block from the main memory replaces the tag of the cache line. It is also the fastest mapping amongst all types. Different block replacement policies are used for replacing the existing cache content by newly read data, however, those are beyond the scope of this unit. This mapping requires most complex circuitry, as it requires all the cache tags to be checked simultaneously with the block address of the access request.

Main Memory Address :

Address of a block of data is same as Tag	**Address bits for identifying a word in a Block**
(n-k) bits	*k bits*

Every line of Associative Cache has the following format:

Tag	Data Block of *k* words
(n-k) bits	Data bits

Figure 6.8: Associative mapping

The following example explains the set associative mapping.

***Example*:** Let us consider a main memory of 16 MB having a word size of a byte and a block size is of 128 bits or 16 words (one word is one byte in this case). The cache memory can store 64 KB of data. Determine the following size of various addresses and fields, if associative mapping is used

Solution:

Size of main memory= 16MB = 2^{24} Bytes

Each block consists of 16 words, thus total number of blocks in main memory would be = $2^{24}/ 16 = 2^{20}$ blocks, thus $n = 24$ and $k = 4$ (as 2^4=16). Therefore, main memory block address (n-r) = 20

Data size of cache is 64 KB = 2^{16} Bytes

Total number of cache lines (M) = cache size/ block size = $2^{16} / 2^4 = 2^{12}$, therefore, number of cache lines = 2^{12} and m = 12

Length of Tag field = (24- 4) = 20 bits.

Size of data = $2^k \times$ Word Size in bit = $2^4 \times 8 = 128$ bits.

Thus, size of one line of cache = 128+20=148 bits.

Set Associative Mapping:

The major disadvantage of direct mapping is that location of the cache line onto which a memory block is going to be mapped is fixed which results in poor hit ratio and unused cache locations. The associative mapping removes these hurdles and any block of memory can be stored anywhere in cache location. But associative cache uses complex matching circuit and big tag size. Set Associative mapping reduces the disadvantages of both the above mentioned cache mapping techniques and is built on their strengths. In set associative mapping scheme, cache memory is divided into *v sets* where each set contains *w cache lines*. So, total number of cache lines *M* is given as:

$$M = v \; x \; w$$

where v is the number of sets and w is the number of cache lines in v

The cache mapping is done using the formula:

$$i = j \; modulo \; v$$

where *i* is the set number and *j* is the block address of word to be accessed.

Cache control logic interprets the address field as a combination of *tag* and *set* fields as shown:

Tag	Set	Word
((n-k)-d) bits	*d bits*	*k bits*

Figure 6.8: Set Associative mapping

Cache mapping logic uses *d-bits* to identify the set as $v = 2^d$ and *((n-k)-d)) bits* are used to represent the *tag field.* In set-associative mapping, a *block j* can be stored at any of the cache line of *set i.* To read a data item, the cache control logic first simultaneously looks into all the cache lines using *((n-k)-d)) bits* of *tag field* of the *set* identified by *d-bits* of the *set field,* otherwise a data item is read from the main memory and corresponding data is copied into the cache accordingly. Set associative mapping is also known as *w*-way set-associative mapping. It uses lesser number of bits (*((n-k)-d) bits)* as compare to (*n-k*) *bits* in associative mapping in *tag field*.
A comprehensive example showing possible locations of main memory blocks in Cache for different cache mapping schemes is discussed next.

Example: Assume a main memory of a computer consists of 256 bytes, with each memory word of one byte. The memory has a block size of 4 words. This system has a cache which can store 32 Byte data. Show how main memory content be mapped to cache if (i) Direct mapping (ii) Associative mapping and (iii) 2 way set associative memory is used.
Solution:

Main memory size = 256 words (a word = one byte) = 2^8 ⇒ n=8 bits
Block Size = 4 words = 2^2 ⇒ k=2 bits

The visual representation of this main memory:

Block Number of memory in equivalent decimal	Memory Location Address		Assume data stored in the location
	Block Address	Word Address	
0	000000	00	1001010
	000000	01	1101010
	000000	10	0001010
	000000	11	0001010
1	000001	00	1111010
	000001	01	0101010
	000001	10	1001010
	000001	11	1101010
2	000010	00	1101010
	000010	01	0001010
	000010	10	0101010
	000010	11	0011010
…	…	…	…
7	000111	00	0000010
	000111	01	0000011
	000111	10	0000011
	000111	11	0001110
…	…	…	…
63	111111	00	1111010
	111111	01	1111011
	111111	10	0101011
	111111	11	0101110

Figure 6.9: An example of Main Memory Blocks

(i) Direct Mapping Cache:
The size of cache = 32 bytes
The block size of main memory = words in one line of cache =4 ⇒ k=2 bits
The cache has = 32 /4 = 8 lines with each line storing 32 bits of data (4 words)

Therefore, m=3 as $2^3 = 8$

Thus, Tag size = (n-k) - m = (8 - 2) - 3 = 3

The address mapping for an address: 11111101

Block Address of Main Memory		**Address of a word in a Block**
111	111	01
111	111	01
Tag	**Line Number**	

Line Number = 111 = 7 in decimal
Tag = 111

The address mapping for an address: 00001011

Block Address of Main Memory		**Address of a word in a Block**
000	010	11
000	010	11
Tag	**Line Number**	

Line Number = 010 = 2 in decimal
Tag = 000

The following cache memory that uses direct mapping shows these two words (along with complete block in the cache)

Line Number of Cache in Decimal	**Contents of Cache Memory**				
	Tag of Data	**Data in Cache = 4 words = 32 bits**			
		Word 11	**Word 10**	**Word 01**	**Word 00**
0					
1					
2	000	0011010	0101010	0001010	1101010
3					
4					
5					
6					
7	111	0101110	0101011	1111011	1111010

Figure 6.10: An example Cache memory with Direct mapping

The access for an address: 00011110

Block Address of Main Memory		**Address of a word in a Block**
000	111	10
000	111	10
Tag	**Line Number**	

In case, a word like 00011110 is to be accessed, which is not in the cache memory and as per mapping should be mapped to line number 111 =7, the cache access logic will compare the tags, which are 000 for this address, and 111 in the cache line 7. This is the situation of cache miss, so accordingly this block will replace the content stored in line 7, which after replacement is shown below:

7	000	0001110	0000011	0000011	0000010

Please note the change in data value in the cache line 7.

(ii) Associative Mapping Cache:
The size of cache = 32 bytes
The block size of main memory = words in one line of cache =4 ⇒ k=2 bits
Therefore, cache has = 32 /4 = 8 lines with each line storing 32 bits of data (4 words)
Tag size = n-k = (8 - 2) = 6
The address mapping for an address: 11111101

Block Address of Main Memory	**Address of a word in a Block**
111111	01
111111	01
Tag	

The address mapping for an address: 00001011

Block Address of Main Memory	**Address of a word in a Block**
000010	11
000010	11
Tag	

The following associative cache shows these two words.

Line Number of Cache in Decimal	**Contents of Cache Memory**				
	Tag of Data	**Data in Cache = 4 words = 32 bits**			
		Word 11	**Word 10**	**Word 01**	**Word 00**
0	111111	0101110	0101011	1111011	1111010
1	000010	0011010	0101010	0001010	1101010
2	000111	0001110	0000011	0000011	0000010
3					
4					
5					
6					
7					

Figure 6.11: An example Cache memory with Associative mapping

The access for an address: 00011110

Block Address of Main Memory	**Address of a word in a Block**
000111	10
000111	10
Tag Number	

A word like 00011110 can in any cache line, for example, in the cache memory shown above it is in line 2 and can be accessed.
(iii) 2way set associative Mapping:
The size of cache = 32 bytes
The block size of main memory = words in one line of cache =4 ⇒ k=2 bits
The number of lines in a set (w) = 2 (this is a 2 way set associative memory)
The number of sets (v) = Size of cache in words/(words per line × w)
= 32/(4×2) =4
Thus, set number can be identified using 2 bits as $2^2 = 4$

Tag size = $(n\text{-}k)\text{-}v$ = (8 - 2) - 2 = 4

The address mapping for an address: 11111101

Block Address of Main Memory		Address of a word in a Block
1111	11	01
1111	11	01
Tag	**Set Number**	

Set number = 11 = 3 in decimal

Tag = 1111

The address mapping for an address: 00001011

Block Address of Main Memory		Address of a word in a Block
0000	10	11
0000	10	11
Tag	**Set Number**	

Set number = 10 = 2 in decimal

Tag = 0000

Contents of Cache Memory Way 0					Set #	Contents of Cache Memory Way 1				
Tag						**Tag**	**Data in Cache = 4 words = 32 bits**			
	Word 11	**Word 10**	**Word 01**	**Word 00**			**Word 11**	**Word 10**	**Word 01**	**Word 00**
					0					
					1					
					2	0000	0011010	0101010	0001010	1101010
1111	0101110	0101011	1111011	1111010	3	0001	0001110	0000011	0000011	0000010

Figure 6.12: An example Cache memory with Set Associative mapping

The access for an address: 00011110

Block Address of Main Memory		Address of a word in a Block
0001	11	10
0001	11	10
Tag	**Set Number**	

Set number = 11= 3 in decimal

Tag = 0001

Word 00011110 can be stored and accessed from the cache set 11 at the second line (way 1).

6.4.3 Write Policy

Many processes read and write data in cache and main memory either by the processor or by the input/ output devices. Multiple read possess no challenge to the state of the data item, as you know, cache maintains a copy of frequently required data items to improve the system performance. Whenever a process writes/ updates the values of the data item in cache or in main memory, it must be updated in the copy as well. Otherwise it will lead to an inconsistent data and cache content may become invalid. Problems associated with writing in cache memories can be summarised as:

- Caches and main memory can be altered by multiple processes which may result in inconsistency in the values of the data item in cache and main memory.

- If there are multiple CPUs with individual cache memories, data item written by one processor in one cache may invalidate the value of the data item in other cache memories.

These issues can be addressed in two different ways:

1. **Write through:** This writing policy ensures that if a CPU updates a cache, then it has to write/ or make the changes in the main memory as well. In multiple processor systems, other CPUs-Cache need to keep an eye over the updates made by other processor's cache into the main memory and make suitable changes accordingly. It creates a bottleneck as many CPUs try to access the main memory.

2. **Write Back:** Cache control logic uses an ***update bit.*** Changes are allowed to write only in cache and whenever a data item is updated in the cache, the update bit of the block is set. As long as data item is in the cache no update is made in the main memory. All those blocks whose update bit is set is replaced in the main memory at the time when the block is being replaced in the cache. This policy ensures that all the accesses to the main memory are only through cache, and this may create a bottleneck.

You may refer to further readings for more details on cache memories.

Check Your Progress 2

1. Assume that a Computer system have following memories:
   RAM 64 words with each word of 16 bits
   Cache memory of 8 Blocks (block size of cache is 32 bits)
   Find in which location of cache memory a decimal address 21 can be found if Associative Mapping is used.

   ……

   ……

2. For the system as given above, find in which location of cache memory a decimal address 27 will be located if Direct Mapping is used.

……

……

3. For the system as given above, find in which location of cache memory a decimal address 12 will be located if two way set associative Mapping is used.

   ……

   ……

6.5 MEMORY INTERLEAVING

As you know that cache memory is used as a buffer memory between processor and the main memory to bridge the difference between the processor speed and access time of the main memory. So, when processor requests a data item, it is first looked into the cache and if data item is not present in the cache (called cache miss), only then main memory is accessed to read the data item. To further enhance the performance of the computer system and to reduce the memory access time of the main memory, in case of cache miss, the concept of memory interleaving is used. Memory interleaving is of three type, viz. lower order memory interleaving, higher order memory interleaving and hybrid memory interleaving. In this section we will discuss the lower order memory interleaving only. Discussion on other memory interleaving techniques is beyond the scope of this unit.

In memory interleaving technique, main memory is partitioned into *n* number of equal sized modules called as memory banks and technique is known as *n-way memory*

interleaving. Where each memory module has its own memory address register, base register and instruction register, thus each memory bank can be accessed individually and simultaneously. Instructions of a process are stored in successive memory banks. So, in a single memory access time *n* successive instructions of the process can be accessed from *n* memory banks. For example, suppose main memory is divided into four modules or memory banks denoted as M1, M2, M3 and M4 then first n instructions of a process will be stored as: first instruction in M1, second instruction in M2, third instruction in M3, fourth instruction in M4 and again fifth instruction in M1 and so on.

When processor issues a memory fetch command during the execution of the program, memory access system creates *n* consecutive memory addresses and places them in respective memory address register of all memory banks in the right order. Instructions are read from all memory modules simultaneously and loads them into *n* instruction registers. Thus, each fetch for a new instruction results in the loading of *n* consecutive instructions in *n* instruction registers of the CPU, in the time of a single memory access. Figure 6.13 shows the structure of 4-way memory interleaving. The address is resolved by interpreting the least significant bits to select the memory module, and rest of the most significant bits are the address in the memory module. For example, in an 8-bit address and 4-way memory interleaving, two least significant bits will be used for module selection and six most significant bits will be used as an address in the module.

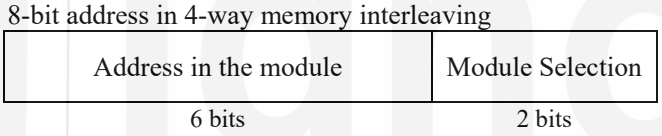


Figure 6.13: Address mapping for Memory interleaving

The following example demonstrates how the main memory words be distributed to different interleaved memory modules. For this example, only a four bit address of main memory is used.

Main Memory	
Address	**Data**
0000	10
0001	20
0010	30
0011	40
0100	50
0101	60
0110	80
0111	76
1000	46
1001	25
1010	58
1011	100
1100	23
1101	78
1110	35
1111	11

Module 00	
Address	**Data**
00	10
01	50
10	46
11	23

Module 01	
Address	**Data**
00	20
01	60
10	25
11	78

Module 10	
Address	**Data**
00	30
01	80
10	58
11	35

Module 11	
Address	**Data**
00	40
01	76
10	100
11	11

Figure 6.14: Example of Memory interleaving

Please note in the figure above how various data values are distributed in the modules.

6.6 ASSOCIATIVE MEMORY

Though cache is a high speed memory but still it needs to search the data item stored in it. Many search algorithms have been developed to reduce the search time in a sequential or random access memory. Searching time of a data item can be reduced further to a significant amount of data item is identified by its content rather by the address. Associative memory is a content addressable memory (CAM), that is memory unit of associative memory is addressed by the content of the data rather by the physical address. The major advantage of this type of memory is that memory can be searched in parallel on the basis of data. When a data item is to be read from an associative memory, the content of the data item, or part of it, is specified. The memory locates all data items, which matches the specified content, and marks them for reading. Because of the architecture of the associative memory, complete data item or a part of it can be searched in parallel.

Hardware Organization

Associative memory consists of a memory array and logic for *m* words with *n* bits per word as shown in block diagram in Figure 6.15. Both argument register (A) and key register (K) have *n* bits each. Each bit of argument and key register is for one bit of a word. The match register M has *m* bits, one each for each memory word.

The key register provides a mask for choosing a particular field or key in the argument word. The entire argument is compared with each memory word only if the key register contains all 1s. Otherwise, only those bits in the argument that have 1s in their corresponding positions of the key register are compared. Thus, the key provides a mask or identifying information, which specifies how reference to memory is made.

The content of argument register is simultaneously matched with every word in the memory. Corresponding bits in the mach register is set by the words that have match with the content of the argument register. Set bits of the matching register indicates that corresponding words have a match. Thereafter, memory is accessed sequentially, to read only those words whose corresponding bits in the match register have been set.

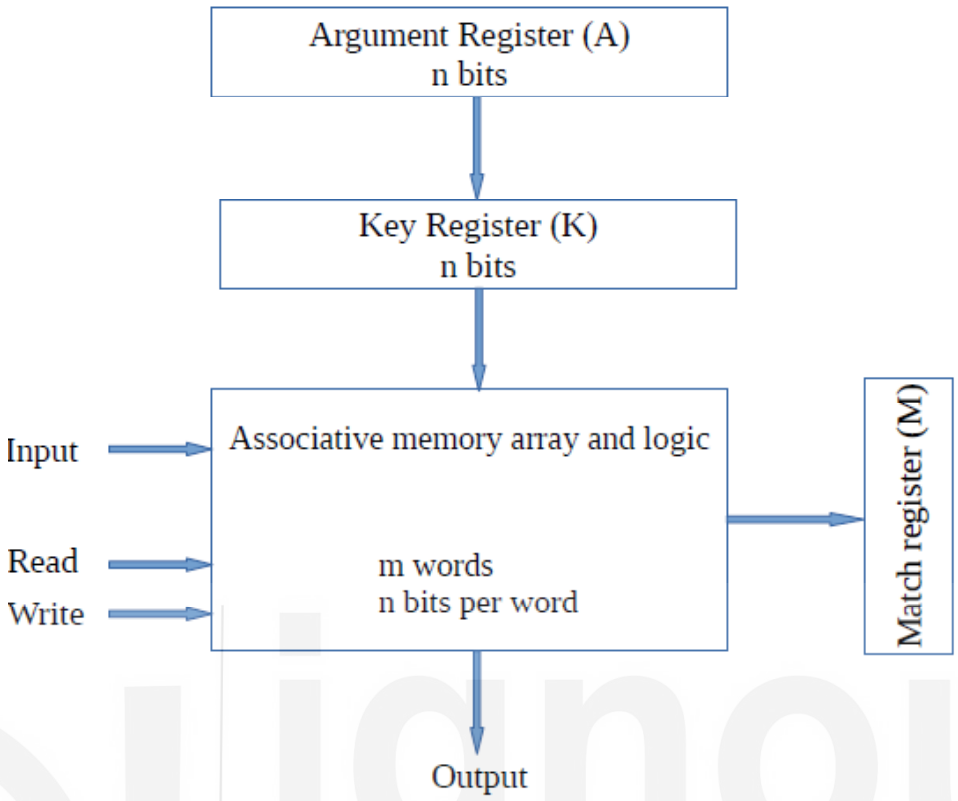


Figure 6.15: Block diagram of associative memory

Example: Consider an associative memory of just 2 bytes. The content register and argument registers are also shown in the diagram.

Description	The content of associative Memory								Match Word
Argument Register	0	1	1	0	0	0	0	1	
Key Register	1	1	1	1	0	0	0	0	
Bits to be matched	0	1	1	0					
Word 1	0	1	1	0	0	1	1	0	Match
Word 2	1	0	0	1	1	0	0	0	Not machted

Figure 6.16: An Example of Associative matching

Please note as four most significant bits of key register are 1, therefore only they are matched.

6.7 VIRTUAL MEMORY

As we know that a program is loaded into the main memory for execution. The size of the program is limited by the size of the main memory i.e. cannot load a program in to the main memory whose size is larger than the size of the main memory. Virtual memory system allows users to write programs even larger than the main memory. Virtual memory system works on the principle that portions of a program or data are loaded into the main memory as per the requirement. This gives an illusion to the programmer that they have very large main memory at their disposal. When an address is generated to reference a data item, virtual address generated by the processor is mapped to a physical address in the main memory. The translation or mapping is handled automatically by the hardware by means of a mapping table.

Let us say, you have a main memory of size 256K (2^{18})words. This requires 18-bits to specify a physical address in main memory. A system also has an auxiliary memory as large as the capacity of 16 main memories. So, the size of the auxiliary memory is 256K ×16 = 4096 K which requires 24 bits to address the auxiliary memory. A 24-bit virtual address will be generated by the processor which will be mapped into an 18-bit physical address by the address mapping mechanism as shown in Figure 6.17.

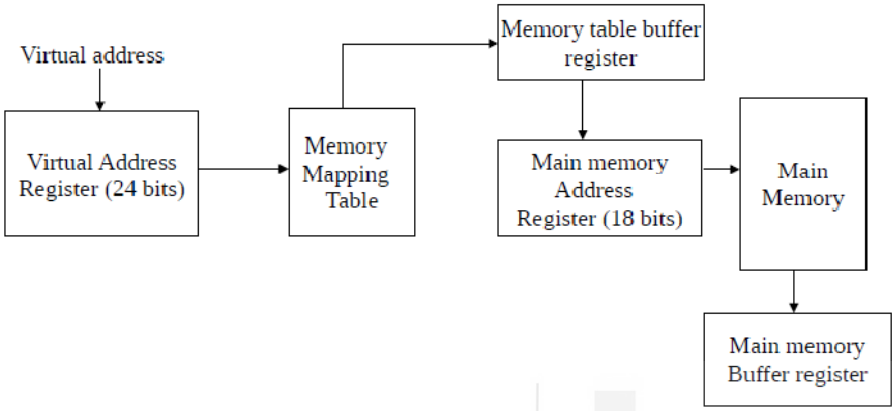


Figure 6.17: Virtual Address Mapping to Physical Address

In a multiprogramming environment, programs and data are transferred to and from auxiliary memory and main memory based on demands imposed by the processor. For example program 1 is currently being executed by the CPU. Only program 1 and a portion of its associated data as demanded by the processor are loaded from secondary memory into the main memory. As programs and data are continuously moving in and out of the main memory, space will be created and thus both program or its portions and data will be scattered throughout the main memory.

Check Your Progress 3

1. How can interleaved memory can be used to improve the speed of main memory access?

   ……

   ……

2. Explain the advantages of using associative memory?

   ……

   ……

3. What is the need of virtual memory.

   ……

   ……

6.8 SUMMARY

This unit introduces you to the concept relating to cache memory. The unit defines some of basic issues of cache design. The concept of cache mapping schemes were explains in details. The direct mapping cache uses simple modulo function, but has limited use. Associative mapping though allows flexibility but uses complex circuitry and more bits for tag field. Set-associative mapping uses the concept of associative and direct mapping cache. The unit also explain the use of memory interleaving, which allows multiple words to be accessed in a single access cycle. The concept of

content addressable memories are also discussed. The cache memory, memory interleaving and associative memories are primarily used to increase the speed of memory access. Finally, the unit discusses the concept of virtual memory, which allows execution of programs requiring more than physical memory space on a computer. You may refer to further readings of the block for more details on memory system.

6.9 ANSWERS

Check Your Progress 1

1. While executing a program during a period of time or during a specific set of instructions, it was found that memory reference to instructions and data tend to cluster to a set of memory locations, which are accessed frequently. This is referred to as locality of reference. This allows you to use a small memory like cache, which stores the most used instructions and data, to enhance the speed of main memory access.
2. Typical block size of main memory for cache transfer may be 1, 2, 4, 8, 16, 32 words.
3. *effective access time = 0.9 (hit ratio) x 10 (cache hit time)*
   + 0.1(miss ratio) x (50+10) (cache miss and memory reference)
   effective access time = 0.9 x 10 + 0.1x 60
   = 9 + 6
   = 15 ns

Check Your Progress 2

1. Main memory size = 64 words (a word = 16 bits) = $2^6 \Rightarrow n$=6 bits
Block Size = 32 bits = 2 words = $2^1 \Rightarrow k$=1 bit
The size of cache = 8 blocks of 32 bits each = 8 lines
Tag size for associative mapping = n-k = (6 - 1) = 5
The address mapping for an address: 21 in decimal that is 010101

Block Address	**Address of a word in a Block**
01010	1
01010	1
Tag	

In set associative memory the given tag can be stored in any of the 8 lines.

2. Main memory size = 64 words (a word = 16 bits) = $2^6 \Rightarrow n$=6 bits
Block Size = 32 bits = 2 words = $2^1 \Rightarrow k$=1 bit
The size of cache = 8 blocks of 32 bits each = 8 lines $\Rightarrow m$=3 bits
Tag size for direct mapping = (n-k) - m = (6 - 1) - 3 = 2
The address mapping for an address: 27 in decimal that is 011011

Block Address of Main Memory		**Address of a word in a Block**
01	101	1
01	101	1
Tag	**Line Number**	

The required word will be found in line number 101 or 5 (decimal)

3. Main memory size = 64 words (a word = 16 bits) = $2^6 \Rightarrow n$=6 bits
Block Size = 32 bits = 2 words = $2^1 \Rightarrow k$=1 bit

The number of sets (v) = 4 sets of 2 lines each, thus, d = 2
Tag size for direct mapping = (n-k) - d = (6 - 1) - 2 = 3
The address mapping for an address: 12 in decimal that is 001100

Block Address of Main Memory		**Address of a word in a Block**
001	10	0
001	10	0
Tag	**Set Number**	

Set number = 10 = 2 in decimal
Thus, required word can be in any of the line in set number 2.

Check Your Progress 3

1. Memory interleaving divides the main memory into modules. Each of these module stores the words of main memory as follows (example uses 4 modules and 16 word main memory.
   Module 0: Words 0, 4, 8, 12 Module 1: Words 1, 5, 9, 13
   Module 2: Words 2, 6, 10, 14 Module 3: Words 3, 7, 11, 15
   Thus, several consecutive memory words can be fetched from the interleaved memory in one access. For example, in a typical access words 4, 5, 6, and 7 can be accessed simultaneously from the Modules 0, 1, 2 and 3 respectively.

2. Associative memory do not use addresses. They are accessed by contents. They are very fast.

3. Virtual memory is useful, when large programs are to be executed by a computer having smaller physical memory.

UNIT 7 – INPUT/OUTPUT ORGANISATION

Structure

7.0 Introduction
7.1 Objectives
7.2 Input/Output (I/O) Devices
7.3 The Input/Output (I/O) Interface
  7.3.1 System Bus and I/O Interface Modules
  7.3.2 I/O and Memory Bus
  7.3.3Isolated and Memory-Mapped I/O
7.4 Device Controllers
  7.4.1 Device Controller and I/O Interface
7.5 Device Drivers
7.6 Asynchronous Data Transfer
  7.6.1 Strobe Control
  7.6.2 Handshaking
7.7 Input/Output (I/O) Techniques
  7.7.1 Programmed I/O
  7.7.2 Interrupt-Driven I/O
  7.7.3 Interrupt Handling
  7.7.4 Direct Memory Access (DMA)
7.8 Input Output Processor (IOP)
  7.8.1 Characteristics of I/O Channels
7.9 External Communication Interfaces
7.10 Summary
7.11 Solutions /Answers

7.0 INTRODUCTION

In the preceding units, you have learned the concepts of the memory system for a computer system. The memory system of a computer includes primary, secondary and auxiliary, and high-speed memories. As discussed, the main memory of thecomputer system is used for storing the instructions and data of the programs, which are getting executed. To execute the program, the computer may need some data which is known as input.The program execution results in the creation of processed data, which is known as output. In addition to the memory system, another important component is the input and output system which is used to receive/send data from/to the external environment. This unit introduces you to various Input/Output (I/O) techniques and controllers, device drivers, structure ofI/O interface, and asynchronous data transfer.This unit also explains the I/O processor which is exclusively used for I/O operations.

7.1 OBJECTIVES

After going though this unit, you should be able to:

- define the structure of input/output (I/O) interface and I/O devices;
- explain the structure of controllers;
- explain different data transfermodes;
- explain various techniques used for Input/Output in a computer system;
- discuss the need of an input/output (I/O) processor;
- explain the role of external serial and parallel communication interfaces;
- explain the concepts of interrupt processing

7.2 INPUT/ OUTPUT (I/O) DEVICES

The input/output devices are used by a computer system to provide an interface with the external environment i.e., humans and other devices, which are connected to a computer. Themajor components of a typical microcomputer system are *microprocessor/CPU, memory*

system and an *I/O interface*.These components are connected through buses. The primary function of the system bus is to transfer control information, addresses, instructions, and data between these components. Figure 7.1 depicts the block diagram of a typical microcomputer system.

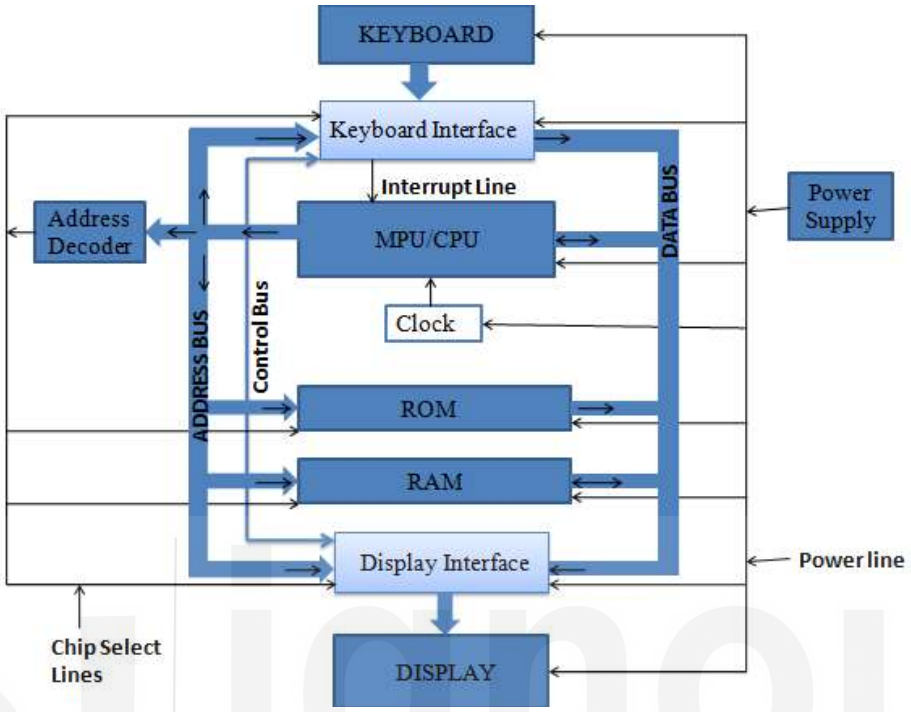


Figure 7.1: A Typical Microcomputer System's Block Diagram

The microcomputer, as shown in Figure 7.1, has a single micro processing unit (MPU). It also has RAM and ROM, which may be constructed by a number of RAM and ROM chips. The diagram also shows two basic interfaces, viz. keyboard interface and display interface, which are connected through the system bus. You may please note that other I/O interface may be connected in a similar way. Please also note that the system bus has been shown in the diagram as three distinct buses, viz. control bus, address bus, and data bus.

An Input/Output (I/O) subsystem includes all the input/output interfaces and connected Input/output devices.The basic objective of an I/O subsystem is to provide an efficient communication medium between the computer system and the external environment (humans and other devices).An I/O interface is used to connect an external I/O device with the computer system. The I/O interface interchanges control, status and data with the external device. The I/O interfaces can also be used to transfer instruction/data/control within the computer units, including processor registers and memory units.An I/O device attached to the computer is also known as **peripheral**or **external device**. The external devices may be categorized on the basis of their communication endpoints as:

- **Human readable**: These devices provide information in human readable form. Example-*display terminals*,*printers,* etc.
- **Machine-readable:**These devices provide information in machine readable form. Example-*magnetic disks*, *CD-RW,* etc.
- **Communication:**These devices provide information to communication devices such as *MODEM*.

7.3 THE INPUT/OUTPUT (I/O) INTERFACE

AnI/O interface (also known as **I/O Module**) is used totransfer information between internal memory and peripheral devices. Each peripheral device has its own set of characteristics. An I/O interface helps in resolving the differences between the processing unit components and peripherals. The following table lists major differences:

Table 7.1: Peripheral devices and processing unit components

Sr. No.	Processing unit components	Peripheral Devices	Remarks
1	Processor and memory are electronic devices.	Peripheral devices are electromagnetic devices.	Hence, they perform operations in a different manner
2	Data transfer rate of processor is very fast.	Data transfer rate of peripheral devices is slow.	Thus, a synchronization mechanism is required.
3	Processor uses word format.	Peripherals, in general, use bytes/blocks format.	
4	Processor may communicate either directly with different manners or indirectly using an interface in similar fashion.	Each peripheral device may have a different operating mode.	Thus, an interface to handle different operating modes is required.
5		Each peripheral device must be controlled in such a way that the operation of one does not disturb the operation peripheral device.	

To address the issues mentioned in Table 7.1, ahardware component between the peripheral devices and CPU is required to control and synchronize all input/output operations in the computer systems. These hardware components, which are used to provide an interface between the peripheral devices and the processor, are known as**interface units**.An I/O interface acts as a bridge between the processor and peripheral devices.

The I/O interface offers an interface which connects the internal components i.e.,processor and main memory as well as external components i.e., peripheral or external devices. In addition to data transfer between processor to I/O devices, it also establishes the coordination between them. Moreover, the I/O interface also hascomponents like **buffer system** and **error detection** mechanism to deal with the speed differences between processor and peripheral devices.

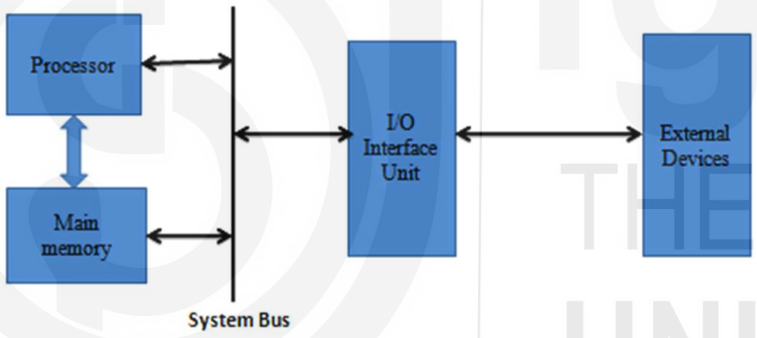


Figure 7.2: A Block Diagram of an I/OInterface connectingwith Processor and External Devices

Roles of I/O Interface

The major roles of I/O interfacesare:

1. Communication with the processor

To provide data transfer between processor and I/O interface, the control signals (Read/Write/Status), address of memory locations and actual data are required to exchange. The system bus works as a communication medium in order to facilitate the exchange of information/data between the processor and I/O interface. In system bus, the addressbus is used to send addresses; the control bus for control signals whereasthe data bus is employed for actual data transfer between processor and I/O interface.

2. Synchronization of control and timing signals

The I/O interface shares system bus with memory in order to provide data transfer. The timing and control signals are required in order to synchronize the flow of data from peripheral devices to processor/memory and vice versa. For instance-An I/O interface may request for control of data bus for data transfer from a peripheral device to the processor.

3. Communication with the I/O device

To complete an I/O operation, the exchange of data between I/O device and I/O interface is necessary. AnI/O operation may require sendingstatus signals, commands, or data.

4. Provision for data buffering

Due to the speed mismatch between the I/O devices and processor, the facility for data storage for a temporary period is required. The I/O interface stores information in registers temporarily which is referred to as data buffering. An I/O operation occurs in short bursts and the data are temporarily stored in a buffer area of I/O interfaceuntil the peripheral device/processoris ready

to receive the data. Therefore, it is not required to hold the system bus for I/O operation due to slower I/O devices.

5. Error detection mechanism

The I/O interface also provides an in-built mechanism for error detection to check the communication errors along withmechanical errors. The errors are reported to the processor using parity bits and other mechanisms for further actions. Some mechanical errors may occur in devices such as mechanical and electrical failures, printer's paper jam etc.

In Figure 7.3, the internal block diagram of I/O interface unit consisting of different signals for processor as well as for I/O device is shown.

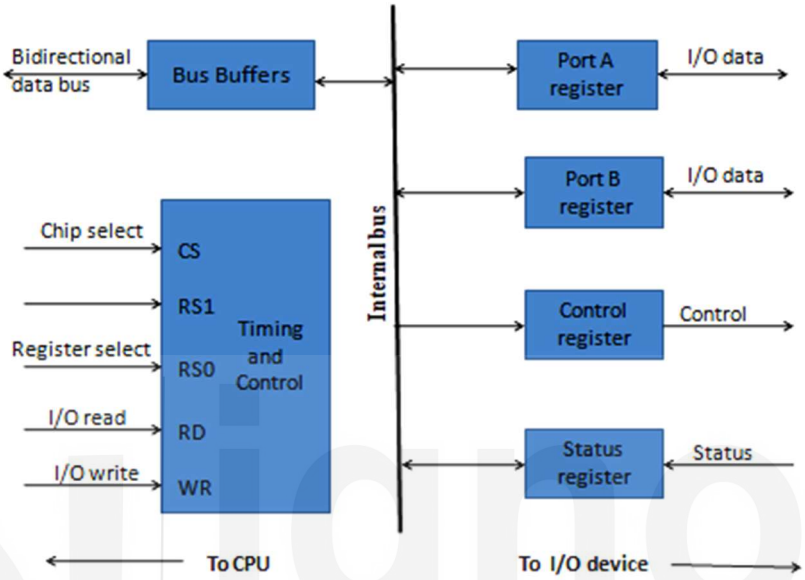


Figure 7.3: I/O Interface Unit's Block Diagram

7.3.1 System Bus and I/O Interface Modules

The control bus, data bus and address bus form a single bus which is known as system bus and it is used to establish the connection between the processor and I/O devices. Figure 7.4 depicts the communication links between the processor and several peripheral devices.

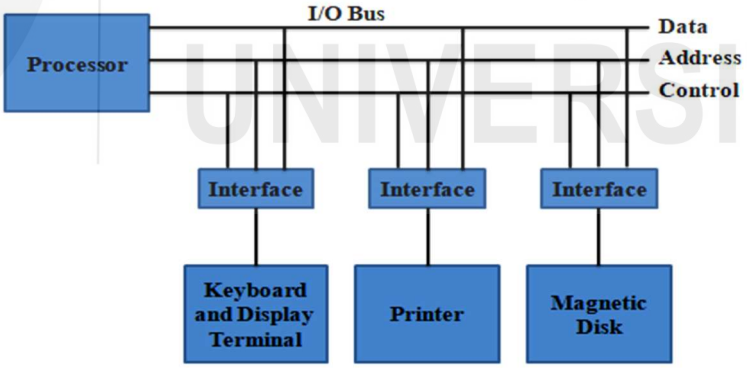


Figure7.4: Communication links between Processor and Peripheral devices

The I/O bus is connected to all peripheral devices through I/O interfaces. In order to communicate with an intended device, the processor sendsthe address of the device using address bus. Every I/O interface consistsof an address decoder to continuously monitor the content of address bus. When anI/O interface observes the address of its own peripheral device, it activates the associated device and the bus; otherwise, the peripheral device is disabled. A control signal (*I/O command*) is also provided simultaneously through the control bus. The four types of I/O commands that are arising out of the processor are as follows:

1. **Control Command:** This is the control signal code which is sent to the corresponding peripheral device and informs it about the action it has to perform.
2. **Status Command:** The is the statussignal which is used to test status conditions of the peripheral devices and interface. Some status commands are BUSY, DATA AVAILABLE, ERROR or NOT IN BUFFER etc.

3. **Data Output Command:** The signal/commandis utilized to activate the I/O interface for data transfer from the processor to buffer of I/O interface. The data from the buffer is ultimately sent to the peripheral device. The data is sent from the CPU to the buffer of interface after this command is provided.
4. **Data Input Command:** The processor sends this signalwhenever there is a need to read data from data any peripheral device. After the issuance of this command, the data from the intended peripheral device are extracted into the interface's buffer and this is followed by the data read operation by the processor.

7.3.2 I/O and Memory Bus

Aprocessor requires communicating with the I/O devices and memory system. The system bus (I/O Bus & Memory Bus) is used to control the exchangeof data among the processor, memory system and I/O devices. The I/O bus and memory bus; both busesconsist of data, address and control lines. To establish communication with the memory and I/O devices, the system bus can be used as follows:

i. One bus for each memory system and I/O.
ii. Shared data and addressbuses for both I/O devices and memory system but exclusive control bus for both.
iii. Shared system bus for both memory system and I/O devices.

7.3.3 Isolated and Memory-Mapped I/O

In case ofisolated I/O, the data and address buses are shared between I/O and memory but separate read/write control lines are used for I/O devices. Whenneverthe processor decodes instruction for an I/O device, it places the address on the address line and activates I/O read or write control line which causes data transfer between CPU and I/O device.

In other alternative, the computer employs the same set of read/write signals for I/O and memory and does not differentiate between I/O and memory addresses. This configuration is known as memory-mapped I/O.

7.4 DEVICE CONTROLLERS

The system bus is used to establish communication between the processor and all components including I/O devices. Some components are directly connected to the processor through system while some components are not directly connected to system bus.The intermediate electronic device, known as *device controller*, is used to connectthe I/O device and the system bus. The I/O device is connected at one end while it is connected with the system bus on another end. Thus, a device controller works as an interface to provide the communication medium between the system bus and I/O device.

Device Controller

It is not necessary that a device controller controls a single peripheral device. It can generally control more than peripheral devices. The device controller is available as an electronic circuit board, which can be directly plugged into the system bus. The device controller is also connected through a cable with a peripheral device. The connectors at the rear panel of a computer are in fact the end points of these cables. These connctors are called the ports and are used to plugin the external peripheral device to a computer. Figure 7.5 depicts the connection of computer system and I/O devices through device controllers.

In Figure 7.5, the following points are important to consider:

i. Each peripheral device is attachedtoa hardware interface known as I/O Port.
ii. A single-port device controller can controlonlyone peripheral device whereas the multi-port device controller controls multiple peripheral devices.
iii. In case of direct memory access (DMA), the communication between I/O controller and memory is established through system bus only, while the communication path passes through the CPU for non-DMA communication.

The followings are themajor benefits for connecting computer system with I/O devices using device controllers:

i. A single device controller may be used to connect multiple I/O devices with the computer system simultaneously.
ii. The device controllers allow I/O devices to upgrade/change without any update/change required in the current configuration of a computer system.
iii. The device controllers allow connecting the I/O devices ofdifferent configurations/manufacturers with the computer system. It allows computer usersto purchase I/O devices of different their configurations/manufacturers.

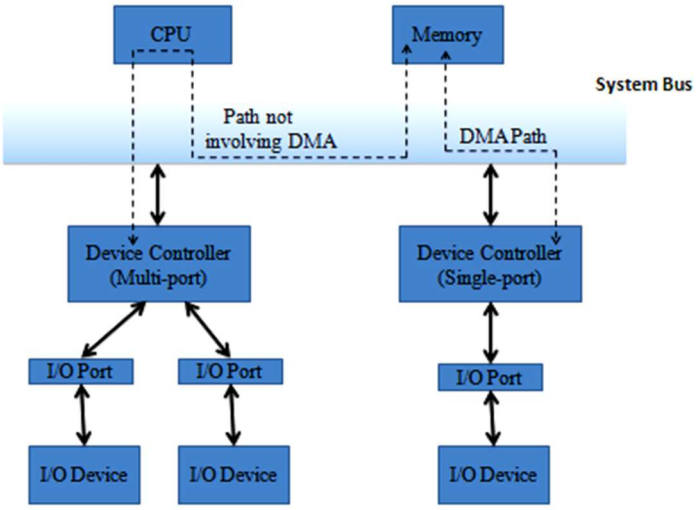


Figure 7.5: Device Controller connecting I/O Devices and Computer System

7.4.1 Device Controller and I/O Interface

Due to the different input/output needs and design complexities, there exist various peripheral deviceswith different structures. Thus,the common/standard structure of I/O interface does not exist. Figure 7.6 shows a generalblock diagram of an I/O interface. The common characteristics of the general structure of an I/O interface are as follows:

i. I/O logicis required in order to decode and execute thedata between the processor and I/O interface. Hence, control lines between the I/O interface and processors are required.
ii. Data lines between the system bus and I/O interface are necessary to facilitate the actual data transfer.
iii. The data registers may behave like a buffer between processor and I/O interface.
iv. To deal with each I/O device with different configuration,the I/O interface must have a separate logic specific.

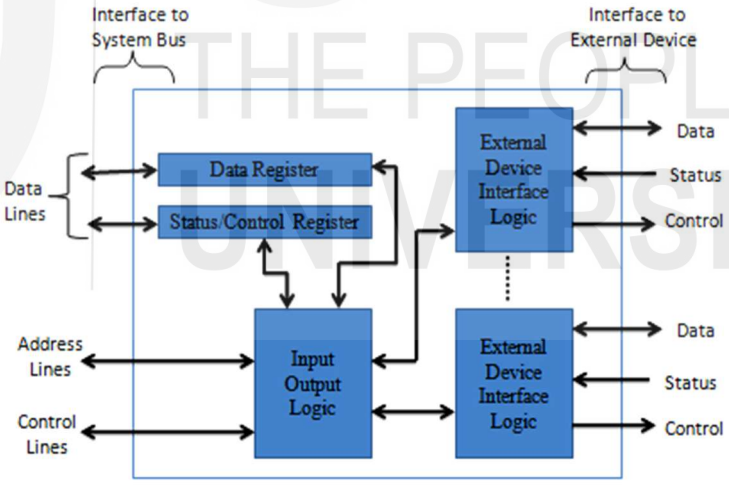


Figure 7.6: Block Diagram of an I/O Interface

7.5 DEVICE DRIVERS

The device driver is a software interface to handle communication with a specific peripheral device. The purpose of the device driver is to decodea logical request from the computer user into a series of specific actions performed by theperipheral device. For instance, a computer user may request to read a record from a disk. The device driver converts this logical request into a series of discrete commandsi.e.,check the status of the intended disk in the drive, locating the desired file, setting the position of R/E head, etc.

Device Drivers in Windows Systems

The device drivers are realized in the form of dynamic linked libraries (DLLs). The DLL consists of shareable code and thus, a single copy of the DLL code is loaded into the mainmemory. The hardware/software vendors can implement adevice driver for new

deviceswithout modifying or affecting the code of Windows operating system. Moreover, it allows multiple optional drivers tobe configured for differentperipheral devices.
For Windows based system, the device installation in the form of***Plug and Play*** is necessaryin order to add new peripheral devices. The objective of ***Plug and Play*** is to convertthe device connecting process from manual to automatic. In automatic device connecting process, the device will be attached which is followed by installation of driver software. After this, the installation process will be automatic and the settings will be changedaccording to the configuration of the host computer system.

Device Drivers for UNIX Systems
The device drivers are generally linked to the object code of the core of operating system which is known as kernel. To use a new device not included in operating system, the device driver object code has been re-linked with UNIX kernel. The advantages of this technique are simplicity and run-time efficiency while the main disadvantage is it requires regeneration of the kernel in order to attach a new device. Each entry of the /dev directory of the UNIX system is associated with a device driver which in turn is attached with the related device. Table 7.2 consists of information of some device as follows:

Table 7.2: Device Name in UNIX System

Device name	Description
/dev/console	console of a system
/dev/lp	line printer
/dev/tty01	user terminal 1
/dev/tty02	user terminal 2

Check Your Progress 1

1. What do you understand by I/O Interface?List major functions of I/O interface?

   ……

   ……

   ……

2. What is the need of device controller? What are the major benefits?

   ……

   ……

   ……

3. What is the need of a device driver? Give examples of device drivers.

   …….

   ……

   ……..

7.6 ASYNCHRONOUS DATA TRANSFER

This section discusses the different data transfer modes.Thereexists two ways to transfer data from sender to receiver in a digital computer system. It is necessary tosynchronize the data transfer operationsusing clock pulses. When the internal registers of the two units i.e. I/O interface and CPU share a common clock, the mode of data transfer between them is known as synchronous data transfer. On the contrary, both of the units i.e., I/O interface and CPU are designed to work independent manner and each unit uses its own private clock in order to establish coordination. This mode of data transfer is knownas asynchronous data transfer.
The asynchronous data transfer faces some problems. Since there exists no fixed time slot to send or receive data. Therefore, there is no guarantee that whether the data is old or new. This problem can be solved using the following two methods:

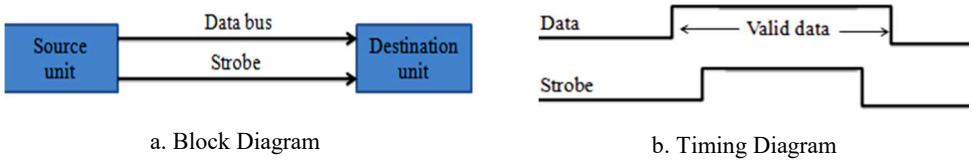
i. Strobe Control Method
ii. Handshaking Method

7.6.1 Strobe Control method

In strobe control method a single control line is used each time for data transfer and this control signal is referred to as *Strobe*. The strobe may be activated in the following two different ways:

A. Source-initiated Strobe – Figure 7.7 depicts the block diagram and timing diagram for source-initiated strobe method. In this method, the data transfer is performed as:

i. Initially, source unit putsthe data on data bus and the strobe signal is switched ON.
ii. The destination unit reads data from the data bus.
iii. After reading data,the strobe gets OFF.



a. Block Diagram b. Timing Diagram

Figure 7.7: Source Initiated Strobe

B. **Destination-initiated Strobe** –Figure 7.8 depicts the block diagram and timing diagram for destination-initiated strobe method. The steps for data transfer are as follows:

i. First, the destination unit switches ON the strobe signal.
ii. After observingstrobe ON, the source unitputs data on the data bus.
iii. The destination unit reads the data from the data bus and the strobe signal gets OFF.



a. Block Diagram b. Timing Diagram

Figure 7.8: Destination Initiated Strobe

Disadvantages of Strobe Method:

The disadvantages of strobe methods are as follows:

- In source-initiatedstrobe method, it is not possible toconfirm the status of data at destination unit i.e., whether the data received is valid or not.
- In destination-initiatedstrobe method,itis not possible toconfirm the status of data at the data bus i.e., whether the data extracted from the data bus is valid or not.

7.6.2 Handshaking method

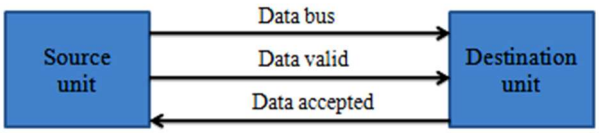
In handshaking method of asynchronous data transfer, the problemsof strobe control method can be handled. The handshaking method can be realized in the following two different ways:

A. Source-Initiated Handshaking – It consists of the following signals:
Data Valid: When activates, itspecifies that data on the data bus is validotherwise invalid.
Data Accepted: When activates, it specifies that data is acceptedotherwise not accepted.

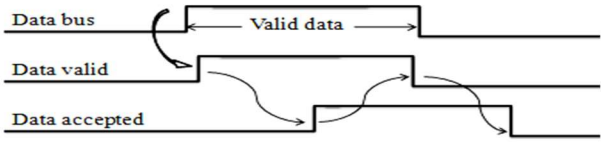
The steps of data transfer in Source Initiated Handshaking method are as follows:

i. First, source unitcopiesdata on the data bus and enables*Data Valid* signal.
ii. The destination unitreceives data from the data bus and enables*Data Accepted* signal.
iii. After this, *Data Valid*signalisdisabled whichimplies that data on data bus is invalid now.
iv. Finally, *Data Accepted* signal is disabled which indicates that the process of data transferhas been completed.

Now,it can be ensured that destination unit has read the valid data from the data bus using*Data Accepted* signal.*F*igure 7.9 depicts the blockdiagram and timing diagram of thesource-initiated handshaking method.



a. Block Diagram



b. Timing Diagram

Figure 7.9: Source Initiated Handshaking

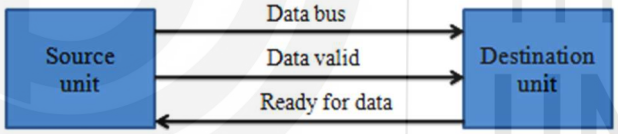
B. Destination-Initiated Handshaking – It employs the following signals:

***Request for Data*:**When activated, it requests for putting data on the data bus.
***Data Valid*:** When activated, itspecifies data is valid on the data bus otherwise invalid data.

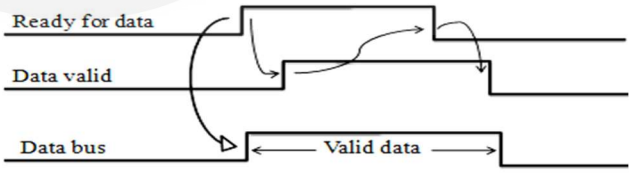
The steps of data transfer in Source Initiated Handshaking method are as follows:

i. When destination unit is ready to receive data, the appropriate signal i.e.,***Request for Data*** gets activated.
ii. In response, the sourceunit places data on the data bus as well asit also enables***Data valid*** signal.
iii. Next, the destination unit accepts data from the data bus. After receiving data, it disables***Request for Data*** signal.
iv. Lastly, the ***Data valid*** signal is disabled implies that the data bus consists of invalid data now.

Using ***data valid*** signal in **Destination Initiated Handshaking**, it is possible to ensure that source has put the data on the data bus.*F*igure 7.10 shows the block diagram and timing signal in destination-initiated handshaking. It consists of ***Request for Data***as well as ***Data valid*** signals.



a. Block Diagram



b. Timing Diagram

Figure 7.10: Destination Initiated Handshaking

7.7 INPUT/OUTPUT (I/O) TECHNIQUES

This section explains the methods to use I/O interface to support data transfer (input/output) from peripheral devices.The data transfer (I/O operation) between the computer system and peripheral devices may be performed using three techniques given as follows:

- Programmed I/O
- Interrupt-driven I/O
- Direct memory access (DMA)

Table 7.3presents important characteristics of the above three techniques.

Table 7.3: Three I/O Techniques

I/O Technique	Data Transfer Path through	Processor Intervention	Interrupt
Programmed I/O	Processor	Continuous	Not-Required
Interrupt-driven I/O	Processor	Discontinuous	Required
Direct memory access (DMA)	Direct to Memory	At the beginning and end	Required

7.7.1 Programmed I/O

In this technique, the processor is completely responsible for managing all I/O operations. The program which requests for I/O operation directly and continuously controls the I/O operation through the processor.The processor runsprogram that starts, continuously monitors and endsan I/O operation.The I/O operation may involve the following data transfer operations:

a. Data transfer from processor registers to I/O device.
b. Data transfer from I/O device to main memory through processor registers.

The major steps in programmed I/O are as follows:

i. The processor issues READ/WRITE command.
ii. The processor requests for I/O operation to I/O interface.
iii. I/O interface sets status bits.
iv. The processor continuously checks the status bits.
v. The processor waits for I/O device.
vi. When I/O device is ready, I/O operation (Read/Write data) is performed.

Figure 7.11 depicts the diagram of programmed I/O technique. When the CPU issues a command for an Input or an output to the I/O interface, it must wait until the I/O operation is completed. This process is known as polling.

Disadvantage: With the programmed I/O method, the processor continuously checks the status of the I/O device whether it is ready or not. The processor has to wait for slower I/O operations to complete. Therefore, this technique is wasteful of processor's time.

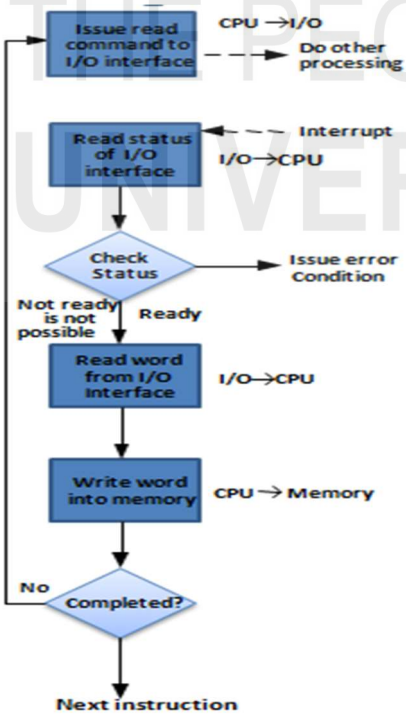


Figure 7.11: Programmed I/O

I/O Commands

Whenever the processor addresses an I/O interface, it sends an address and I/O command. There are four I/O commands which a processor may send to I/O interface for I/O operation.The I/O commands are given as follows:

- **Control:**Control commands are used to activate the device and also specify what operation is to perform. For example- aUSE command to make a specific device as current device for read/write operation.
- **Test:** The Test I/O command is used to check the status of a device. For instance- Whether the device is in **error condition**, **ready state**, or **notreadystate**.
- **Read:** This command is used to receiveone item of input data from the I/O device which is in communication.
- **Write:**This command is used to sendone item of output data to the respective output device.

7.7.2 Interrupt-driven I/O

For an I/O operation inprogrammed I/O, the processor continuously waits for the operation to be completed and the data transfer occurs when the device is ready. This process is known as polling which leads to slow performance of the processor. Interrupt-driven I/O can reduce the polling time efficiently.In interrupt-driven I/O, the processor issues a read/write command and it starts the execution of some other program/instructions. Whenever the desired device is ready or has completed the assigned I/O task, an interrupt signal is sent to processor for further actions. Figure 7.12 depicts the procedure of interrupt-driven I/O technique. The complete list of steps in interrupt-driven I/O is as follows:

i. The processor issues read /write command
ii. The processor executes some other program/instructions
iii. The I/O interface reads data from the desiredI/O device
iv. I/O interface interrupts the processor
v. The processor checksthe interrupt after completingeach instruction cycle
vi. The processor saves the context of the program in execution
vii. The processor requests for the desired data
viii. The I/O interface transfers data
ix. The processor resumes the previous program it had been executing before the interrupt.

Advantage: This technique reduces the overall waiting time of the processor.

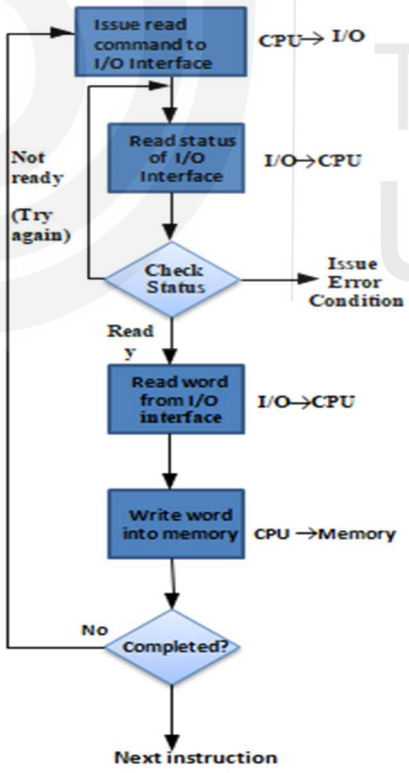


Figure 7.12: Interrupt Driven I/O

7.7.3 Interrupt Handling

An interrupt is an event which may occur to receive some service by a device or due to some error. After an interrupt occurs, it causes multiple tasks in the software as well as in the hardware.

Figure 7.13 represents the series of hardware tasks which occurred in order to I/O operation using an I/O device. The complete lists of tasks required for interrupt processing are as follows:

1. An I/O device sends an interrupt signal to processor.
2. The processor completes the execution of currently running instruction and next, it checks and takes action accordingly.
3. An acknowledgement signal is also sent by the processor to the corresponding I/O device.
4. The processor saves the context of the program that is being executed currently. It saves the following registers:
   (a) The status of the processor: program status word (PSW) register
   (b) The next instruction to be executed: program counter (PC) register.
5. In addition to thecontents of PC & PSW for the interrupted program, it is required to store the contents of the registers of processor on the stack. Figure 7.14 depicts an example of interrupt occurrence. According to theFigure 7.14, a user running program is interrupted after the execution of the instruction at location N. To handle the interrupt, the processor saves the contents of all general-purpose registers, PSW, PC along with the address of the next instruction at location N+1.
6. The processor loads the PC with the address of first location of the interrupt and then, the processor starts executing instruction from the interrupt-handler program. It means that the interrupt handler handles the interrupt.
7. When the processing of the interruptis completed. The context of the previous program is restored i.e.contents stored at stack are restored in the processor registers, which is shown in Figure 7.15.
8. Finally, the values of PC and PSW are retrieved from the stack and restored on the corresponding registers. It leadsto execution of the instruction whichwas interrupted from the previously interrupted program.

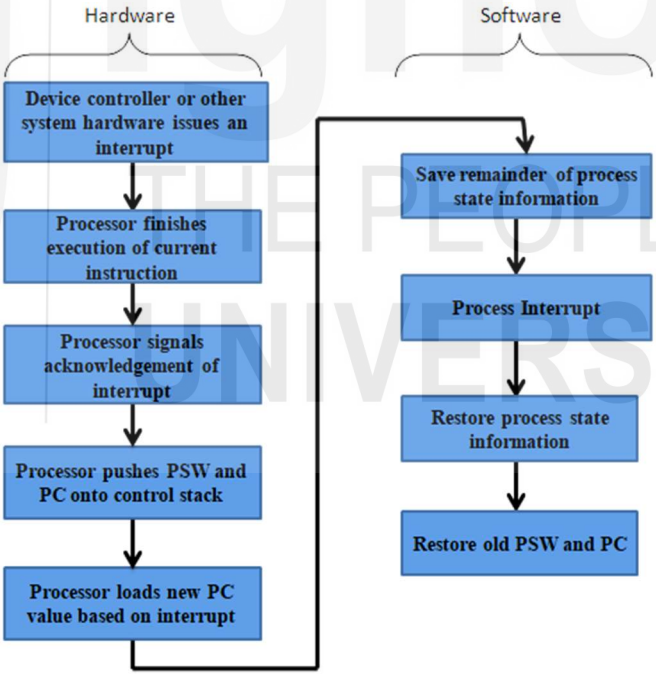


Figure 7.13: The sequence of steps in Interrupt-Processing

In simplewords, the interrupt handling consists of the followings:

i. Interruptthe current program
ii. Saves the context of the interrupted program
iii. Transfer control to interrupt servicing program
iv. Execute the interrupt servicing program
v. Restore the context of interrupted program
vi. Resume interrupted program from the point where it was interrupting.

Design Issues

To realize the interrupt-driven I/O, there exist two design issues given as follows:

1) How does the processor identify the I/O device issuing the interrupt?
2) How does the processor handlemultiple interrupts occurred simultaneously?

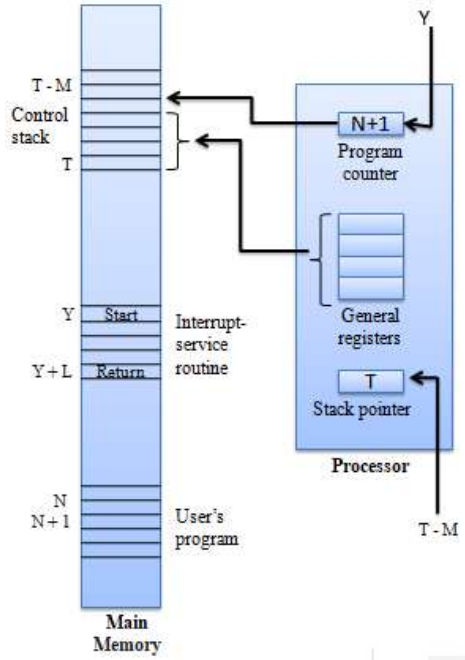


Figure 7.14: Interrupt occur for instruction at memory location N

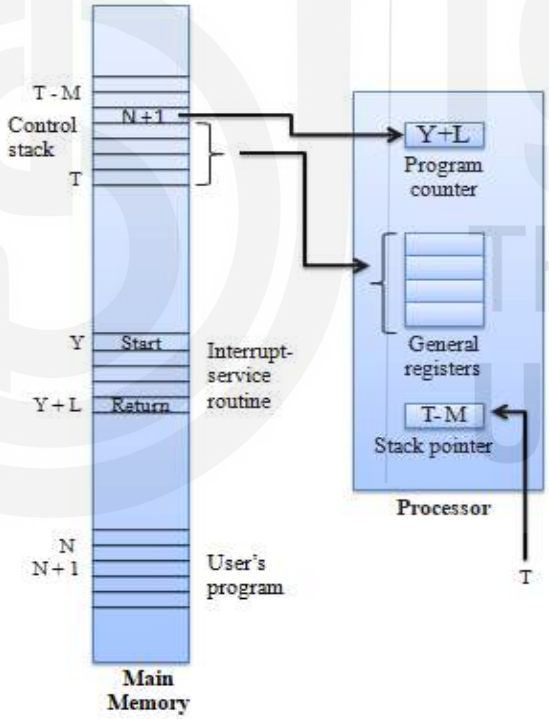


Figure 7.15: Return after handling an Interrupt

The following four general techniques can be used to solve the above design issues:

i. **Multiple Interrupt Lines:** Multiple interrupt lines can be used to handle multiple interrupts. In this technique, the priorities are assigned to various interrupts. Whenever an interrupt occurs, the highest priority interruptwill be handled first among all interrupts. However, the practical realization of this technique is difficult because computer system provides only a few lines for the interrupt.

ii. **Software Poll:** The processor jumps to an interrupt service routine after an interrupt has occurred. The processor polls at I/O interface by observing the status register in order to identify the I/O interface which caused the interrupt. The software poll technique is time consuming due to polling process.

iii. **Daisy Chain or Hardware Poll:** In this technique, the interrupt request line is shared by all I/O interfaces. Whenever an interrupt occurs, the processor sends an

acknowledgement of interrupt. This acknowledgement goes to all the I/O devices through I/O interface. When it reaches to the I/O device which sends interrupt, the I/O device sends a response by specifying an address or unique identifier through the data bus. This unique identifier helps in deciding the appropriate interrupt servicing program. The hardware poll consists of an in-built priority mechanism and this priority is based on the sequence of devices on interrupt acknowledge line.

iv. **Bus Arbitration**: The bus arbitration technique allows only one I/O interface to control the bus and this I/O interface is known as bus master. It means only one interrupt request can be fulfilled at the same time. After acknowledgement of interrupt by processor, the I/O interface sends the interrupt vector to processor through data lines. The interrupt is handled according to the number in interrupt vector.

7.7.4 Direct Memory Access (DMA)

The programmed I/O as well as interrupt-driven I/O techniques require the processor's interventions for data transfer to/from the main memory. Since processor may involve in the execution of multiple programs due to multiprogramming, which restricts the processor time for testing the I/O device and servicing the I/O request by transferring I/O data over the system bus. Is it possible to store/retrieve data to/from main memory without involving the processor in the data transfer? Yes! There exists an alternative approach which is referredto as **direct memory access (DMA)**. In DMA, the main memory andI/O interface exchange data directly without the active involvement of processor. Figure 7.16 depicts the block diagram of direct memory access (DMA) technique. Now, an obvious question arises: What is the use of DMA interface? The DMA interface is mainly used to transfer a large quantity of data which isexchanged between I/O device andmainmemory.

Figure 7.16 depicts the block diagram of DMA technique. The steps of the DMA technique aregiven as follows:

i. Processor issues read/write command to DMA module for transferring the block of data
ii. The processor sends the following information to DMA interface-
    a. To perform read/write operation, the Read /Write signal is sent using control lines
    b. I/O Device address is communicated through a data bus
    c. Beginning address of memory block using data bus
    d. "The number of words to be transferred" is communicated through data bus. This information is stored in a count register.
iii. The processor executes some other program.
iv. Now, DMA controller performs the data transfer
v. When DMA controller finishes the data transfer, it sends an interrupt signal to processor

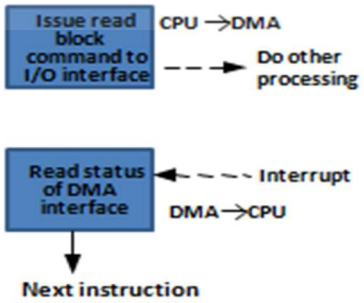


Figure 7.16: DMA Technique

Figure 7.17 shows the block diagram of DMA module along with its different signal lines.

The DMA module handles data transfers to send the entire block of data and one word at a time is transferred directly to or from memory without going through the registers of processor.After transferring the entire block, an interrupt signal is sent to processor by DMA module.Therefore, the processor involvement is limited only at the start and end of the I/O operation.

In DMA, the processor intervention is minimized but it must use the path through system bus. Thus, DMA interface requests for system bus to transfer one data word at a time and the control of the system bus is returned to the processor after transferring on word of data. This process is known as **cycle stealing**. The CPU cycle stealing causes the delay to the operation

of processor for one memory cycle. In DMA technique, the active involvement of processor can be restricted at the beginning and at the completion of the I/O operation.
Figure 7.18 depicts the five cycles of typical instruction execution. In Figure 7.18, three points are marked where a processor can respond to DMA request, and; a point is also marked where the processor can respond tointerrupt request. The point at which the interrupt request can be acknowledged is called the interrupt cycle.

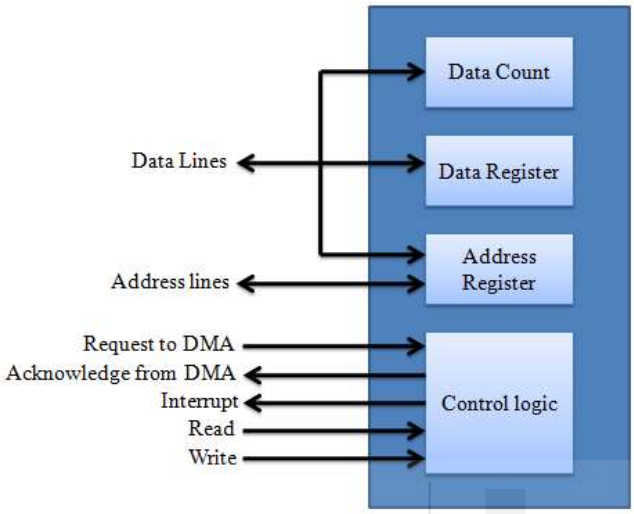


Figure 7.17: DMA's Block Diagram

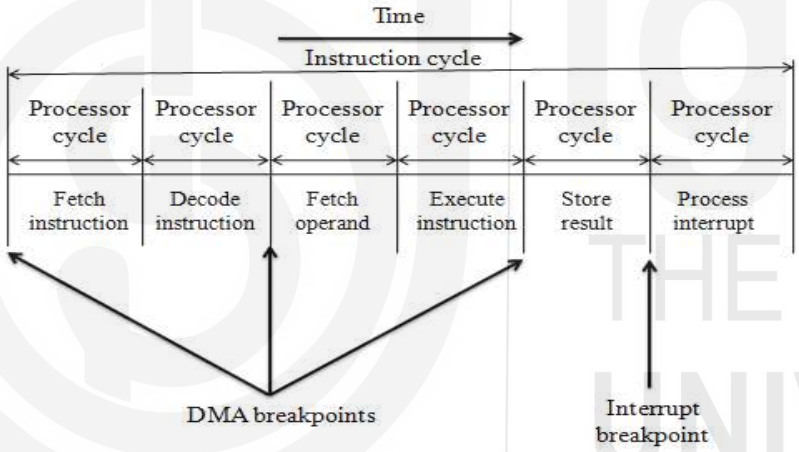
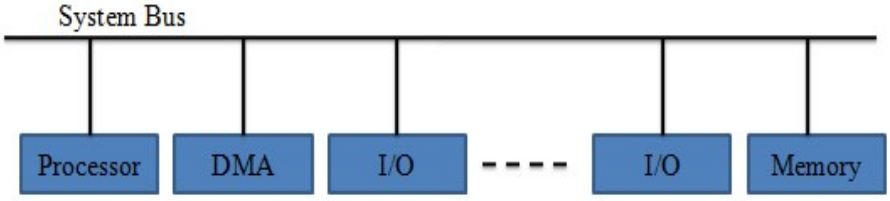
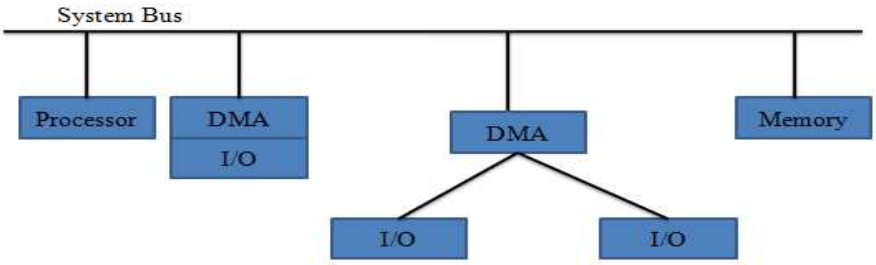


Figure 7.18: Breakpoints for DMA and Interrupt in an Instruction Cycle

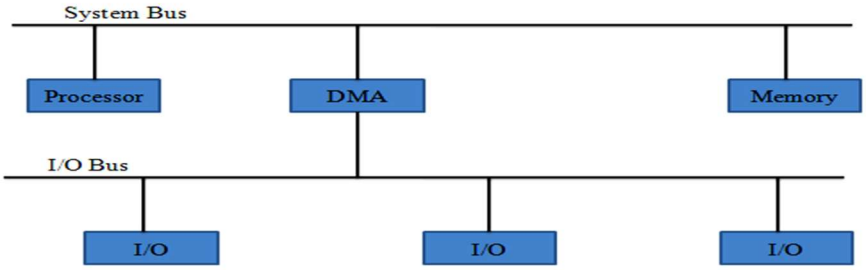
There exist different configurations to realize the DMA mechanism. Figure 7.19(a) shows one of the several configurations. In this configuration, the system bus is shared by all interfaces. The DMA works as a supportive component and may employ programmed I/O fordata transfer between memory and I/O interface using DMA interface. The advantage of this configuration and programmed I/O is that DMA does not require extra system bus cycles to transfer information from DMA to/from I/O interface as well as from main memory to/from DMA.



(a) Single Shared System Bus with detached DMA



(b) Single Shared SystemBus with integrated DMA-I/O



(c) I/O bus

Figure 7.19: Three Different DMA Configurations

Figure 7.19(b) depicts one more DMA configuration which has some advantages over the first configuration shown in Figure 7.19(a). In this configuration, adirect path is provided between the DMA interface and I/O interface and this path is different than the system bus. Furthermore, DMA logic may actas aconstituent of I/O interface and single or multiple I/O interfaces may be controlled by it. One more flexible configuration is shown in Figure 7.19(c) in which the DMA interface is connected to I/O bus. This configuration isextendable in an easy manner. The additional benefit in both configurations is that the data transfer can be performed from DMA interface to I/O interface without the involvement of the system bus.

Check Your Progress 2

1. What do you understand by Programmed I/O technique?
   ……
   ……
   ……
2. Explain interrupt?What processing is performed on the occurrence of an interrupt?
   ……
   ……
   ……..
3. Why is DMA needed? What are its advantage and disadvantages?
   ……
   ……
   …….

7.8 INPUT-OUTPUT PROCESSOR (IOP)

The evolution of I/O techniques passes through many development stages and this evolution can be helpful to understand the idea of I/O processors. Various stages of evolutionaresummarizedas follows:

1. Direct Control: The processor directly controls I/O device.
2. Control using I/O controllers without interrupts: In this configuration, processor and I/O interfaces are different with each other and programmed I/O without interrupts was employed by I/O interface for I/O data transfer.

3. Control using I/O controllers with interrupts: In this configuration, the processor need not to wait for an I/O operation to be completed which leads to improved efficiency of the processor.
4. Control transferred to DMA: In this configuration, the involvement of the processor is restricted at the beginning and on completion of DMA operation. The I/O interface and DMA module control the data transfer directly without the involvement of the processor.
5. Control transferred to I/O processor: On the request of main processor, the I/O processor takes control of I/O operation and performs I/O operation with the intervention of the main processor. This approach allows controllingvarious I/O devices with minimum involvement of processor.

Through various evolution stages, it is evident that the responsibility of I/O tasks has been shifted from CPU to I/O-processor and it leads to improvedperformance of the processor.
It is also evident from the recent developments that a major shift occurs due to the introduction of I/O interface which is capable to executethe I/O instructions. Hence, the *'I/O interface'* is often called an**I/O processor** or **I/O channel**.

Input Output Processor (IOP)
An IOP is a processor that handles only the I/O processing.The block diagram of a computer system with input output processor is shown in Figure 7.20. The computer system with I/O processor consists of three major components: (a) *a memory unit* (ii) *the processor* and (iii) *the IOP*. The IOP controls and manages the input-output tasks. On the command of CPU, the IOP can fetch, decode and execute I/O instructions which are loaded to perform the I/O operations.

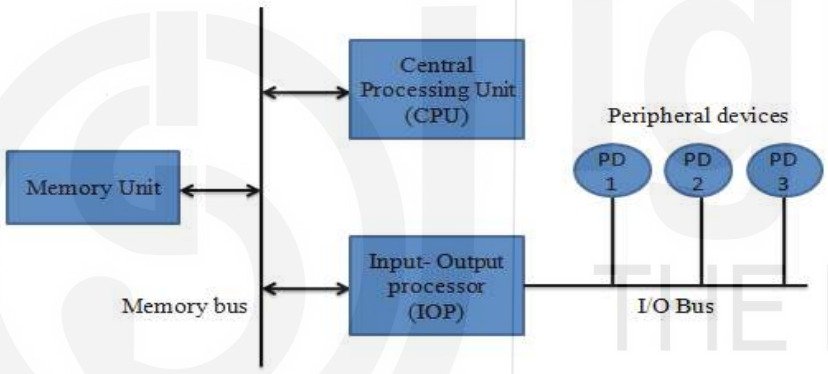


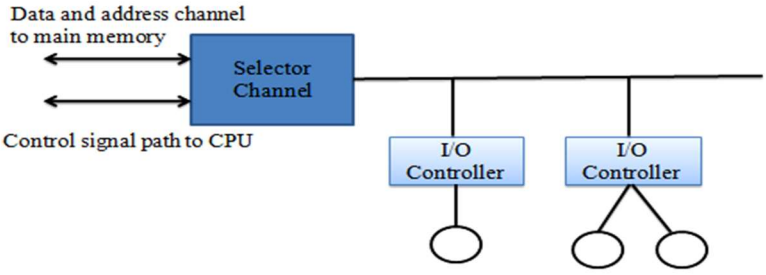
Figure 7.20: Block Diagram of a Computer System with I/O Processor

7.8.1 I/O Processor's Characteristics

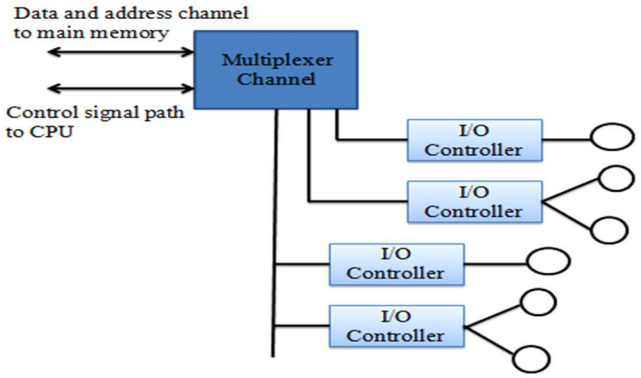
In simple words, anI/O processor (or I/O channel)is theextension of the concept of DMA technique. An I/O channel holds complete control of the I/O operation and is also capable to execute I/O instructions. The CPU initiates the I/O operations andhandovers the execution of I/O commands to I/O processor. In response, the I/O processor executes the I/O instructions stored into the main memory.

The I/O channels can be divided into two categories:

i. Selector I/O channels
ii. Multiplexer I/O channels.



(a) Selector I/O Channel



(b) Multiplexer I/O Channel

Figure 7.21: Structures of I/O Channels

A. **Selector I/O channel:** The selector I/O channel isa dedicated channel to perform data transfer exclusively with a high speed I/O device connected to it. Each I/O controller may controla single or multiple I/O devices. Thus, I/O channelsreduce the burden ofthe CPU by controlling the I/O controllers. The selector I/O channel is shown in Figure 7.21(a).

B. **Multiplexer I/O channel:** Figure 7.21(b) depicts themultiplexer I/O channel. This I/O channel is capable to handle I/O operations with more than oneI/O device altogether. For example-if, there are three slow devices interested to send the following bytes given as:

Ist I/O device:	A1	A2	A3	A4	A5 ……
IIndI/O device:	B1	B2	B3	B4	B5……
IIIrd I/O device:	C1	C2	C3	C4	C5……

Then, the byte multiplexer I/O channel may send the bytes as follows:

C1 B1 A1 C2 B2 A2 C3 B3 A3……

For high-speed devices, block multiplexer I/O channels can be used which transfers data in form of blocks.

7.9 EXTERNAL COMMUNICATION INTERFACES

The external communication interface is used to provide an interface between the peripheral devices and I/O interface. There exist two main categories of external communication interfaces:
(a) Serial Communication Interface
(b) Parallel Communication Interface

The serial communication interfaceemploys only one line for transferring one bit of data at a time. Serial interfaces are used for *terminals*and*printers*.
Theparallel communication interfacecan transfer several bits at the same time. The parallel interfacesareusuallyutilized for I/O deviceshigh-speed.For instance, *disks* and *tapes* use parallel interfaces. The communication that takes place across the communicationinterface includes the exchange of data and control information.
In both serial and parallel communication, the I/O interface must engage in communication with the peripheral. The communication steps for the read/write operation are given as follows:

- The I/O interface sends a control signal to I/O device to requestpermission to receive or send data.
- TheI/O device accepts the Read/Write request and sends an acknowledgement.
- The data transfer operation either from I/O device to I/O interface (Read operation) or from I/O interface to I/O device (Write operation) is performed.
- The I/O device sends an acknowledgement signal after the data transfer operation.

Point-to-Point andMultipoint Configuration
There are two types of connection between the external I/O devices and I/O interface in a computer system: (i) point-to-point (ii) multipoint. In a point-to-point interface, a dedicated line is provided between the external device and I/O interface in the computer system. The

examples of use of point-to-point interfaces point are: keyboard, printer and external modems. Some common examples of point-to-point interfaces are **EAI-232** and **RS-232C**.
On the other hand, the multipoint external interfacesare used to support external multimedia devices (such as audio, video, CD-ROM) and mass storage devices (such as tape drives and disk). Some common examples of multipoint external interfaces are **Infini Band** and **FireWire**.

Check Your Progress 3

1. List majorcharacteristics of I/O processor.

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   ……

   ……

2. Differentiate selector and multiplexer I/O channels.

   ……

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   ……

3. Compare serial and parallel external interfaces.

   ……

   ……

   ……

7.10 SUMMARY

This unit explains exclusively the I/Oorganisation of a computer system. This unit presents the description of I/O devices, the concept of I/O interface, the description and structure of device controllers, and asynchronous data transfer modes. It also discusses three I/O techniques i.e.,programmed I/O, interrupt-driven I/O, and direct memory access (DMA). The threeI/O techniques involve processor at different levels and help processor to behave differently. The interrupt processing is also discussed in detail.The evolution of I/O processor is also discussed along with the brief explanation of the input/output processor as well as the external communication interfaces. The I/O processors are the most recent I/O interfaces that are capable to execute the I/O instructions without the involvement of the processor.

7.11 SOLUTIONS /ANSWERS

Check Your Progress 1

1. An I/O interface acts as a bridge between the I/O devices and processor. It provides a communication interface to connect the computer with the external environment. The I/O interfaces are used to send/receive data from the external environment using external device or peripheral. The I/O interface offers the following major functions:
   - Synchronization of control and Timing signals
   - Establish communication between peripheral devices and processor
   - Data buffering to due to the speed mismatch between memory and processor
   - A mechanism for error detection
2. Device controllers work as a connecting mediumto connect the peripheral devices to a computer system rather than connecting them directly to the system bus. On device controller, the I/O device is connected at one end while it is connected with the system bus on another end. Thus, a device controller works as an interface between the system bus and the I/O device. The major benefits for connecting I/O devices to a computer system using device controllers are as follows:

- A single device controller can be used to connect multiple I/O devices with the computer system simultaneously.
- The device controllers allow I/O devices to upgrade/change without any update/change required in the current configuration of computer system.
- The device controllers allow connecting the I/O devices with different configurations/manufacturers with the computer system. This feature offers more flexibility to the computer users in order to buy I/O devices of different configurations/manufacturers

3. A device driver is a software module which manages the communication with a specific I/O device. It provides a software interface to hardware devices.Some examples of different device drivers are printer driver, sound card driver, graphics driver, network card driver, USB driver etc.

Check Your Progress 2

1. Programmed I/O technique is used to perform the I/O operation and it does not need an interrupt for I/O operation.The main purpose of the programmed I/O is to perform data transfer between processor and external environment. This technique is very inefficient, especially when multiprogramming environment is used, as Programmed I/O requires continuous involvement of the processor for the I/O to complete. This wasted time could have been used for exection of other programs.

2. An interrupt is an event which may occur to receive some service by a device or due to some error. After an interrupt occurs, it causes multiple tasks in the software as well in the hardware. The abstract steps for the interrupt handling are as follows:
   - Interrupt the current program
   - Saves the context of the interrupted program
   - Transfer control to interrupt servicing program
   - Execute the interrupt servicing program
   - Restore the context of the interrupted program
   - Resume interrupted program from the point of interruption.

3. DMA is needed to enable a device to transfer data without exposing the CPU which results in much less CPU overhead.The advantages of DMA are:
   - DMA allows a peripheral device to read and write to/from memory without passing through the CPU.
   - DMA allows for faster processing as the processor can be working on something else while the peripheral can perform memory related work.

The disadvantages of DMA are:

- requires a DMA controller to carry out the operation, which increases the cost of the system
- problem of cache coherence

Check Your Progress 3

1. Some of the characteristics of I/O channels are:
   - An I/O processor (I/O channel)is the extension of the DMA concept.
   - An I/O channel holds complete control of the I/O operation and is also capable to execute I/O instructions.
   - The CPU initiates the I/O operations and handovers the execution I/O commands to I/O channel. In response, the I/O channel executes the I/O instructions stored into the main memory.

2. The difference between Selector Channel and Multiplexer Channel is that the Selector channel communicates the data with only one dedicated high-speed device at one time while the multiplexer channel is capable to handle I/O with multiple devices at the same time.

3. The difference between serial and parallel interfaces is given below:
   - The serial communication interface employs only one line to transfer one bit of data at a time. The parallel communication interface can transfer several bits at the same timeis usually utilized for I/O devices high-speed.
   - Serial interfaces are used for terminals and printers while high-speed peripherals such as disks and tapes use parallel interfaces.

UNIT 8 I/O TECHNOLOGY

Structure

8.0 Introduction
8.1 Objectives
8.2 Mouse
    8.2.1 Classifications of Mice
8.3 Keyboard
    8.3.1 Features of Keyboard
8.4 Monitors
    8.4.1 Cathode Ray Tube
    8.4.2 Liquid Crystal Display
    8.4.3 Light-Emitting diode
8.5 Video Cards
    8.5.1 Resolution
    8.5.2 Color Depth
    8.5.3 Video Memory
    8.5.4 Refresh Rates
    8.5.5 Graphic Accelerators
    8.5.6 Video Card Interfaces
8.6 Sound Cards
8.7 Digital Camera
    8.7.1 Webcam
8.8 Voice Based Input Devices
    8.8.1 Siri
    8.8.2 Alexa
8.9 Printers
    8.9.1 Impact Printers
    8.9.2 Non-impact Printers
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    8.10.1 Resolution
8.11 Modems
8.12 Summary
8.13 Solutions /Answers
    References

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8.0 INTRODUCTION

In the previous unit, you have studied the concept of input/output interfaces and I/O techniques. The previous unit discussed three I/O techniques i.e., programmed I/O, Interrupt-driven I/O and DMA were discussed along with the evolution of I/O processor. A computer supports a number of I/O devices in order to perform data transfer with external environment. This unit provides a brief introduction to the various I/O devices such as mouse, keyboard, monitor, printer, scanner, video & sound cards etc. It also discusses the modern voice-based input devices. The unit does not attempt to provide all the details of these devices, but attempts to introduce you the characteristics, basic functions and use of the devices in the context of the processor.

8.1 OBJECTIVES

After study of this unit, the students ought to be able to:

- Explain the features of mouse and its classifications;
- List the basic characteristics, functioning and interfacing requirements of keyboard;
- Explain different types of monitors
- Explain video Cards, sound cards, and digital camera
- Explain different types of printers;
- Explain the basic characteristics of Modems and scanners;
- Explain the concept of voice-based input

8.2 MOUSE

Douglas C. Engelbart at Stanford Research Institute (now SRI International) proposed the basic concept of mouse in order to use it with computer system. Xerox Corporation is first organization which developed the first Mouse. It is hand-held hardware input pointing device, which gives user a cursor (pointing mark) on monitor screen and this cursor is used to send the input to computer system. The purpose of mouse is to detect two-dimensional movement relative to surface. Typically, mouse is available with two or three buttons but a single button is sufficient to control the movement of cursor. There exist different types of mice namely Wired, Wireless, Bluetooth, Trackball, Optical, Laser, Magic, USB etc.

The unit of mouse resolution is Counts Per Inch (CPI) which represents *the number of signals per inch of physical travel of mouse*. The value of CPI may range from 400 to 1600. The mouse also sends CPI data to computer with some frequency which is known as polling rate. The polling rate may range from 60 Hz to 1000 Hz. The large value of CPI will result in faster movement of cursor which requires sending much data to computer demanding high polling rate. Therefore, it will be difficult to control the accuracy for large value of CPI.

8.2.1 Classifications of Mice

The classifications of mice are based on connectors, number of buttons and position sensing technologies. Two classifications are discussed-

1. **Connectors:** This category deals with categorization of computer mice based on ports/physical channels which are used to connect the mouse and computers.
   a) **Bus Mouse**-Bus was used to connect the first mouse with PC. Thus, it has been called as the bus mouse. It was used with IBM-compatible personal computers in its early days. A specialized bus interface was used to connect them with PC which was implemented via an ISA add-in card.

   b) **Serial Mouse**-In Serial mouse, serial port was used for connection. It is basically an interface present physically on computer for communication. Bit by bit information goes in and taken out of the computer through this port. It is a male port of D-type having 9 pin (DB9M) which is found at the back of the motherboard. However, this category of mice is no longer in use.

   c) **PS/2 Mouse**-The green colored PS/2 port is used to connect the mouse. Introduced in 1987, PS/2 uses 6-pin mini-din connector. It is the successor of serial connectors. PS/2 ports were first used in the PS/2 systems and they are still being used in modern designs. Green color of PS/2 port is for mouse and purple colored is for keyboard.

   d) **USB Mouse**-USB mouse are same in terms of shape and appearance but the difference lies in terms of connector. They are connected to a USB port. USB stands

for universal serial bus has superseded the PS/2 ports, though some of the computers still have the PS/2 ports. This standard defines the cables, connectors and communication protocols for connection and communication between computers and attached peripheral devices. The objective of this standard was to standardize computer devices connection.

e) **Wireless Mouse**-These are the modern mouse that does not require any cable for connection. Eliminating the clutter of cables, it provides a neat type of mouse to use. Some of its key features are- comfortable ergonomic design, improved battery life, Plug-and-Play, multi-function and wide compatibility

2. **Sensing Technology:** There are two types of mice based on sensing technologies i.e., mechanical mouse and optical mouse.
   a) Mechanical mouse has a rubber or metal ball in middle, which is used to control the movement of cursor. The sensors inside the ball detect the rotation of ball. When the ball rolls with the movement of mouse, it causes sensors to detect the rotation of ball along the two axes which consequently send signals to monitor screen. Figure 8.1(a) depicts the mechanical mouse.
   b) Optical mice use light emitting diodes (LEDs), optical sensors and digital image processing. The optical mouse detects by sensing the changes in the reflected light. The change in reflected light is measured by analyzing the images and the cursor moves on screen accordingly. Figure 8.1(b) shows the optical mouse.

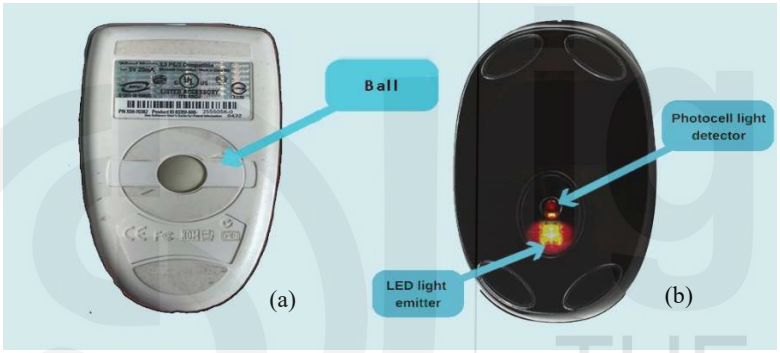


Figure-8.1: Difference between (a) Mechanical Mouse and (b) Optical Mouse

8.3 KEYBOARD

A keyboard is an input device, which is used since the inception of the computer systems. The keyboard allows manual input of alphabets, numbers, special characters, which are available as keys on a board. Figure 8.2 depicts a keyboard. In general, users use a keyboard to transfer a meaningful sequence of characters or numbers to a computer. Thus, a keyboard can be used to send input data into a computer from the external world.



Figure-8.2: Overview of how Keyboard transfers data to the computer

8.3.1 Features of Keyboard

Some of the basic characteristics of keyboard are given as follows.

- **Keyboard Layout**

The layout defines the arrangement of the keys on the keyboard. This arrangement is mostly influenced by the typewriter. The keyboard layout is now available for many different languages of the world. A good layout is the one, which allows faster data input. Thus, in most cases, computer keyboard layout is identical to typewriter key layout, which was designed for enhancing the speed of data input. The standard keyboard layout for English is called QWERTY layout. QWERTY layout stands for the six top alphabets in the keyboard. The

QWERY arrangement was created by Sholes, who invented typewriter. However, a computer in addition to alphabets is required to input numbers, special characters, and several shortcut commands. Therefore, keyboard has a very detailed arrangement of keys.
One popular keyboard designed by IBM for the personal computer had 101 keys. This keyboard's key layout is shown in Figure 8.3. This keyboard consists of key sets for alphabets in the middle, numeric key pad in the right for easy entry of numbers, function keys at the top name F1, F2, which are used as shortcut to various functions in different software, and a set of cursor keys to move the cursor on the screen. Later, a set of windows function keys were added to this design.

Another keyboard layout was designed by A Dvorak and William Dealey in 1936. Their layout was primality designed for people who find typing with all fingers difficult and would like to type using only two fingers. It was expected that typing would be faster by using Dvorak-Dealey keyboard, as you can use both the hands while typing. You can find more details on this keyboard in the further readings. With the availability of newer mobile smart devices, there are many possibilities of designing new keyboards for specific areas, including better designed layout for regional scripts.
For the languages of our nation, Indic keyboard layout is a standard layout. This layout supports 12 Indian scripts.



Figure-8.3: IBM 101-key Keyboard layout

- **Keyboard Touch**

In addition to layout, the other important characteristic of a keyboard is the keyboard touch. The keys should be sensitive enough to capture the data being entered by the user. A good keyboard must be able to send data with speed. These days, in addition, to physical keyboards, touch screen keyboards are also available. Most of these keyboards provide features of predictive text and autocorrect, which facilitate data entry by the user.

- **Scan Codes**

When a key is pressed on a keyboard, it transfers the scan code relating to those keys to the processor. Scan code of every key is unique. The scan codes are used to communicate the desired data or action to the processor. A keyboard of processor is connected through interrupt driven I/O mechanism. Therefore, when a key or several keys are pressed together on the keyboard, it interrupts the processor, provided processor has enabled interrupts. The processor receives the scan code/codes and identifies the key or keys that were pressed using the scan code table stored in the ROM BIOS. In addition, the status byte that is associated with the keyboard informs the processor about the status of keys that are used a toggles, like, Caps lock, Num Lock, etc. But how does a keyboard identified that more than one keys are pressed together, such as CTRL & ALT & DEL. Interestingly, a keyboard sends two scan codes to the processor - one when key is pressed and second key is released, which were called Make and Break scan codes respectively. Thus, by knowing the timing of these make and break scan codes, processor determines, which keys are pressed together. A detailed discussion on scan codes is beyond the scope of this Unit.

8.4 MONITORS

A monitor is an output display device connected to processor and it displays the vision into the brain of the processor. It allows a user to graphically interact with the processor which is helpful to send output as well as to receive input to/from the user. Technically speaking, it is display device which provides a graphical vision by converting the digital/analog signals into the visual form.

The monitor looks like a television set but both the devices are different with each other. The monitors have greater sharpness, lower input lag, higher refresh rates, color purity, and operate at higher frequencies in comparison to TV sets. The TV set consists of tuner or demodulator circuit to convert the signals

Whenever users are interested to buy a monitor, they search for the better configurations in minimum possible budget. The configuration of monitors consists of display size, resolution, supported frequencies, the size of the picture tube and the type of connector used to connect to the computer. Monitors are manufactured by many manufacturers like LG, Samsung, Acer, Dell, HP, Lenovo, Sony, Asus, BenQ, etc. The monitors are available with different sizes i.e., 14'', 15'', 17", 19'', 21.9'', 24'' or even higher. The monitors are also available with different screen form factor i.e., flat and curved screens. The monitors can be categorized into three categories based on the design technology. These categories are discussed in next sub-sections.

8.4.1 Cathode Ray Tubes

The cathode ray tubes monitors and television sets are based on the technology of Cathode ray tube (CRT). A CRT is a partly empty glass tube which consists of inert gas at low pressure. A negatively charged electrode which is known as Cathode/Electron gun is used to shoot beams of electrons at high speed towards a positively charged electrode (anode). The high-speed electrons impinge on the small phosphor coated screen. The screen consists of dots with three primary colors i.e., Red, Green and Blue. Indeed, there exists either one electron gun for the three colors (Red, Green and Blue) or one different electron gun for each color. Figure-8.4 depicts the cathode ray tube (CRT). The quality of image on CRT screen is influenced by following four factors:

1. **Phosphor coating**: The monitor screen is coated with phosphor (fluorescent material) which emits light when bombarded by electron gun. The phosphor coating is provided in inner surface of cathode ray tube. The coating affects the color and the persistence. The term persistence in the context of monitor is defined as the time for which the effect of a single hit on a dot on the monitor surface lasts.
2. **Shadow Mask/Aperture Grill:** It is the manufacturing technology for CRT monitors to produce clear and focused color images. It determines the resolution of the screen in color monitors. In shadow mask CRT, each pixel position consists of 3 phosphor color dots one for each red, green and blue. The Triad and inline arrangements are used for the alignment of color dots to produce good quality images.
   Another technology for same purpose is the aperture grille
3. **Electron Gun:** The electron gun must be efficient in its working. The high-quality electron gun affects the quality/sharpness of the image.
4. The screen glare and lighting of the monitor are also major factors to influence the quality of the image.

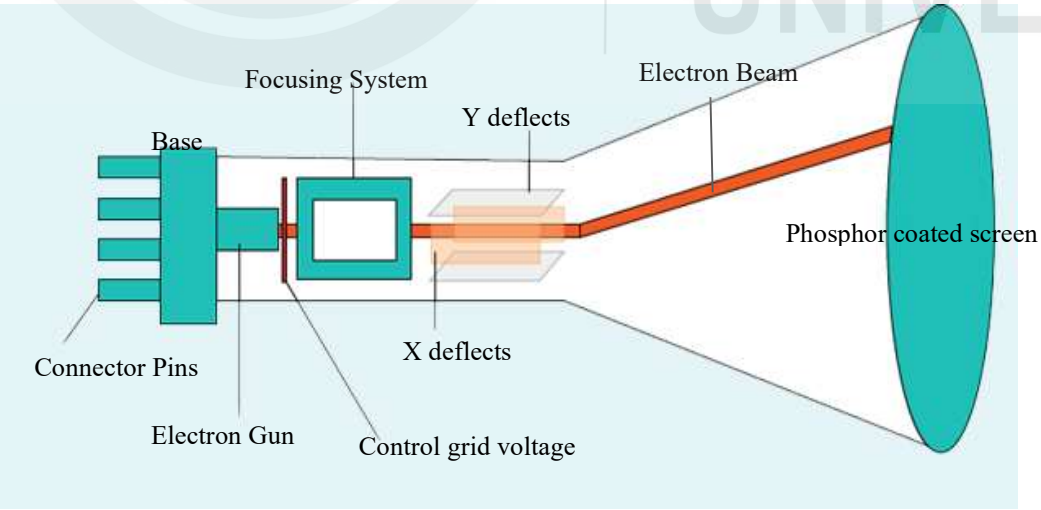


Figure-8.4: Cathode Ray Tube

8.4.2 LIQUID CRYSTAL DISPLAYS (LCDs)

LCDs were developed by the company RCA in the year 1960. An LCD is an electronically modulated optical device which employs light-modulating properties of liquid crystals combined with polarizers. The light is not emitted directly from the liquid crystals rather a

reflector is used to produce images in color or monochrome. An LCD blocks the light to display patterns. LCDs are lightweight screens and are mainly used for portable computers. They are known for low power consumption, good resolution and bright colors. The LCDs can be divided into following three categories based on display generation techniques.

1. **Reflective LCDs**: The display is generated by selectively blocking reflected light.
2. **Backlit LCDs**: The display is generated due to a light source behind LCD panel.
3. **Edgelite LCDs**: The display is generated due to a light source that is adjacent to LCD panel.

LCD Technology

To manufacture the LCD screens, Nematic technology is used. The molecules of liquid crystals (rod-shaped crystals) which are known as Nematic cells are used. Figure 8.5 depicts Nematic cells. The Nematic cells are packed (sandwich) between two thin plastic membranes. The Nematic cells have special properties i.e., these cells can change the polarity and bend of the light. The electric current is used to control these properties by applying the electric on grooves in the plastic membranes.

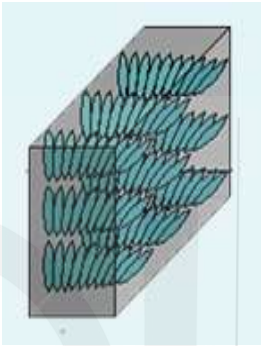


Figure-8.5: Nematic Cells

There exist two types of LCDs i.e., Passive matrix and Active matrix.

1. **Passive Matrix**- The passive matrix arrangement is most widely used technology due to low weight, high image quality, low cost and high response time. LCD panel consists of a grid of horizontal and vertical conductors. The conductors consist of Indium Tin Oxide to create a picture. Each pixel is located at the intersection of two conductors in the grid. Whenever current is passed through a pixel, it becomes dark
2. **Active Matrix**-It employs Thin Film Transistors (TFT) and that's why known as TFT technology. In active-matrix arrangement, TFTs are arranged in a matrix on a glass surface and these TFTs are considered as pixels. These TFTs receives a small amount of white light, which is then enhanced by TFT to activate a pixel. The advantage of using TFTs is that they have faster response times, as they use smaller amount of light. However, the disadvantage of using TFTs are that they are difficult to fabricate, therefore, are costly. TFT LCD Display Technology is shown in the below Figure-8.6.

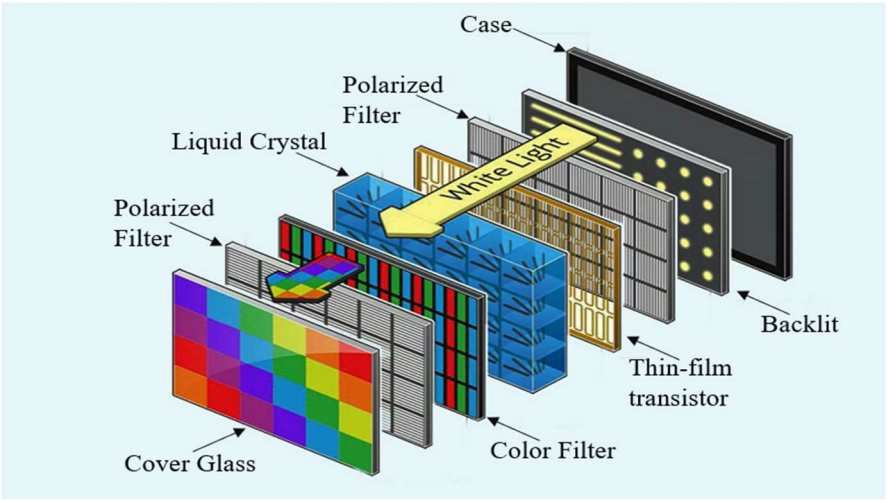


Figure-8.6: TFT LCD Display Technology

8.4.3 Light Emitting Diodes (LED)

The light-emitting diode (LED) monitors display is a flat screen, flat-panel computer monitors or television. It uses array of LEDs as pixels for displaying the videos. It is light weight and has a very short depth. LCD monitors and LED monitors differ only in terms of backlighting; typical LCD monitors uses fluorescent backlights whereas an LED monitor uses light-emitting diodes. The earlier LCD monitors used CCFL instead of LEDs to illuminate the screen. LED monitors offer many features/benefits namely slim design, flicker-free & brighter images, longer lifespan, broader dimming range, low power consumption, better color and picture quality etc. Figure 8.7 lists the benefits of LED monitors.

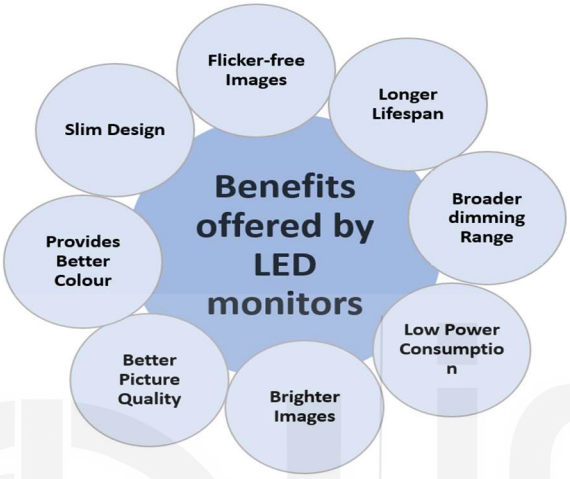


Figure-8.7 Benefits offered by LED monitors

Check Your Progress 1

1. Explain mechanical and optical mice.

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2. Discuss scan codes.

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3. What are the differences between LCDs and LEDs?

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8.5 VIDEO CARDS

First, this section discusses a brief overview of graphic display technology with the primary focus on CRT monitors, before jumping to video hardware. The graphic display system is responsible for displaying bit-mapped graphics on monitor. Every image is formed using small dots which are known as ***picture elements*** or ***pixels***. Figure 8.8 shows the pixels of an image. The description of each pixel is stored in the memory which is taken care by video system.

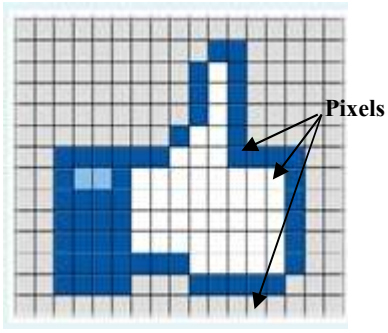


Figure-8.8 Bit-mapped Graphics Image

The display memory which is used to store the data for images is known as **frame buffer**. At any moment, the frame buffer consists of data for bit-map representation of current image on screen and the next image. The frames are read dozens of times per second and sent to the monitor using a cable in serial manner. Upon receiving the stream of data, the monitor forms and displays it on the screen by scanning raster movement from first up to down one row at a time. Based on this raster movement CRT, the monitor will illuminate its small phosphor dots. It is shown in Figure 8.9 and Figure 8.10.

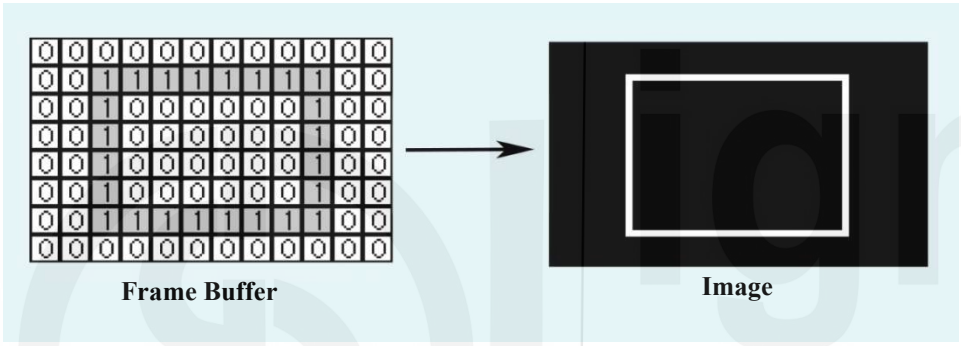


Figure-8.9: Frame Buffer and the corresponding image displayed on the system

The greater number of dots leads to better resolution of the image as well as the sharper the picture. The number of dots directly correspond to the richness of the image (or gray levels for a monochrome display) displayed by the system. The higher the number of colors, the more is the information required for each dot. Therefore, higher resolution and color depth of the system required bigger memory storage by the system to store the frame buffers.

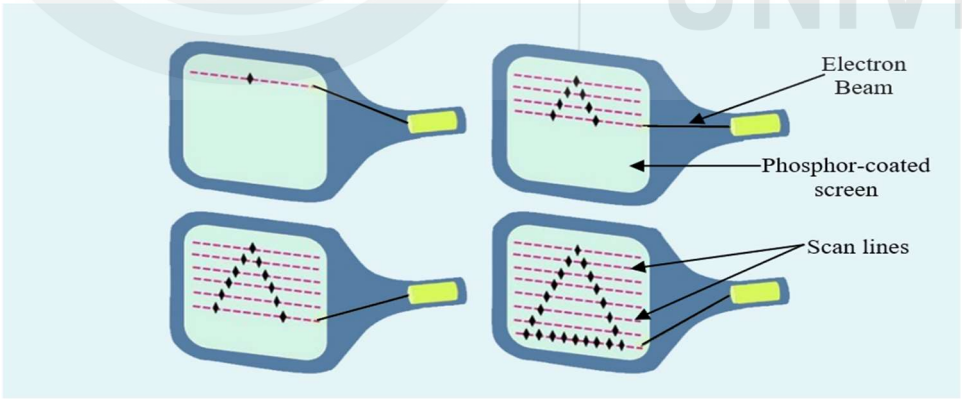


Figure 8.10: Raster Display

8.5.1 Resolution

The resolution is defined as the possible *sharpness or clarity of an image*. The resolution does not depend upon the physical characteristics of the monitor. It is measured in terms of number of pixels on a monitor. For instance, a standard VGA graphic display with resolution 640×480 consists of 640 and 480 pixels on horizontal and vertical axes respectively. In order to construct an image, different numbers of pixels are spread across both the axes of monitor screen. Higher is the resolution, sharper is the image due to large number of pixels.

The sharpness of an image on actual live-screen does not depend only on resolution but it is measured in the unit of dots-per-inch. These dots-per-inch are dependent on (i) size of the

image and (ii) resolution of the image. An image will be sharper on a smaller screen in comparison to bigger screen. For instance, an image may appear sharp on a 15" monitor and may be a little jagged on a 12" monitor display. Figure 8.11 shows a circle with different sharpness on different size monitor screens.

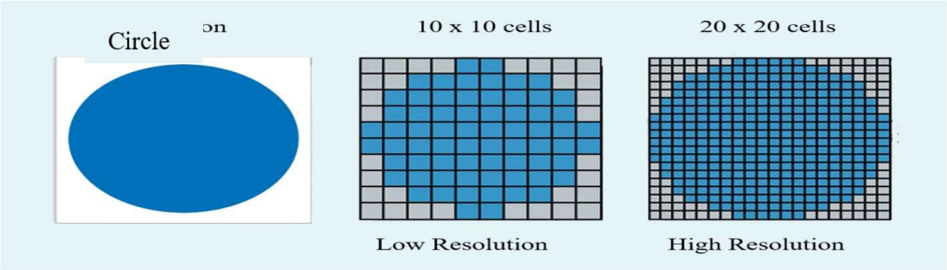


Figure-8.11: Circle with low and high resolutions

8.5.2 Color Depth

The image is constructed using stream of pixels. If the value of pixel is 'ON' and 'OFF', the pixel will be displayed in image on the screen as a pure black and white respectively. If single bit is assigned to a pixel, the image will be black and white. This system is known as **two-color system**. The pure black and white images can be converted to gray levels, which are different levels between white and black. This requires a greater number of bits to code each pixel. For instance-if you assign two bits to each pixel, four color levels are possible: White, Light Gray, Dark Gray and Black. In general, you need more than one bit to describe a pixel. Hence, one bit per pixel implies 2 colors or 2 gray-levels, 2 bits per pixel implies 4 colors or 4 gray-levels, and 3 bits per pixel implies 8 colors and so on. It means n bits per pixel imply 2^n gray-levels. For colored images color codes for the intensity of the three primary colors, viz. Red, Green and Blue, for each pixel are stored.

Color Depth can be understood as *the number of bits allocated to every pixel in order to store color code information*. Since every bit of a pixel corresponds to a specific color i.e., all bits at the same position for all pixels corresponds to the same color. Thus, the bits corresponding to same color can be regarded to form a plane and these planes are known as color planes. It is considered the color planes are stacked on top of each other which are helpful in deciding final color at each pixel. Thus, depending upon number of bits required for each pixel, there are 3 *Color Planes* (one each for Red, Green and Blue). Figure-8.12 depicts the 3-bits color display and 3 color planes.

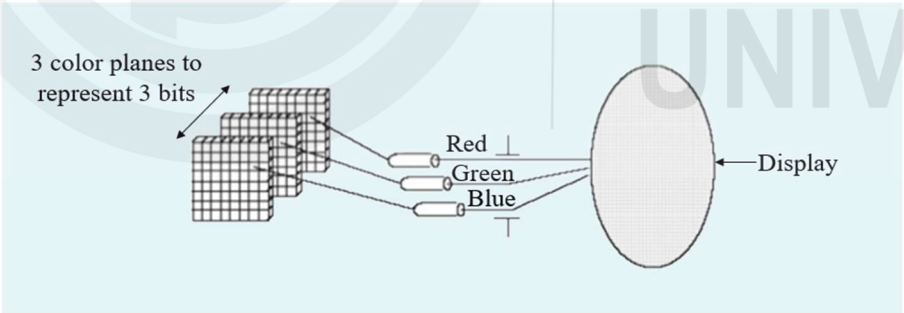


Figure-8.12: 3-bits Color Display with 3 color planes

The computer system with a 3-bit RGB color planes utilized 1 bit for each of the red, green and blue color components. Therefore, every color component can exist only in "ON" or "OFF" state. The three-bit RGB 'ON' or 'OFF" color components result in 8-colors consisting of three primary RGB colors i.e., red, green and blue; two pure colors i.e., white and black; and three complementary colors i.e., magenta, cyan, and yellow colors. The RGB values (ON" or "OFF") of 3-bits color are given in Table 1 and the colors are displayed in Figure 8.13.



Figure-8.13: 8 possible colors for 3-bits Color Display

Table-1: 3-bit Color Display RGB values

Bit-values ('ON' or 'OFF')			Color
0	0	0	**Black**
0	0	1	**Blue**
0	1	0	**Green**
0	1	1	**Cyan**
1	0	0	**Red**
1	0	1	**Magenta**
1	1	0	**Yellow**
1	1	1	**White**

What Color depths are practically used?

If n is the color depth representing the total number of bits used to store one pixel, the number of colors will be 2^n (2 to the power n). Table 2 puts the most popular color modes.

Table 2: Popular Color Depths

Sr. No.	Color Depth(bits/pixel)	Color Mode
1	1	Monochrome
2	4	16-Colours
3	8	256-Colours
4	16	High Color
5	24	True Color

Human & Color Depth of Monitor Screen

The modern monitors can display up to a maximum of 262,144 (2^{18}) colors for 18 bits/pixel *Color Depth*. If different tones (color pitch) are allowed and eight bits are used for each RGB color, a total of 16.77 million colors (256 tones (R) x 256 tones (G) x 256 tones (B) = 16,777,216 colors) can be generated. On the other hand, human eyes are capable to distinguish maximum few million different colors. Thus, even if the monitor screens display colors more than few million, they would not be distinguished by humans. It also implies that the practical upper limit the 24-bit per pixel color depth. Since the number of possible colors produced by this color depth is more than the colors that could be distinguished by human eye, these colors are called the true colors. However, 24-bits color or true color systems have more color than possibly useful; the extra 8-bits are used by designers to store special effects information of the image. These extra bits formed a channel which is known as Alpha channel.

8.5.3 Video Memory

The video memory/frame buffer is used to store the video to be displayed. The quality of video display depends on the efficiency of the video system i.e., how quickly the frames are accessed and updated. Initially, a fixed area of RAM is allocated to the video memory. Later, video RAM along with video cards was introduced and it can be increased by placing additional video RAM under the unified memory architecture (UMA). UMA is helpful in reducing the cost of computer system. In UMA supported systems, an area of main memory is used as frame buffer/video memory which results is elimination of bus for video processing. Therefore, the computer system with UMA may be less costly.

Basically, UMA comes with on-board video card in the modern low-cost motherboards. The required resolution and color-depths are the deciding factor for the size of video memory/frame buffer. The minimum size of video memory can be calculated simply by multiplying the *Color Depths* and resolution of the monitor screen.

Let us solve a simple exercise. Assume a standard VGA monitor screen with resolution 800 × 600 and color depth value 8.

Number of Pixels = 800 × 600=480000
Color Depth (8 or 2^3) = 3-bits
Minimum Memory required = 1,440,000 bits (180,000 bytes)
= 180 KB

It implies that minimum RAM required for resolution 800 × 600 and 8 color depth is 180 KB. But the memory is available in exponential power of 2 and the next minimum size of memory is 256 KB. It implies that minimum size RAM required for resolution 800 × 600 and 8 color depth is 256 KB.

Now-a-days, a very odd-looking resolution i.e., 1152×864 has become popular. Could you guess why this is so? The following are the reasons behind its popularity. There are nearly one million (9,95,328) pixels for VGA with 1152×864 resolution. For color depth value 8, nearly 8 million bits or 1 MB memory is required. Further, human eyes perceive only a few million colors and this resolution is more suitable. In addition, a square pixel that has a ratio of 4: 3 allows easier programming.

Please note that the calculations shown above are not applicable for 3-D displays, which requires more memory due to the issues like "Double Buffering" and/or "Z-Buffering".

8.5.4 Refresh Rates

The Video Controller (a special circuit) scans the frame buffer and reads rows one by one followed by sending this data in serial manner. On monitor screen, the electron beam starts scanning one-line at a time from left to right direction in order to create images. The *horizontal refresh rate or horizontal frequency* is the rate at which horizontal sweeps take place, while *vertical refresh rate* or *vertical frequency* is the rate at which vertical sweeps take place. The vertical frequency is also known as *refresh rate* or *frame rate*, as during a vertical sweep one complete frame is displayed. There exist several hundred rows in each frame and thus, horizontal frequency is hundreds of times higher than vertical frequency. The unit of horizontal frequency is KHz while the unit of vertical frequency is Hz.

Note: It is necessary to maintain the same frequencies between the monitor and video system for better quality of images. The compatible refresh rates are provided with the manual of the monitor.

8.5.5 Graphic Accelerators

An important chip associated with video card is known as Graphic Accelerator which is the replacement older technology known as *Graphic Co-Processor*. The graphic accelerator chip is a dedicated unit that executes in-built video functions of image construction and rendering, thus, releasing the microprocessor (main processor) from this work. The accelerator chips are optional but they are required due to noticeable impact on the performance of the computer, especially in graphics-intensive tasks such as- Rendering of 3D models and images, Video editing and Gaming. The graphic accelerator are needed if you need the following:

- Good support to 3-D graphics.
- Better resolution of graphics.
- Larger size of memory in the frame buffer.
- Better speed of display of drop-down menu.
- Good quality video playback.

The Graphics accelerators are widely used in industries such as- Motion pictures for special effects, Computer-aided design (CAD), Video games, 3D-effect etc.

What is a 3-D Accelerator?

The accelerator chip that has built-in ability to perform the mathematical calculations and execute the algorithms required for 3-D image generation and rendering, are called 3-D Accelerator. A 3-D image is just an illusion for human eyes which basically represents a projection of 3-D images/videos on 2-D monitor screens. This conversion takes place by projection and transparency effects, perspective effects, color depth and lighting effects. In addition, the following techniques can be used for creating 3-D images on 2-D screens: (i) Ray-Tracing, which traces the path of light rays emitted by a light source; (ii) Z-buffering, which uses a buffer to store the third axis, i.e., Z-axis positions and (iii) Double-Buffering, which uses two buffers in place of a single buffer.

8.5.6 Video Card Interfaces

A video card interface connects the video display to the computer system in order to improve the performance of the visual data you see on your screen. The video card can either be a separate component which is plugged into a slot on the motherboard of the computer or it may be integrated into the motherboard known as "onboard". For isolated video cards, the connection is realized using either *Peripheral Connect Interface (PCI)* or *Accelerated Graphics Port (AGP)* bus.

- **PCI-** It is introduced by Intel and also known as *Peripheral Component Interconnect*. It is a high-speed common bus which is used to attach the computer peripherals to the motherboard. It is used to attach sound cards, network cards and video cards. The computers may use now some modern technologies like PCI-Express (PCIe), USB and AGP.
- **AGP-**It is also known as *Advanced Graphics Port*. It is a standard connector port used to connect the video card with the microprocessor and the main memory. It is a dedicated high-speed connection interface which is used by only graphics subsystem. AGP employs pipelining, isolated data and address buses and high-speed mode to improve the performance of graphics card.
  In specific computers, the video card is directly connected with the microprocessor and may use direct memory access (DMA) I/O technique to send data from main memory to frame buffer.

8.6 SOUND CARDS

Multimedia has become an indispensable component of personnel computers to play different music files like MP3, MP4, WAV (Waveform audio file), WMA (Windows media audio), AAC (Advanced audio coding), FALC (Free lossless audio codec), OGG (The latest Free Sound format standard) etc.

The Sound card can either integrated into motherboard (built-in sound card) or connected through expansion slot. As you may study in computer networks, the analog sound waves could be converted into electrical form using electrical signals, which is used to compute the strength of sound. Usually, the analog audio signal is converted into digital audio (or digital signals) in the form of bits using sampling process. The microprocessor manipulates the digital audio bits and this data is sent to the sound card. The sound card converts this data into analog audio in order to play back through the speakers or headphones. The major functions of a modern sound card are as follows:

1. Conversion from digital sound signals to analog form to play back the sound.
2. Amplifiers to augment the strength of sound signals
3. Sound recording.
4. Sound synthesis.
5. Mixing of sound from various resources.

The three basic issues relating to sound cards are - Compatibility, Connections and Quality.

- **Compatibility:** Sound cards must be compatible for hardware as well as for software to meet the current industry standards/protocols. Some specific software like games need sound cards to be compatible with industry standards. You may refer to further readings to know about these standards.
- **Connections:** The sound card must provide different connections in order to perform various functions. It should provide MIDI port (Musical Instrument Device Interface) which allows user to produce music directly by using synthesizer circuit in the sound card. It also allows connecting a Piano keyboard to the computer system.
- **Quality:** There exist different sound cards which provide sounds with different qualities. The quality of sound differs due to the noise control, digital quality and the ranges of frequency supported by the sound card.

Check Your Progress 2

1. Explain the concept of a frame buffer in the context of Video Card interfaces.
   .....
   .....
   .....
2. What do you understand by horizontal and vertical frequencies?
   .....
   .....
   .....
3. Compute the minimum required video memory for 16 color depth and a monitor screen with the highest possible resolution 7680x4320.

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4. Explain sound card. What are the functions of sound card?

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8.7 DIGITAL CAMERA

The first digital camera was invented in the year 1975 by Steven Sasson at Eastman Kodak. Digital camera is a hardware device that takes images or record videos and stores them on memory as digital data on memory card instead of on photographic film in analog camera. In digital Camera, the images are stored in digital form and thus they can be reused later for different purposes like printing, editing etc. Since the digital camera takes images (input) and sends them to computer (output), it is considered as input as well as output device.

Figure 8.14 depicts the digital camera which is taking an image of subject (scene) under consideration. A digital camera consists of a sequence of lenses which focuses light on to a semiconductor device to create an image of a scene under consideration. The semiconductor device, in turn, records this light as digital images by using an in-built processor.

The semiconductor device is known as an Image sensor which converts light into electrical charges. Two types of Image sensors exist: Charge Coupled Devices (CCD) and Complementary Metal Oxide Semiconductor (CMOS). CCD is more popular and powerful kind of sensor in comparison to CMOS image sensors.

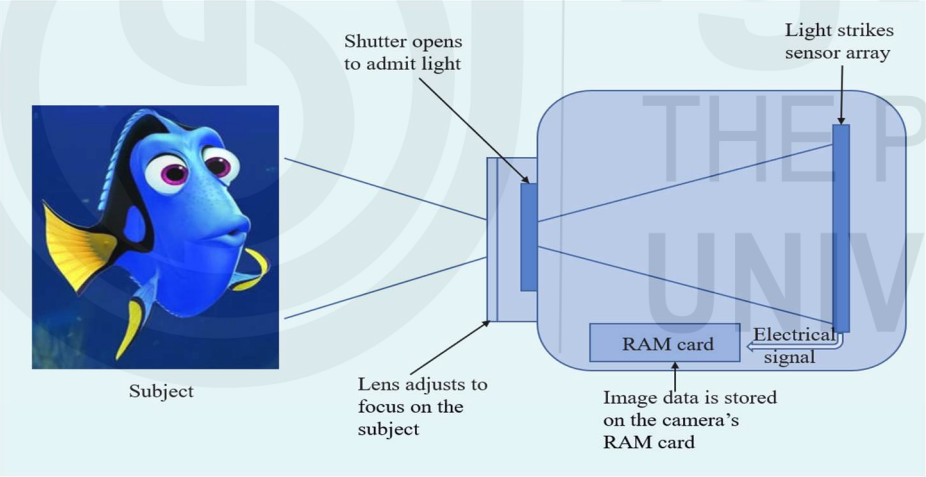


Figure-8.14: Working of a Digital Camera

The resolution (pixels) of digital camera is major deciding factor for the quality images. The higher the resolution, the better the digital camera is. The major benefits of a digital camera are as follows

- Allows user to see the videos and images immediately
- Allows user to store thousands of images/videos due to in-built memory
- Digital cameras are portable
- Allows user to edit images directly
- Allows flexibility in printing of desired images

8.7.1 Webcam

A digital camera without storage connected to computer system or network is referred as Webcam. In modern computers, the webcam can either be a separate component which is plugged into the computer or be an in-built integrated camera. In order to use webcams, it is necessary to install the required software. A webcam is an input device which is used to capture the images/videos and then send it to the computer. Webcams are used for videoconference or video calling or online meeting using Google Meet, Zoom, MS Team and others services.

8.8 Voice Based Input Device

Modern devices are capable to take human voice as input using speech recognition processes and execute applications accordingly. These devices are known as Voice Based Input devices. As compared to microphone, the speech recognition process of these devices recognizes human voice; converts it into machine-language and execute programs/applications accordingly. Figure 8.15 two devices which use speech recognition process to recognize human voice.



Figure-8.15: (a) Siri (b) Echo Dot 3 Smart Speaker with Alexa

The Voice Based Input Devices can recognize spoken words in two ways. The spoken words can either be recognized from a pre-defined vocabulary or be recognized from a known speaker after training of the input device. Whenever speaker utters a word from the pre-defined vocabulary, the Voice Based Input device may display the characters of monitor screen for verification by the speaker. However, some of these devices may process the speech without verification from the speaker. The process of speech recognition compares each uttered word with the words stored in pre-defined vocabulary table.

8.8.1 Siri

Apple Inc. offers a built-in, voice-controlled virtual assistant with most of products i.e., iPhone, iPad, Apple Watch or Mac (macOS Sierra and later) etc. This voice-controlled personal assistant is known as Siri. The users may talk to Siri as they talk to their friends. Siri allows a seamless interaction with Apple devices such that user speaks to Siri and Siri speaks to user. Siri helps users to get their job done after receiving user commands. Siri can help to open a file, send messages, open a web browser, open a website, booking a ticket, watch movies, and many other activities.

Siri works based on the Artificial Intelligence and Natural Language Processing fields. It consists of three components -Conversational interface, personal context awareness and service delegation systems. The conversational interface understands the user word-for-word manner and the semantic of text is produced using personal context awareness which is based on habit and language of the user. The service delegation helps to deliver services using built-in apps and their inner workings.

8.8.2 Alexa

Amazon offers virtual interactive voice-based AI powered digital assistant known as Alexa. This device has been designed in association with Alexa Voice Service (AVS) in order to simulate real conversations. “Alexa” is basically the “wake word” which is used to alert the device to start listening the voice to perform some tasks. Alexa employs intuitive voice commands to provide services to perform some specific tasks. Figure 8.15(b) depicts the Amazon Alexa. It is available as Echo speakers, smart thermostats, lamps and lights, and right on your phone through the Alexa app. Alexa can do quick math, play music, check news and weather updates, read emails and control many of the smart products.

Alexa also works based on the Artificial Intelligence and Natural Language Processing fields. It Alexa consists of speakers, microphone and a processor which is used to activate the device. It receives input and sends it to cloud where Alexa Voice services (AVS) interprets and understands the user input. Accordingly, AVS sends the appropriate output back to user device. The internet connection is the basic requirement to use Alexa.

8.9 PRINTERS

Printers are devices that accept textual and graphical contents as output from a computer system and print contents on paper in a controlled manner. The text and photographic images are produced by printers. Printers differ in technology used, memory, speed, resolution, color supported, size, hardware compatibility, cost and others factors. The present-day printer technologies include the dot matrix printer, Inkjet or tank printers, Laser Printers etc. to serve different needs. The available printers can be divided into two classifications-Impact and Non-impact printers. Figure 8.16 shows the classifications of printers.

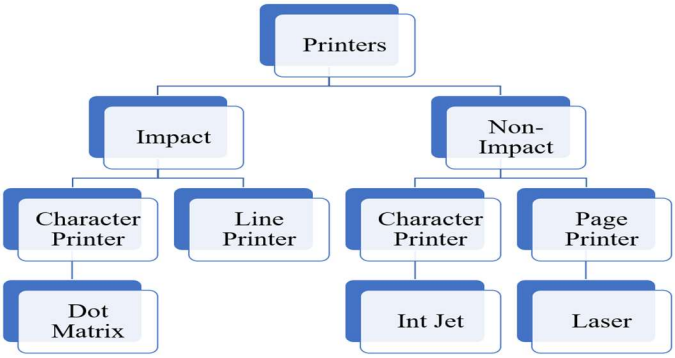


Figure-8.16: Classification of Printers

8.9.1 Impact Printers

Impact printer uses mechanical components for printing i.e., physical contact between printing head and paper. In order to print, the characters and graphics are produced on a paper by striking. They produce banging noise during printing. Impact printers can be divided further two categories-Character and line printers. The character printers print only one at a time by striking on ribbon whereas line printers print one line at time. The line printers are fast and costly printer in comparison to character printers. Different types of impact printers are line printers, dot-matrix printers and, daisy-wheel printers.



(a)



(b)

Figure-8.17: (a) Dot-matrix printer (b) Local Railway ticket

Dot Matrix Printer

A Dot Matrix Printer (DMP) employs fixed number of pins to print on paper. The print head with many pins runs back and forth on the page and prints by striking against a socked cloth ribbon in order to make a mark on paper. In DMP, the characters are using matrix of dots and the printed character is basically an accumulation of several dots on the paper. Therefore, the arbitrary font or graphics will be generated in each printing. Figure 8.17 shows a DMP and the local railway ticket printed by DMP.

8.9.2 Non-Impact Printers

In non-impact printer, no mechanical moving component is employed for printing. These printers don't strike or impact the ribbon for print. The technologies used by non-impact printers are chemical, inkjet, electrostatic, xerographic and laser. These printers work silently. Non-impact printers can be further divided into two categories-Character and Page printers. The non-impact character printers spray tiny drops of ink on to the without striking/physical contact with paper. The page printers print one full page at once. Different types of non-impact printers are inkjet, photo, and laser printers.

Laser Printer

Laser printers are very common page printers and print one page at once. Laser printers employ a focused light beam to transfer image or text onto paper. The modern laser printer use Resolution Enhancement Technology (RET) which is introduced by Hewlett-Packard. This technology smoothens the edges of character, diagonal lines etc. to produce better quality printouts. To produce high quality print, the basic requirement is the memory which increases as a square of resolution i.e., dots per inch (DPI). For 600 dpi, approximately 3.5 MB (600x600 bits) memory is required whereas 14 MB (1200x1200 bits) is required for 1200 dpi. Figure 8.18 depicts a single function monochrome laser printer.



Figure-8.18: Single function Monochrome Laser Printer

8.10 SCANNERS

A scanner is an electronic device which is used to capture images from tangible sources like photographic images, paper, posters, slides and others. The scanner converts the captured images into electronic form and stores them in computer memory in order to view/modify later. The scanner employs light sensors arranged in the form of an array in scan-able area. The light sensors detect differences in brightness of reflections from an image and then scan the source.

The existing scanners differ in many factors such as compatibility, resolution, support for different media and interfaces, etc. Two popular types of scanners are - *Flatbed Scanners* and *Handheld Scanners*.

Flatbed Scanners are used to scan high-resolution tangible images into detailed and sharp electronic images. The images are placed on a flat glass tray and movable sensors are used to scan the images. Figure 8.19(a) shows a flatbed scanner. Handheld scanners are used to scan the physical documents, and require good hand control for high quality scanning. These are the most portable and cheapest scanners and shown in Figure-8.19(b).

Scanning is used for many different applications. The scanners are used as Magnetic Ink Character Recognition (MICR) scanner in order to scan cheques and Bar-Code readers to identify different objects. One more application is *Optical Recognition of Characters (OCR*). The *OCR* software use character/pattern matching algorithms to recognize characters and converts the scanned text to a text file. The OCR technology is very much useful in digitizing the ancient text written in old scripts.



Figure-8.19: (a) Flatbed Scanner (b) Handheld Scanner

8.10.1 Resolution

The resolution of scanner is the quality of image achieved by scanner. It is measured in *dots per inch (dpi)* and it indicates the number of dots per inch scanned horizontally and vertically.

It implies that the more is the dpi of a scanner, the more details a scanned electronic image will have. The scanned file size increases with increased resolution. There are various ways to measure the resolution.

Optical Resolution - The upper resolution limit of a scanner which is used to scan the images is known as optical resolution (hardware resolution). For example- if the optical resolution of a scanner is 300 dpi, it means 90000 (300x300) pixels per square inch can be captured by the scanner. The scanners may be available with optical resolutions of 300, 600, 1200, 2400 dpi or even more.

Interpolated Resolution- The resolution of image can be augmented using interpolation algorithms and this resolution is known as Interpolation resolution. The interpolation technique employing complex algorithms is used to add intermediate pixels based on the properties of surrounding pixels. The interpolation technique results in increased size of scanned images but it provides smoother and high-quality images without adding any additional information. For instance- if the optical and interpolated resolutions are 300x300 dpi and 4800x4800 dpi respectively. This implies 90000 pixels per square inch can be captured by the scanner while the interpolation algorithm can add 15 pixels between every pair of pixels to increase the dpi of image.

8.11 MODEMS

Modem (i.e., modulator-demodulator) is a device that connects two computers using telephone lines in order to exchange data with each other. The modem receives digital signals from computer, puts them into analog circuit by modifying a constant wave (known as carries) and then analog signals are transmitted over the telephone lines. This process is known as modulation. It occurs whenever user connects to the Internet. Demodulation is the inverse process of modulation in which the digital signals are derived from the modulated wave. It occurs whenever user receives data from a website, which is then displayed by your browser. Figure 8.20 shows the process of modulation and demodulation performed by the modem. You may refer to further readings for more details on modulation and demodulation techniques.

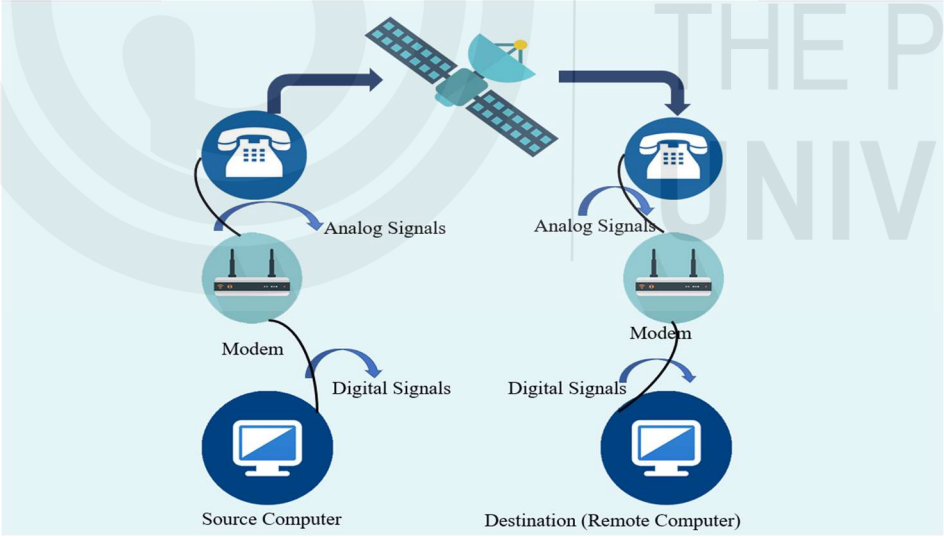


Figure-8.20: Modulation and Demodulation process by Modem

Check Your Progress 3

1. Explain webcam and its major benefits.

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2. How characters are recognized by voice-based input devices?
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   .....
3. Compare impact and non-impact printers.
   .....
   .....
   .....
4. Explain Interpolated resolution.
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   .....
   .....
5. How many pixels can be captured by a scanner with 600 dpi optical resolution?
   .....
   .....
   .....

8.12 SUMMARY

This unit discussed several input/output devices and the technologies behind them. This unit covers different input device along with different components or types or features. It is discussed Mouse and classifications of Mice, Keyboard along with its features, voice-based input devices, scanners and webcam its different types. This unit also discussed output devices along with other components or types or features. It discussed computer monitors with three different categories i.e., CRT, LCD and TFT screens. The printers with their different categories have also been discussed. The video cards have also been discussed with their characteristics like resolution, color depth, video memory, refresh rates, graphic accelerators and video card interfaces. It also discussed sound cards with its functions and different characteristics. At last, the modem device is discussed in brief manner.

8.13 SOLUTIONS/ ANSWERS

Check Your Progress 1

1. Mechanical mouse has a rubber or metal ball in middle, which is used to control the movement of cursor. The sensors inside the ball detect the rotation of ball. When the ball rolls with the movement of mouse, it causes sensors to detect the rotation of ball along the two axes which consequently send signals to monitor screen. Figure 8.1(a) depicts the mechanical mouse.
   Optical mice use light emitting diodes (LEDs), optical sensors and digital image processing. The optical mouse detects by sensing the changes in the reflected light. The change in reflected light is measured by analyzing the images and the cursor moves on screen accordingly. Figure 8.1(b) shows the optical mouse.

2. Scan Codes-When a key is pressed on a keyboard, it transfers the scan code relating to those keys to the processor. Scan code of every key is unique. The scan codes are used to communicate the desired data or action to the processor. A keyboard of processor is connected through interrupt driven I/O mechanism. Therefore, when a key or several keys are pressed together on the keyboard, it interrupts the processor, provided processor has enabled interrupts. The processor receives the scan code/codes and identifies the key or keys that were pressed using the scan code table stored in the ROM BIOS.

3. LCD monitors and LED monitors differ only in terms of backlighting; typical LCD monitors uses fluorescent backlights whereas an LED monitor uses light-emitting diodes. The earlier LCD monitors used CCFL instead of LEDs to illuminate the screen.

Check Your Progress 2

1. The display memory which is used to store the data for images is known as frame buffer. At any moment, the frame buffer consists of data for bit-map representation of current image on screen and the next image. The frames are read dozens of times per second and sent to the monitor using a cable in serial manner. Upon receiving the stream of data, the monitor forms and displays it on the screen by scanning raster movement from first up to down one row at a time. Based on this raster movement CRT, the monitor will illuminate its small phosphor dots.

2. Refer text 8.5.4

3. The minimum required video memory is computed as follows-

Number of Pixels	$= 7680 \times 4320 = 33,177,600$
Color Depth (16-colours = 2^4)	= 4-bits
Minimum Memory	= 132,710,400 bits (16,588,800 bytes)
	= 16,200 KB = 15.82 MB

4. Sound card is used to convert digital audio data into analog audio in order to play back through the speakers or headphones. The Sound card can either integrated into motherboard (built-in sound card) or connected through expansion slot. The major functions of a modern sound card are as follows:
   a) Conversion from digital sound signals to analog form to play back the sound.
   b) Amplifiers to augment the strength of sound signals
   c) Sound recording.
   d) Built-in synthesizer
   e) Sound mixer circuits.

Check Your Progress 3

1. A digital camera without storage connected to computer system or network is referred as Webcam. In modern computers, the webcam can either be a separate component which is plugged into the computer or be an in-built integrated camera. The webcam can be used for video conference or video calling or online meeting using Google Meet, Zoom, MS Team and others services.

2. These devices recognize spoken words in two ways. The spoken words can either be recognized from a pre-defined vocabulary or be recognized from a known speaker after training of the input device. Whenever speaker utters a word from the pre-defined vocabulary, the Voice Based Input device may display the characters on monitor screen for verification by the speaker. However, some of these devices may process the speech without verification from the speaker.

3. Impact printer uses mechanical components for printing i.e., the characters and graphics are produced on a paper by striking whereas non-impact printer don't strike or impact the ribbon to print on paper. Impact printers produce banging noise during printing while non-impact printers work silently.

4. Refer text 8.10.1

5. The scanner can capture 360000 pixels per square inch